A COMPLETE STUDY GUIDE

Business Valuation

Concepts, Methods and Worked Applications — from Asset and Income Approaches to DCF, EVA, Start-ups, Digital Platforms and ESG

Built around the syllabus of ICAI Advanced Financial Management, Chapter 13 "Business Valuation" — with every illustration re-derived, every practical question solved, and new practice material

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Preface

This book is a study guide to **business valuation**. It follows, section by section, the syllabus of the chapter on Business Valuation in the Advanced Financial Management paper of the Institute of Chartered Accountants of India (ICAI): the conceptual framework of valuation, the approaches and methods (asset-based, earnings-based and cash-flow-based), the measurement of the cost of equity, relative valuation, other approaches (LBO, chop-shop, EVA, MVA and SVA), fair value, going-concern and non-going-concern valuation, and the valuation of distressed companies, start-ups, digital platforms, professional firms and the impact of ESG.

It is an independent study aid. It is **not** an official ICAI publication and it does not replace the Institute's study material or the latest amendments to the syllabus; the structure of the source chapter is used as a map, and the explanations, derivations, extra examples and exercises are original to this book.

What this book adds to a syllabus chapter

A syllabus chapter must be compact. A student who meets a formula in two lines, or a numerical illustration without its derivation, often memorises rather than understands. This guide therefore:

1. **Explains the reasoning** behind each method — why it exists, what it assumes, when it fails.
2. **Re-derives every worked illustration** from the source chapter step by step, and checks the arithmetic. Where the original compresses a step (for example, in its illustrative DCF the line labelled "terminal value" is already a present value), the book spells the step out and shows the alternative presentation as well. Knowing exactly what each line represents is part of mastering the method.
3. **Solves every practical question** in the source chapter in full (Chapter 21), with commentary on the technique being tested.
4. **Integrates the methods** in one running case (Chapter 19): a single company valued by the asset, income, DCF, EVA and market approaches, with reconciliation of the answers into a range.
5. **Supplies new practice problems** with solutions (Chapter 22 and Appendix D), a self-test bank (Appendix E) and guidance on how to answer theory questions (Chapter 23).
6. **Extends the syllabus** with supplementary chapters on securities, M&A, cash-flow construction, intangibles, the Indian context and valuation reports

(Chapters 24–30).

7. **Adds reference material**: a formula sheet, glossary, discount-factor tables and an index.

Structure

Part	Chapters	Content
I — Foundations	1–2	Why we value; standards of value; the vocabulary of valuation (PV, IRR, ROI, Gordon model, terminal value)
II — The Three Approaches	3–5	Asset-based, income-based, cash-flow-based
III — Discount Rates	6–7	CAPM, APT, beta estimation for unlisted firms and new businesses
IV — Relative Valuation and Other Approaches	8–12	Multiples, LBO, chop-shop, EVA, MVA, SVA
V — Fair Value and Special Entities	13–18	Fair value, going concern, distressed firms, start-ups, digital platforms, professional firms, ESG
VI — Cases, Practice and Integration	19–23	Integrated case, real-life case studies, solutions, practice, exam technique
VII — Extensions	24–30	Valuing securities, M&A and synergies, building FCFF from accounts, intangibles and goodwill, the Indian regulatory context, scenarios and valuation reports, a valuer's FAQ
Appendices	A–E	Formula sheet, glossary, discount tables, a second problem set, a self-test question bank; index

A note on numbers

Amounts follow the source illustrations (₹ lakh, ₹ crore, ₹ thousand, or foreign currency as the question states). Figures are rounded for readability; recomputation may differ in the last decimal. Companies in the new examples are fictional. Market inputs (rates, premiums, multiples) are illustrations, not forecasts. Where the book refers to law, standards or regulations, the references are for orientation: **check the current text** before relying on them in an examination or engagement.

How to Use This Book

For first reading. Read Parts I–III in order; they are the foundation. Work each chapter's "Check your understanding" questions before turning to the answers at the end of the chapter.

For examination revision. Use the *Exam focus* box at the end of each chapter (what examiners typically test, and the traps), the formula sheet in Appendix A, Chapters 21–23 for practice, and the chapter summaries. Examination questions on this topic are mostly of five kinds: (i) compute value by a named method (NAV, capitalisation of earnings, DCF); (ii) compute cost of equity/WACC with un-levering and re-levering; (iii) compute EVA/MVA; (iv) compute an exchange ratio for a merger; (v) write short notes on a theory topic (distressed firms, start-ups, ESG, going vs non-going concern).

For practitioners. The Pitfall boxes and Chapters 13, 14, 19 and 23 are the most relevant.

Conventions. Formulas are in **bold**. *Key idea* boxes state the central point. *Pitfall* boxes flag common errors. *Worked example* boxes give fully solved numbers. *Exam focus* boxes list what to remember for the paper.

Part I — Foundations

Chapter 1 — The Conceptual Framework of Valuation

1.1 Why valuation matters

The basic purpose of any enterprise is to earn profits in order to sustain itself and promote growth. Managements everywhere — in a sole proprietorship or a multinational — work toward this. As an enterprise grows, so does the number of its stakeholders: shareholders, lenders, employees, suppliers, customers, regulators. Each of them, sooner or later, asks the same question: **what is this enterprise really worth?**

Corporate valuation has a long history. The (United) East India Company, often described as the first corporation to be valued and floated through an initial public offering, is also an early example of a corporatised way of doing world trade — and of an institution focused on wealth maximisation, though not always by ethical means. Today almost any enterprise that generates positive cash flow and employs people feels the need to know its "value", whether to approach the local bank for debt financing or to prepare for an initial public offering.

Valuation is not a single technique but a body of approaches developed by practitioners, analysts, intermediaries and academics. It has become an inseparable part of **strategic financial management**. In a true sense it imbibes both the **science** (mathematics, models, data) and the **art** (judgement about the future) of finance.

1.2 Reasons for valuing an enterprise

The need for a proper assessment of an enterprise's value typically arises for:

#	Purpose	Explanation
(a)	**Information for internal stakeholders**	Owners and managers want to know the worth of what they run
(b)	**Comparison with similar enterprises**	Understanding relative management efficiency
(c)	**Future public listing**	Setting an issue price; assessing readiness
(d)	**Strategic planning**	Finding the *value drivers* of the enterprise; deciding how to deploy surplus cash
(e)	**Ball-park price for acquisition**	An approximate price to start negotiations in a merger or acquisition

The source chapter emphasises mergers and acquisitions: valuing the *target* is essential to gauge the price to be offered. A target may be **listed** or **unlisted**:

- If **unlisted**, the price is *negotiated* and must be acceptable to both parties.

- If **listed**, stock-market value (market capitalisation) forms a basis. But the market price is only a guide: it does not necessarily tell what the target is *worth*, because market prices depend on **market efficiency**, sentiment and liquidity. Also, in some cases only a small portion of total shares is quoted, so the price reflects a marginal slice of capital. This calls for further analysis.

1.3 Value, price and cost

Three words are often used interchangeably but must be kept apart.

- **Cost** — what was paid to acquire or build an asset (historical).
- **Price** — what the asset trades for in a market on a given day (observed).
- **Value** — what the asset is *worth* given the benefits it will produce and the risk of those benefits (estimated).

A balance-sheet figure is a *cost-based* number; a stock quote is a *price*; a DCF result is an *estimate of value*. Investing is the business of spotting when price and value diverge; valuation is the business of estimating the second.

> **Key idea.** Price is what you pay; value is what you get. A valuation is a *reasoned estimate*, expressed with an honest range, not a single "true" number.

1.4 Standards of value

The meaning of "value" depends on the purpose; professional practice therefore defines *standards of value*.

Standard	Meaning	Typical use
Fair market value	Price at which property would change hands between a willing buyer and willing seller, neither compelled and both reasonably informed	Tax and legal contexts
Fair value (reporting)	Price to sell an asset or transfer a liability in an orderly transaction between market participants at the measurement date	Financial reporting (Ind AS 113)
Investment value	Value to a specific investor given that investor's own synergies and required return	Acquisitions
Intrinsic value	Value implied by a rational analysis of fundamentals (present value of expected benefits)	DCF analysis
Liquidation value	Net proceeds from a break-up sale after paying liabilities	Distress, insolvency
Going-concern value	Value of a business expected to continue operating	Most valuations

Always state **the standard of value** and **the premise** (going concern or liquidation) at the start of any valuation.

1.5 Who needs which kind of value?

The source chapter notes that fair value looks different to different observers: a chartered accountant would think of a transaction **at arm's length**; a financial analyst, of the **present value of an entity in cash terms**; a speculative investor, of the **arbitrage opportunities** that arise when similar entities carry dissimilar values. In upward boom periods valuations can defy fair value (the dot-com boom is the standard example): when the downturn arrived, some of those companies vanished and others barely held their ground.

1.6 Three families of approach

All valuation methods fall into three broad families. The source chapter calls them:

1. **Asset-based valuation models** — value from the balance sheet of assets less liabilities.
2. **Earning-based models** — value from the earning capacity of the business.
3. **Cash-flow-based models** — value from the free cash flows the business will generate, discounted at an appropriate rate.

In addition there are **relative valuation** (comparing with similar firms) and several **other approaches** (LBO, chop-shop, EVA, MVA, SVA). The next chapters take them one by one.

Approach	What it asks	Strength	Weakness
Asset-based	What are the net assets worth?	Objective; gives a floor	Ignores future earning power
Earnings-based	What are maintainable earnings worth?	Simple; market-linked	Needs a reliable multiple/capitalisation rate
Cash-flow-based	What is the present value of future free cash flows?	Most complete; intrinsic	Sensitive to assumptions (WACC, terminal value)
Relative	What do comparable assets sell for?	Reflects market conditions	Peers are never identical

1.7 There is no single "correct" value

An important message of the source chapter is that, because every method has strengths and weaknesses, there is **no single answer** to which method is correct. It is better to determine a **range** of values — the **minimum acceptable to the seller** and the **maximum payable by the buyer** — and let the final price be settled by negotiation. Valuation supplies the boundaries of the negotiation, not its outcome.

1.8 The valuation process

1. **Define** the purpose, standard and premise of value, the valuation date and the interest being valued (100%, a minority stake, specific shares).
2. **Understand** the business, its industry and its competitive position.
3. **Gather and analyse** information: audited financial statements, tax returns, management projections, industry data, market data.
4. **Normalise** earnings and balance-sheet items (remove one-offs, non-operating items, related-party distortions).
5. **Select methods** and apply them (as many as the data allow).
6. **Reconcile** the results into a range; consider adjustments (control, marketability) where the standard of value requires.
7. **Report** with assumptions, limitations and sensitivity.

1.9 Information: the key to valuation

"Information holds the key to a successful valuation." The approaches in this book can be applied freely to *publicly traded* companies, where earnings, assets and the board's views on growth are readily available, and where audited statements are widely used by analysts to prepare review scorecards. In a developing economy such as India, many *privately held* firms in sunrise sectors (e-retail, hospitality, service management) attract venture capitalists and angel investors; directors approach chartered accountants for a valuation, and the qualified accountant is involved in the process. Valuing such firms without public information is demanding but, if done meticulously, yields real benefits — one reason why the cost-of-capital and beta issues of Chapter 7 matter.

1.10 Worked mini-example: three views of one company

Anand Foods Ltd has net assets (book) of ₹500 crore, maintainable annual profit after tax of ₹70 crore, and its shares are quoted such that market capitalisation is ₹620 crore.

- **Asset view:** ₹500 crore (book) — a floor, if assets are fairly stated.
- **Earnings view:** if a comparable company's P/E is 10 and Anand's shares are less liquid (adjust to 9), value ≈ 70 × 9 = ₹630 crore.
- **Market view:** ₹620 crore — but this reflects today's sentiment and the free float.

The three numbers are close, which gives comfort; if they were far apart (say ₹500 crore vs ₹900 crore), the valuer would have to explain *why* before choosing. This habit — **triangulation** — is the heart of professional valuation.

1.11 Common misconceptions

1. *"The balance sheet shows what the company is worth."* — It shows historical cost (adjusted for depreciation), not market worth or earning power.
2. *"The market price is the value."* — It is the price; it may be above or below value.
3. *"A higher valuation is always better."* — Not for a buyer; and a seller who insists on an unsupportable price loses credibility.
4. *"DCF is exact."* — It is exact arithmetic on uncertain assumptions.
5. *"One method is enough."* — Use several and reconcile.

Chapter 1 summary

- Valuation serves internal information, comparison, listing, strategy and acquisition pricing.
- Value ≠ price ≠ cost; state the standard and premise of value.
- Three core approaches (asset, earnings, cash flow) plus relative and other approaches.
- Aim for a *range* — seller's minimum to buyer's maximum — not one number.
- Information quality determines valuation quality.

Check your understanding

1. List five reasons for valuing an enterprise.
2. Why is the stock-market price of a target an imperfect guide to its worth?
3. Distinguish fair market value from investment value.
4. Explain why a range of values is preferable to a single value.
5. Why is information described as the "key" to valuation, especially for private firms?

Answers. (1) Internal information; comparison with similar enterprises; future listing; strategic planning; ball-park acquisition price. (2) It depends on market efficiency and sentiment, and only a marginal portion of shares may be traded. (3) Fair market value assumes a hypothetical willing buyer and seller; investment value reflects one investor's synergies and required return. (4) Each method has strengths and weaknesses; the range shows seller's minimum and buyer's maximum for negotiation. (5) Without reliable earnings, asset and growth information there is nothing to value; private firms lack public disclosure.

> **Exam focus.** Be ready to list the purposes of valuation, name the three approaches, and explain why there is no single correct method. Short-note questions often ask "fair value" and "going vs non-going concern".

Chapter 2 — Important Terms Associated with Valuation

Before applying any valuation model one must master the vocabulary the models rest on. Some of these terms evolved over time and now stand on their own as the precursors to the models themselves. This chapter covers present value, internal rate of return, return on investment, the perpetual growth (Gordon) model and terminal value.

2.1 Present value (PV) of cash flows

A rupee received a year from now is worth less than a rupee today, because money today can be invested to earn a return — the **time value of money**.

$PV = FV \div (1 + r)^n$

where r is the discount rate and n the number of periods. The factor $1/(1 + r)^n$ is the **discount factor**.

Worked example 2.1. ₹1,000 received twelve months from now, discounted at 10%: PV = 1,000 ÷ 1.10 = **₹909** (approximately). If received in two years: 1,000 ÷ 1.21 = ₹826.

Why it matters. Every cash-flow valuation is a sum of present values. The *choice of discount rate* is the central judgement: it must reflect the risk of the cash flows (Chapters 6–7).

Annuities and perpetuities

Pattern	Formula
Level payment C for n years (annuity)	$PV = C \times [1 - (1 + r)^{-n}] \div r$
Level payment C for ever (perpetuity)	$PV = C \div r$
Payment growing at g for ever	$PV = C_1 \div (r - g)$, $r > g$

Example. ₹100 per year for 5 years at 10%: 100 × (1 − 0.6209) ÷ 0.10 = **₹379.08**.

2.2 Internal rate of return (IRR)

IRR is the discount rate that makes the present value of cash **inflows** equal to the present value of cash **outflows**; equivalently, the rate at which **NPV = 0**. It can be pictured as the discount rate that equates the PV of inflows to the investment.

Worked example 2.2. A project costs ₹800 and returns ₹250, ₹300, ₹350 and ₹400 at the end of years 1 to 4. At 11% the NPV is
−800 + 225.23 + 243.49 + 255.92 + 263.49 = **₹188.13**.
Increasing the rate to about 20.7% makes NPV zero; the **IRR is about 20.7%**.

> **Decision rule (source chapter).** The higher the IRR of a project, the more likely it is selected for further investment. (A project is acceptable if IRR exceeds the required return.)

Limitations. IRR (i) assumes interim cash flows are reinvested at the IRR itself; (ii) ignores scale; (iii) can give multiple answers when cash flows change sign repeatedly. NPV is the safer criterion for ranking mutually exclusive projects.

2.3 Return on investment (ROI)

ROI = (Return ÷ Investment), expressed as a percentage: the return over the investment made, from a stakeholder's point of view.

2.3.1 A simple standalone transaction

A stakeholder sells shares valued at 1,400 that he bought for 1,000:

ROI = (1,400 − 1,000) ÷ 1,000 = **40%**

This is the *absolute rate of return on cash investment* for a single transaction.

2.3.2 When holdings change during the year

If a stakeholder's holdings were worth 1,000 at the beginning of the year, he invested an additional 200 during the year, and the investment is valued at 2,000 at the end, his ROI is calculated on the **average investment** held during the year:

- Return = 2,000 − 1,200 = 800
- Average investment = (1,000 + 1,200) ÷ 2 = 1,100
- **ROI = 800 ÷ 1,100 = 72.7%** (the source chapter rounds to 72%)

The average cost of investments recognises the *timing* of the additional investment. A more precise stakeholder may weight by actual days of the additional investment; those with several investments and portfolios use the average-cost approach.

2.3.3 ROI from the entity's point of view

For a company the most significant use of ROI is to calculate the return generated by each individual or incremental investment or project. A company that has launched two new business lines during the year can compute ROI on

each and decide whether to correct course or channel more funds into the more successful line. Note that **ROI is a historical ratio**: it supports corrective action, not prediction.

2.3.4 From ROI on projects to returns on total assets

A natural next question is how to measure returns against total investments — the total assets held by the enterprise. After all, all assets exist for wealth maximisation. This leads to the concepts of ROCE/ROIC and eventually EVA (Chapter 11).

2.4 Perpetual growth rate: the Gordon model

As in the cost-of-capital topic of the Intermediate paper, **Gordon's model** assumes dividends grow at a constant rate g forever. A potential investor looking for stable inflows takes the latest dividend, grows it, and relates it to the expected rate of return:

$P_0 = D_1 \div (Ke - g)$ (also written $D_0(1 + g) \div (Ke - g)$)

Worked example 2.3. Next year's dividend ₹12, Ke = 13%, g = 5%: **P_0 = 12 ÷ 0.08 = ₹150.** If the market price were ₹180, the implied growth would be g = Ke − D_1/P = 13% − 6.67% = 6.33%.

Why the Gordon model is rarely used directly

The source chapter notes that potential investors rarely use this model because:

1. there are more parameters to be factored in, and
2. dividends rarely grow perpetually at a steady rate.

It remains the darling of academicians because it fits neatly into a "constant rate" model for deliberation. Its real importance in valuation is as the engine of the **terminal value** (next section).

2.5 Terminal value (TV)

"Terminal" means the end of something. In the valuation world, to terminate is to exit from a particular investment or line of business. When an investor decides to pull out and book profits, he expects a fair value of the value created, and he also looks to the **horizon** to evaluate future cash flows, incorporating them into the selling price. Hence terminal value is also called **horizon value**: the value that an investor forecasts for his investment at the exit point.

Estimating TV. Mostly the perpetual growth (Gordon) model:

$TV_n = CF_{n+1} \div (r - g) = CF_n \times (1 + g) \div (r - g)$

Why TV matters. In most DCF valuations it is a large share — often more than half — of total value; the growth rate and assumptions must therefore be tested carefully (Chapter 5).

Illustration of TV

Free cash flows: Year 1 ₹120, Year 2 ₹140, Year 3 ₹160; WACC 10%; stable growth 4% afterwards.

- TV at the end of year 3 = 160 × 1.04 ÷ (0.10 − 0.04) = **₹2,773.3**
- Present value of TV = 2,773.3 ÷ 1.10^3 = **₹2,083.7**
- PV of explicit flows = 109.09 + 115.70 + 120.21 = 345.0
- **Value = 345.0 + 2,083.7 = ₹2,428.7**

TV share = 2,083.7 ÷ 2,428.7 = **85.8%**. Such dominance is typical and explains why the growth assumption deserves most scrutiny.

> **Pitfall.** The TV is a value *at the end of year n* and must be discounted back to today by $(1 + r)^n$. Adding an undiscounted TV to the PV of explicit flows overstates value.

2.6 Other basic ideas used throughout

Term	Meaning
Going concern	An enterprise is assumed to continue in operation for the foreseeable future
Free cash flow (FCF)	Cash left after operating costs, taxes and reinvestment
WACC	Weighted average cost of capital — blended required return of all capital providers
Capitalisation rate	Rate at which maintainable earnings are converted into value (reciprocal of a multiple)
EBIT / EBITDA	Operating profit before interest and tax / plus depreciation and amortisation
Enterprise value (EV)	Market value of equity plus debt minus cash
P/E ratio	Price per share ÷ earnings per share
Beta	Measure of a security's market (systematic) risk

Chapter 2 summary

- PV = FV/$(1 + r)^n$; NPV sums PVs; IRR is the rate that gives NPV = 0.
- ROI is historical; use the *average* investment where capital changes during the year.
- Gordon model: $P_0 = D_1/(Ke - g)$; a theoretical workhorse, rarely used raw.
- Terminal (horizon) value = value at exit; discount it to today.

Check your understanding

1. Compute the PV of ₹5,000 receivable in 3 years at 12%.

2. A stakeholder starts the year with investments worth ₹2,000, adds ₹500 during the year, and ends with ₹3,300. Compute ROI on the average-investment basis.
3. State the decision rule for IRR and two of its limitations.
4. Why is the Gordon model rarely used directly by investors?
5. A project's cash flows after year 4 will be ₹200, growing at 3% forever; the discount rate is 11%. Compute the TV at year 4.

Answers. (1) 5,000 ÷ 1.4049 = **₹3,559**. (2) Return = 3,300 − 2,500 = 800; average investment = (2,000 + 2,500) ÷ 2 = 2,250; ROI = **35.6%**. (3) Accept if IRR exceeds the required return; reinvestment assumption, multiple IRRs, scale-blindness. (4) Many parameters; dividends rarely grow at a steady rate forever. (5) 200 × 1.03 ÷ (0.11 − 0.03) = **₹2,575**.

> **Exam focus.** Expect short numerical questions on PV/IRR/ROI and the use of TV inside DCF. Remember: ROI on average investment; TV must be discounted; g must be below the discount rate.

Part II — The Three Approaches

Chapter 3 — The Asset-Based Approach

3.1 The idea

The simplest way to value a business is to ask what its **assets** are worth and subtract what it **owes**. Under the asset-based approach, the value of the shares of a target company is computed in terms of the **net assets acquired**. The method is not based on income generation; it is based on income-*generating assets*.

Its natural users are holding companies, real-estate and investment firms, asset-heavy manufacturers, companies in liquidation, and acquirers who intend to break up and sell parts of a business. It is **least important for IT and knowledge-based companies**, where "hard" assets matter little because the real assets are intellectual property rights and human resources.

The approach has three variants: **net asset value** (book value), **net realisable value** (liquidation value, or adjusted book value) and **replaceable value**.

3.2 Net asset value (book value) method

Value of the firm (as per the source chapter) = Fixed assets + Net current assets − Long-term debt

(The source writes "Net Fixed Asset = Fixed Assets + Net Current Assets − Long Term Debt"; this is the **net assets / net worth** attributable to shareholders.)

Advantage. It is the simplest method, using historical costs that are easily available.

Limitations.

1. A **balance sheet is not a valuation device**. Historical cost less depreciation has little relevance to current worth.
2. It ignores current values of assets, **even for intangibles** such as brand and intellectual property.
3. Therefore it offers only a **lower limit** to the value of shares of the target.

Worked example 3.1. Fixed assets ₹600, net current assets ₹250, long-term debt ₹300 (all in crore). Net asset value = 600 + 250 − 300 = **₹550 crore**. With 10 crore shares: ₹55 per share.

Adjusted book value

Often the starting point is improved by (i) removing fictitious assets (preliminary expenses, accumulated losses), (ii) adding unrecorded assets and (iii) deducting unrecorded liabilities. This produces an *adjusted* NAV that still uses book values for the ordinary items.

3.3 Net realisable value (NRV)

Also called **liquidation value** or **adjusted book value**, NRV is the realisable value of all assets after deducting liquidation expenses and paying off liabilities. In some cases, where the target is a going concern, liquidation expenses can be ignored.

Although it looks simple, computing NRV can be harder than it seems: a buyer in an off-market (distress) purchase is likely to offer the lowest prices. The method is **not popular** because it involves a **total break-up** of the target. It is generally useful where the acquirer wants to **sell one part of the business and integrate the remaining part** with its existing operations.

3.3.1 The source chapter's NRV example

The realisable values differ from book values as follows:

	Book values (₹)	Net realisable values (₹)
Long-term debt (term loan from ZB Bank)	10,000	10,000
Current liabilities	10,000	10,000
Total liabilities (A)	**20,000**	**20,000**
Non-current assets (B)		
PPE	50,000	40,000
Licences	10,000	30,000
	60,000	70,000
Current assets (C)		
Sundry debtors	50,000	45,000
Cash	10,000	10,000
	60,000	55,000
Net assets (B) + (C) − (A)	**1,00,000**	**1,05,000**

Thus the net realisable value of ₹1,05,000 exceeds the book net worth of ₹1,00,000 because the licences are worth far more than their cost, outweighing the shortfalls on plant and debtors. If there are **5,000 equity shares**, value per share = 1,05,000 ÷ 5,000 = **₹21** (versus ₹20 on book value).

Lesson. Realisable values can be higher or lower than book values; the direction depends on the assets.

3.3.2 Realisation percentages: a practical guide

Asset	Typical recovery as a % of book value
Cash and bank	100%
Quoted investments	Market value (less tax and costs)
Debtors	60–95% depending on age and quality
Inventory	40–80% (raw materials higher; work-in-progress and specialised stock lower)
Land and buildings	At market value (may exceed book substantially)
Plant and machinery	20–60% in a forced sale; higher in an orderly sale
Licences/IP	Depends on transferability; can exceed book if internally generated and unrecorded
Goodwill	Usually nil in liquidation

3.4 Replaceable value

This method values the business by the **cost to acquire a group of equivalent assets and liabilities in the open market**. It has an advantage over book value because it takes into account **proper current valuation**, and it is generally **slightly higher than net realisable value**, because quick asset disposal is not encouraged. For this reason many authors believe it is the **maximum price an acquirer would pay** for an equivalent business — since it could rebuild or buy the equivalent assets instead.

Limitation: only hard assets are taken into consideration; the **loyalty of staff**, relationships and organisational know-how cannot be replaced by buying equipment.

The three values compared

For a firm with net assets (book) 550:

Basis	Net assets	Per share (10 crore shares)
Net asset value (book)	550	55
Net realisable value (break-up)	680	68
Replaceable value (rebuild)	790	79

The ordering *book ≤ realisable ≤ replacement* is typical for an asset-heavy firm with undervalued land, though it is not a law.

3.5 When is the asset approach appropriate?

Situation	Reason
Liquidation or break-up planned	Going concern value is irrelevant
Holding/investment company	Value is the underlying assets (add a holding-company discount for tax/overhead leakage)
Real-estate and land-rich firms	Market values of land are the main value
Loss-making firm with no earning power	Assets are worth more than earnings

Situation	Reason
Floor value in negotiations	No rational seller accepts less than liquidation value
Companies in the early years with no earnings	A cost-based indication

3.6 Strengths and weaknesses

Strengths: objective; based on observable values; straightforward; provides a floor; useful for cross-checks.

Weaknesses: (i) *ignores earning power* — two firms with the same net assets but different profitability get the same value; (ii) ignores **goodwill** and unrecorded intangibles; (iii) uses historical cost (in the book-value variant); (iv) liquidation value ignores the going-concern premium; (v) requires valuation of each asset (time-consuming).

Conclusion (source chapter). The asset-based approach can depict the enterprise's net worth fairly correctly using the fundamental principle of **going concern**. However, it suffers from a major drawback: **it fails to consider the ability of the enterprise to generate future revenues and how market dynamics will affect future operations and cash flows.** That is the gap the income and cash-flow approaches fill.

3.7 Worked example: break-up of a business

Valmiki Engineering Ltd has these book values (₹ lakh): land 400, building 250, plant 600, inventory 300, debtors 350, cash 50, total 1,950. Liabilities: bank loan 700, creditors 250, other 50 (total 1,000). Estimated realisable values: land 640, building 200, plant 330, inventory 195, debtors 280, cash 50; liquidation costs 60.

- Realisable assets: 640 + 200 + 330 + 195 + 280 + 50 = **1,695**
- Less liquidation costs: 60 → 1,635
- Less liabilities: 1,000 → **NRV to equity = ₹635 lakh**

Book equity = 1,950 − 1,000 = 950. The NRV is 33% lower because plant and stocks fetch much less than book, while land fetches more. An acquirer who values Valmiki as a *going concern* by earnings would compare that figure with this floor of 635.

3.8 Pitfalls

1. **Deducting liabilities twice or omitting contingent liabilities** (litigation, guarantees).
2. **Ignoring tax** on gains when assets are sold at above book value.

3. **Mixing going-concern and liquidation values** within one calculation.
4. **Double-counting** revaluation reserves.
5. **Ignoring unrecorded intangibles** (brands, licences) in the book-value variant — as the NRV example shows, licences can be worth three times book.

Chapter 3 summary

- Net assets = assets − liabilities, valued on book (NAV), realisable (NRV) or replacement basis.
- Book value is a lower limit; replaceable value is often the upper limit an acquirer would pay.
- NRV matters for break-up; it is not popular because it implies total dismantling.
- The approach ignores future earning power — its main limitation.

Check your understanding

1. State the three variants of the asset approach and the typical order of their values.
2. Why is NAV only a "lower limit"?
3. When is NRV useful to an acquirer?
4. Compute NAV per share: fixed assets 800, net current assets 300, long-term debt 400, 20 lakh shares (₹ lakh).
5. Why is the approach least useful for an IT company?

Answers. (1) NAV (book), NRV (realisable), replaceable value; typically book ≤ NRV ≤ replacement for asset-rich firms. (2) It uses historical cost and ignores current values of assets and intangibles. (3) When it plans to sell one part of the business and integrate the rest. (4) 800 + 300 − 400 = 700 lakh; ÷ 20 lakh = **₹35**. (5) Its value lies in intellectual property and people, which the balance sheet does not capture.

> **Exam focus.** Remember the **NRV example** (₹1,00,000 → ₹1,05,000; ₹21 per share) and the statement that replaceable value is "slightly higher than NRV". The standard conclusion — fails to consider future revenue-generating ability — appears in "short notes" answers.

Chapter 4 — The Income-Based (Earnings) Approach

4.1 Overcoming the limits of the asset approach

The asset approach tells us what the enterprise *has*; the income approach tells us what it *earns*. The income-based approach attempts to overcome the drawbacks of asset-backed valuation by referring to **earning potential**. It is more suitable where the acquiring company intends to **continue the business of the target** for the foreseeable future, without selling or liquidating its assets. If any **additional earning** arises from the acquisition (synergy, economies of scale), it should also be considered in the valuation.

Two versions of the same idea are in use:

1. the **P/E ratio** (price-earnings multiplier, or its reciprocal, the *earning yield*);
2. the **capitalisation rate** applied to maintainable earnings.

4.2 PE ratio (earning yield multiplier) method

This method is generally used for valuing **listed companies** whose PE ratios are available. Its benefit: it takes into account the **expected growth rate** of the company as well as **market expectations**, because the market's P/E already embeds growth and risk.

Price per share = EPS × PE ratio

Although mainly followed for listed companies, the PE ratio of **equivalent companies or the industry** can be used to value shares of **unlisted** companies. The method serves as the **minimum acceptable price** to the shareholders of the target company.

Steps for an unlisted target

1. Choose the **PE ratio of an equivalent quoted company**.
2. Make an **adjustment downward** for the additional risk due to non-listing of shares (illiquidity, thinner information).
3. Determine the **future maintainable EPS**.
4. **Multiply** the EPS by the adjusted PE ratio.

Worked example 4.1. A comparable listed firm trades at P/E 12. The target is unlisted; the valuer applies a 20% reduction for non-listing: adjusted P/E = 12 × 0.80 = **9.6**. Future maintainable EPS = ₹8.16. Price per share = 8.16 × 9.6 = **₹78.34**. With 10 crore shares, equity value = **₹783 crore**.

Strengths and limits of PE valuation

- **Strength:** simple, market-linked, captures growth and risk expectations.
- **Weakness:** depends on the comparability of the peer; P/E is distorted by leverage and by one-off items in EPS; cannot be used when earnings are negative; the discount for non-listing (here 20%) is judgemental.

4.3 Capitalisation of earnings

In this method the value of the business is computed by **capitalising the company's expected annual maintainable profit** using an appropriate **required rate of return** (yield or discount rate).

Capitalised earning value = Expected annual maintainable profit ÷ Capitalisation rate (required earning yield)

4.3.1 Maintainable profit

Annual expected maintainable profit can be calculated using the **weighted average of previous years' profits**, after adjusting for synergy benefits or economies of scale. Steps:

1. Collect 3–5 years of profit after tax.
2. **Adjust** for extraordinary and non-recurring items, non-operating income, and accounting differences (normalisation, Chapter 19).
3. Compute a **weighted average**, giving higher weights to recent years if the trend is rising.
4. Adjust for **future** changes: new product, additional capacity, synergy, loss of a customer.

Worked example 4.2. Profits after tax (₹ crore): Year 1: 62; Year 2: 70; Year 3: 78; Year 4: 85. Weights 1, 2, 3, 4:
(62×1 + 70×2 + 78×3 + 85×4) ÷ 10 = (62 + 140 + 234 + 340) ÷ 10 = **77.6**.
If the acquirer expects ₹4 crore of additional after-tax profit from economies of scale, maintainable profit = **81.6**.

4.3.2 The capitalisation rate

The capitalisation rate depends on many factors. It can be approximated as follows:

Required earning yield = EPS ÷ Share price
or
Reciprocal of PE ratio = 1 ÷ PE ratio

Alternatively, **CAPM** can be used to give a required return (Chapter 6): Ke = Rf + β(Rm − Rf).

Worked example 4.3. Maintainable profit ₹81.6 crore; required yield 14%. Value = 81.6 ÷ 0.14 = **₹582.9 crore**. At 13% the value is ₹627.7 crore; at 15% ₹544.0 crore — notice the sensitivity: a 1-point change in the rate moves value by roughly 7%.

4.3.3 Link between the two versions

A P/E of 10 corresponds to a capitalisation rate (earning yield) of 10%; a rate of 14% corresponds to P/E ≈ 7.1. The two are two ways of expressing the same thing. The capitalisation method is *explicit* about the required return; the P/E method takes it from the market.

4.4 Capitalisation of earnings and the Gordon model

Capitalising a constant earning stream is a **zero-growth perpetuity**: value = E ÷ r. If earnings grow at g, the corresponding formula is $E_1 \div (r - g)$. So the capitalisation method silently assumes **no growth** unless the rate is adjusted: the rate used is effectively **(r − g)**. A valuer using a low rate for a growing company is doing exactly that. State it explicitly.

4.5 Advantages and disadvantages (source chapter)

- **Advantage:** it is a **forward-looking** approach (based on future maintainable profit) and is simple.
- **Disadvantages:** (i) estimation of expected future profit is subjective; (ii) differences in treatment of **extraordinary and exceptional items** across years and firms; (iii) no explicit view of cash flows and reinvestment; (iv) sensitive to the capitalisation rate.

4.6 Worked example: exchange ratio by earnings capitalisation (merger)

Hari Ltd and *Bala Ltd* plan a merger. Weighted average maintainable post-tax profits: Hari ₹300 crore; Bala ₹10 crore. Capitalisation rate for both: 8%. Hari has 3.5 crore shares; Bala 0.65 crore shares.

- Hari: 300 ÷ 0.08 = ₹3,750 crore → ₹1,071.43 per share.
- Bala: 10 ÷ 0.08 = ₹125 crore → ₹192.31 per share.

Earnings-based exchange ratio = 192.31 ÷ 1,071.43 = 0.1795 Hari shares per Bala share. (In Chapter 21 this is combined with the net asset approach to arrive at the fair-value ratio of 0.1787 as in the source question.)

4.7 Combining asset and earnings values

Often valuers compute value on **both** asset and income bases, then take a **weighted average**. Weights reflect the reliability of each (e.g., 1 : 3 for NAV : earnings capitalisation in the source chapter's merger question). If the earnings value is higher than the net asset value, the difference is **goodwill**:

Goodwill (super-profit method) = (Maintainable profit − Normal return on net assets) × Capitalisation factor

Example. Net assets 550; normal return 12% → 66; maintainable profit 81.6; super profit 15.6; capitalised at 14%: goodwill = 15.6 ÷ 0.14 = **₹111.4 crore**. Value = net assets + goodwill = 550 + 111.4 = **₹661.4 crore** — between the pure asset (550) and pure earnings (583–745) values.

4.8 Common pitfalls

1. Using *reported* profit including one-offs without normalising.
2. Applying a peer P/E without adjusting for growth/risk differences.
3. Using EPS after preference dividends inconsistently (preference dividends must be deducted to get earnings for equity; see the Eagle Ltd. question in Chapter 21).
4. Capitalising profit *before* tax at an after-tax rate (or vice-versa).
5. Forgetting that capitalisation values the *equity* (profit after interest) while profit before interest values the *enterprise*.

Chapter 4 summary

- The income approach values the business on maintainable earnings: **P/E × EPS** or **profit ÷ capitalisation rate**.
- For unlisted targets use a listed peer's P/E, adjusted downward for non-listing.
- Maintainable profit = weighted average of past profits adjusted for synergies and one-offs.
- Capitalisation rate ≈ EPS/price = 1/PE, or CAPM.
- Forward-looking but subjective; no view of reinvestment.

Check your understanding

1. State the four steps for valuing an unlisted company by the P/E method.
2. Compute the capitalised value: maintainable profit ₹90 crore, required yield 12.5%.

3. Compute weighted-average profit for years 1–3 of 40, 50, 60 with weights 1, 2, 3.
4. What does the capitalisation method assume about growth?
5. Why should preference dividends be deducted before applying a P/E to get equity price?

Answers. (1) Choose peer PE; adjust downward for non-listing; determine maintainable EPS; multiply. (2) 90 ÷ 0.125 = **₹720 crore**. (3) (40 + 100 + 180) ÷ 6 = **53.33**. (4) Zero growth unless the rate is net of growth. (5) PE × EPS gives price for *equity* shares; earnings available to equity are after preference dividends.

> **Exam focus.** Know the four steps of the unlisted-company P/E method and the capitalisation formula. The Eagle Ltd. and H Ltd./B Ltd. questions (Chapter 21) are standard applications.

Chapter 5 — The Cash-Flow-Based (DCF) Approach

5.1 From profit to cash

Profit is an accounting concept; owners are paid in **cash**. As opposed to the asset-based and income-based approaches, the cash-flow approach takes into account the **quantum of free cash** that is available in future periods and **discounts** it appropriately to match the **risk** of the flow. If the present value arrived at after applying the discount rate is more than the current cost of investment, the valuation of the enterprise is attractive to stakeholders as well as to externally interested parties such as stock analysts.

The DCF approach attempts to overcome the problem of **over-reliance on historical data** seen in the previous two methods. It is forward-looking and focuses on cash, not accruals.

5.2 The five steps

According to the source chapter there are essentially five steps in DCF-based valuation:

Step	Task
(a)	Arrive at the **free cash flows**
(b)	**Forecast** future cash flows (projected free cash flows)
(c)	Determine the **discount rate** based on the cost of capital
(d)	Find the **terminal value (TV)** of the enterprise
(e)	Find the **present values** of both the free cash flows and the TV, and interpret the results

5.3 Free cash flow

Free cash flow to the firm (FCFF) is the cash generated by operations available to all providers of capital (equity and debt):

FCFF = EBIT × (1 − t) + Depreciation − Capital expenditure − Increase in working capital

Alternatively, **FCFF = Profit after tax + Depreciation + One-off expenses − One-off incomes (+ after-tax interest if starting from net income) − Capex − Increase in working capital**.

Free cash flow to equity (FCFE) adds net borrowing and deducts interest:

FCFE = Net income + Depreciation − Capex − Increase in working capital + Net borrowing

FCFF is discounted at **WACC**; FCFE at the **cost of equity**.

5.3.1 Eliminating one-time items

In the source chapter's illustration the starting point is earnings after tax (EAT), from which **one-time incomes are eliminated** (they will not recur) and **one-time expenses are added back**, and **depreciation is added back** because it is a book entry (non-cash). A good practice, in line with Chapter 19 (normalisation), is to start from normalised operating profit.

5.4 The source chapter's DCF illustration, re-derived

Step (a): free cash flow for the base year (₹ thousand)

Item	2016-17	Remark
EAT	600	
Less: one-time incomes	(200)	one-time events eliminated
Add: one-time expenses	100	one-time events eliminated
Add: depreciation	100	book entry (non-cash)
Free cash flow	**600**	

Step (b): projections. Free cash flow estimated to grow at 5% a year; suitable assumptions are made for changes in working capital and investments in fixed assets:

	2017-18	2018-19	2019-20
Free cash flow (5% growth)	600.00	630.00	661.50
Less: changes in working capital	(50.00)	(30.00)	10.00
Less: investment in fixed assets	(50.00)	(50.00)	(20.00)
Adjusted free cash flow	**500.00**	**550.00**	**651.50**

Step (c): discounting at WACC 8%. Present-value factors 0.926, 0.857, 0.794:

	2017-18	2018-19	2019-20
Adjusted FCF	500.00	550.00	651.50
PVF @ 8%	0.926	0.857	0.794
PV	**463.00**	**471.35**	**517.29**

Sum of PVs = **₹1,451.64 thousand**.

Step (d): terminal value. Perpetual growth after year 3 is assumed at 3%. The source chapter writes TV = (CF at year 3 × growth) ÷ (WACC − growth) = (517.29 × 1.03) ÷ (0.08 − 0.03) = **10,656.17**.

Step (e): total. The source chapter then adds this TV to the PV of cash flows: **Total DCF of enterprise = 12,107.81 thousand** — "the value of the enterprise for a potential acquisition is approximately 12,108 thousand".

5.4.1 A careful reading of the illustration

The line labelled "TV" in the source (10,656.17) deserves a careful reading. It is computed as (517.29 × 1.03) ÷ (0.08 − 0.03), where 517.29 is the **present value of the year-3 cash flow** (651.50 × 0.794). It is therefore *already* the **present value** of the terminal value, and adding it to the PV of the explicit cash flows is correct. The longer way:

- TV at the end of year 3 = 651.50 × 1.03 ÷ (0.08 − 0.03) = **13,420.9**
- PV of TV = 13,420.9 × 0.794 = **10,656.2** (the same figure as the source's "TV")
- Enterprise value = 1,451.64 + 10,656.2 = **12,107.8** ✔ (with unrounded factors: 12,105.6)

So the source's answer of ₹12,107.81 thousand is right. What a student must not do is the *mixed* version — apply the growth formula to the **un-discounted** cash flow (651.50 × 1.03 ÷ 0.05 = 13,420.9) and then **add it without discounting**; that would overstate value by about 23%. When in doubt, show the terminal value *at the end of year 3* and its *present value* on separate lines.

> **Key idea.** Whatever the presentation, the correct enterprise value equals PV of explicit flows + PV of TV, and TV-at-horizon must be discounted back by n years.

5.5 The weakness: WACC and TV

"The DCF is indeed a revolutionary model for valuation as FCFs truly represent the intrinsic value of an entity. However, the whole calculation gravitates heavily on the **WACC** and the **TV**. In fact in many cases the TV is found to be a significant portion in the final value arrived by DCF. This means that the growth rate and underlying assumptions need to be thoroughly validated to deny any room for margin of error of judgment."

In the illustration: PV of explicit flows = 1,451.64; PV of TV = 10,656.17; **TV share = 88%**.

Sensitivity to the discount rate and growth (illustration)

Using the consistent formula on the same cash flows:

WACC \ g	2%	3%	4%
7%	12,329	15,174	19,916
8%	10,244	**12,106**	14,898
9%	8,755	10,061	11,889

(₹ thousand, enterprise value; exact discounting at $(1 + \text{WACC})^n$, TV from year-3 cash flow of 651.5.) A one-point change in WACC moves value by roughly −17% / +25% (8% → 9% / 7%). This sensitivity is the central reason to present

DCF values as ranges.

5.6 Deriving WACC

WACC = Ke × E/(D + E) + Kd × (1 − t) × D/(D + E)

where Ke is the cost of equity (Chapter 6), Kd the pre-tax cost of debt, t the tax rate, and E and D market values of equity and debt. In Chapter 7 we see how to compute Ke for a private firm or a new business using the un-levered (asset) beta.

5.7 Equity value and per-share value

DCF of FCFF gives **enterprise value**. To reach the value for shareholders:

Equity value = Enterprise value − Debt (and debt-like items) − Preference capital + Cash and non-operating assets

Value per share = Equity value ÷ Number of equity shares

Worked example 5.1. Enterprise value ₹1,478 crore; debt ₹300 crore; surplus cash and investments ₹40 crore; 10 crore shares. Equity = 1,478 − 300 + 40 = ₹1,218 crore → ₹121.8 per share.

5.8 Two-stage and multi-stage DCF

Companies rarely grow at one rate forever. A **two-stage model** uses a high growth rate for n years, then a stable rate. The source chapter's practical questions (the XYZ Ltd. and WXY Ltd. questions) use exactly this approach, with a different WACC in the high-growth and stable phases because the beta and capital structure change. Chapter 21 solves both.

Skeleton

1. Forecast FCFF for each year in the high-growth period (revenue → EBIT → NOPAT → reinvestment).
2. Discount at the high-growth WACC.
3. Compute the terminal-year FCFF with **stable-growth assumptions** (capex offset by depreciation; working-capital growth at stable rate).
4. TV = FCFF(n+1) ÷ (WACC_stable − g).
5. Discount TV by the cumulative factor for n years.

5.9 Strengths and weaknesses

Strengths: based on cash; forward-looking; explicit assumptions; theoretically sound intrinsic value; flexible (scenarios).

Weaknesses: sensitivity to WACC and TV; forecasting difficulty; false precision; inapplicable (without adaptation) to start-ups with no cash flow history, distressed firms and financial institutions.

5.10 Checklist for a defensible DCF

1. Cash flows normalised for one-offs, with working capital and capex tied to growth.
2. Discount rate consistent with risk, currency and capital structure.
3. Terminal growth ≤ long-run growth of the economy; terminal reinvestment consistent with growth.
4. TV share of value disclosed.
5. Bridge to equity complete (debt, cash, minorities, options).
6. Sensitivity table on WACC and g; cross-check against multiples.

Chapter 5 summary

- DCF = PV of free cash flows + PV of terminal value.
- Five steps: FCF → forecast → discount rate → TV → PV and interpret.
- The source illustration gives ₹12,107.81 thousand; its "TV" line of 10,656.17 is already a present value (TV at year 3 is 13,420.9).
- Value hinges on WACC and TV; the TV share is often very large.
- Equity value requires the bridge from enterprise value.

Check your understanding

1. List the five steps of DCF valuation.
2. Compute FCFF: EAT 400, one-time income 50, one-time expense 20, depreciation 60, capex 80, increase in working capital 30.
3. Why is TV a large share of value, and what should the valuer do about it?
4. Explain why WACC differs between the high-growth and stable phases.
5. Value: FCFF 100 (year 1), 110 (year 2); WACC 10%; TV growth 4% after year 2.

Answers. (1) FCF; forecast; discount rate; TV; PV and interpretation. (2) 400 − 50 + 20 + 60 − 80 − 30 = **320** (if starting from EAT as the source does). (3) Because cash flows beyond the explicit period are perpetual; validate growth, ROIC and WACC, show sensitivity. (4) Beta, debt ratio and cost of debt change as the firm matures. (5) PV explicit = 90.91 + 90.91 = 181.8; TV = 110 × 1.04 ÷ 0.06 = 1,906.7; PV = 1,575.8; value = **₹1,757.6**.

Exam focus. Reproduce the five steps; know that TV and WACC dominate; practise two-stage DCF (Chapter 21, questions 7 and 8) and remember to discount TV by the cumulative factor.

Part III — Discount Rates

Chapter 6 — Measuring the Cost of Equity: CAPM and APT

The discount rate is the price of risk. The DCF and income approaches are only as good as the rate applied. This chapter covers the two models the syllabus names for measuring the cost of equity: the **Capital Asset Pricing Model (CAPM)** and the **Arbitrage Pricing Theory (APT)**.

6.1 Cost of equity: what it is

The cost of equity (Ke) is the **return shareholders require** for bearing the risk of the company's shares. It is an *opportunity cost* — what investors could earn elsewhere at equal risk — not an accounting item and not necessarily a payment the company makes. A company creates value only if it earns more than this on equity-financed investments.

A useful way to see it: investing is "assuming risk for return". An alternative way to look at the value of an investment or portfolio is to view returns as the direct benefit of assuming risks.

6.2 The Capital Asset Pricing Model (CAPM)

R = Rf + β × (Rm − Rf)

Symbol	Meaning
R	Expected rate of return (the cost of equity, Ke)
Rf	Risk-free rate of return
β	Beta value of the stock
Rm	Market rate of return
(Rm − Rf)	Market risk premium (equity risk premium)

Worked example 6.1. Rf = 5%, Rm = 11%, β = 2.64. Ke = 5% + 2.64 × (11% − 5%) = 5% + 15.84% = **20.84%**. (This is the cost of equity in the source chapter's Illustration 1.)

6.2.1 Intuition

CAPM divides risk into two buckets: (i) a **risk-free part** — the reward for waiting (time value); and (ii) a **risk part relative to the market index** — the reward for bearing market-wide risk, scaled by the stock's sensitivity (beta). Only *systematic* risk is rewarded, because specific risk can be diversified away.

6.2.2 Inputs

- **Risk-free rate:** yield on a long-dated government security in the currency of the cash flows. For rupee cash flows, the G-Sec yield; for a valuation date,

the yield on that date.

- **Market risk premium:** estimated from long-run history (excess of equity return over government bonds), from the implied premium in the current market, or from surveys. Expect a range, not a point.
- **Beta:** from regression (stock returns on market returns) or, for unlisted firms, built bottom-up from comparables (Chapter 7).

6.2.3 Sanity checks on Ke

- Ke must exceed the risk-free rate and the firm's pre-tax cost of debt.
- Compare with peers' cost of equity and with the market return.
- A defensive stock ($\beta < 1$) should have Ke below the market return; a cyclical stock above.

6.2.4 Using CAPM for the capitalisation rate

In the income approach (Chapter 4), the capitalisation rate can be CAPM-based. In Illustration 2 of the source chapter: capitalisation rate = 4.5% + 0.9 × (12% − 4.5%) = **11.25%**; earning value = EBITDA ÷ 0.1125.

6.3 Strengths and weaknesses of CAPM

Strengths	Weaknesses
Intuitive; separates risk-free and risk parts	**Oversimplifies risk** — one factor (market)
Easy to implement; widely used	Beta is noisy and backward-looking
Consistent with portfolio theory	Market portfolio unobservable; the premium is uncertain
Transparent	Assumes diversified investors, frictionless markets

The source chapter identifies the greatest weakness precisely: "CAPM has gained massive popularity due to its intuitive-based approach of classifying risks into 2 buckets... However, this is also its greatest inherent weakness — the oversimplification of risks."

6.4 Arbitrage Pricing Theory (APT)

In the 1970s **Stephen Alan Ross**, professor and economist, introduced the idea that risk comes from **multiple factors** — a motley of **macro-economic factors** that can influence the return of an asset. The basic idea of APT is to **break risk into individual identifiable elements** that influence the overall risk in a proportion (called a **factor**); each factor is assigned its own beta, and the **sum of all the asset's sensitivities to n factors** gives the expected rate of return.

Formula:

$$\mathbf{R = Rf + \beta_1(RP_1) + \beta_2(RP_2) + \ldots + \beta_n(RP_n)}$$

where RP_i is the risk premium for factor i and β_i the asset's sensitivity to that factor. APT models E(R) as a **linear function of various macro-economic factors**, with sensitivity to each factor represented by a factor-specific beta coefficient.

6.4.1 Worked example 6.2

A stock's major influencers are the **interest rate** and the **sectoral growth rate**. Estimated betas and premia:

Factor	β	Risk premium
Interest-rate fluctuation	0.8	2.0%
Sectoral growth	1.2	3.5%

Rf = 6%. Expected return = 6% + 0.8 × 2.0% + 1.2 × 3.5% = 6% + 1.6% + 4.2% = **11.8%**.

Steps (from the source chapter): (a) calculate the risk premium for both factors (beta for factor 1 — interest rate; beta for factor 2 — sector growth); (b) add the risk-free rate.

6.4.2 Limitations of APT

APT by itself **does not specify the macro-economic factors** that must be tested for sensitivity; these have to be **judiciously developed by the financial analysts**, keeping in mind the economy they operate in. Estimating each factor's premium and each beta needs rich data. Hence in practice APT is more common in research than in routine valuations.

6.5 CAPM versus APT

Feature	CAPM	APT
Factors	One — market	Several macro-economic factors
Inputs	Rf, β, market premium	Rf, factor betas, factor premia
Theory	Equilibrium (mean–variance)	No-arbitrage
Practicality	High	Lower: factors must be chosen and estimated
Typical use	Corporate valuation	Research, portfolio analysis

CAPM is a special case of APT with a single factor (the market).

6.6 Other ways to estimate Ke

1. **Dividend-growth (Gordon) model:** $Ke = D_1/P_0 + g$. Useful where dividends are stable.
2. **Build-up approach:** Rf + equity premium + size premium + industry premium + specific-risk premium; common for private firms (but add-ons are judgemental).

3. **Earnings yield:** for a stable, no-growth firm, Ke ≈ E/P.
4. **Bond yield plus premium:** Kd + 3–5% as a rough check.

6.7 From Ke to WACC

Once Ke is known, the firm's **weighted average cost of capital** is **WACC = Ke × E/(E + D) + Kd × (1 − t) × D/(E + D)**.
In the source chapter's Illustration 1, WACC = 20.84% × 60% + 12% × (1 − 0.30) × 40% = 12.504% + 3.36% = **15.864%**.

6.8 Common mistakes

1. Mixing a rupee Rf with a dollar-based market premium.
2. Using a stale or one-day Rf and Rm.
3. Using a firm's own raw regression beta without checking its error.
4. Applying CAPM to cash flows of different currency.
5. Adding a "safety margin" to Ke while also using prudent cash flows (double-counting risk).

Chapter 6 summary

- Ke = Rf + β(Rm − Rf) (CAPM); APT: Rf + $\Sigma\beta_i \cdot RP_i$.
- CAPM's strength is simplicity; its weakness, oversimplification of risk.
- APT uses multiple macro-economic factors (Ross, 1970s) but leaves factor choice to the analyst.
- Ke feeds into WACC and capitalisation rates.

Check your understanding

1. Compute Ke: Rf 7%, Rm 13%, β 1.1.
2. APT: Rf 6%; factor 1 β 0.5, premium 3%; factor 2 β 1.5, premium 2%. Compute expected return.
3. State the greatest weakness of CAPM per the source.
4. Why is APT less used in routine valuations?
5. Compute WACC: Ke 18%, Kd 10%, tax 30%, E:D = 70:30.

Answers. (1) 7 + 1.1 × 6 = **13.6%**. (2) 6 + 1.5 + 3.0 = **10.5%**. (3) Oversimplification of risk into two buckets. (4) Factors are not specified and need many estimates. (5) 0.7 × 18 + 0.3 × 10 × 0.7 = 12.6 + 2.1 = **14.7%**.

> **Exam focus.** Write both formulas and be ready to compare CAPM and APT in a short note. The cost of equity from CAPM feeds Illustration 1 and Q10, Q16 in Chapter 21.

Chapter 7 — Estimating Beta and Valuing Unlisted Companies

7.1 The information problem

The valuation approaches described so far — asset-based, earnings-based and cash-flow-based — can be applied freely to publicly traded companies, where key information about earnings, assets employed and the board's opinion on future potential is readily available. The difficulty arises for **privately held enterprises**, especially in sunrise sectors such as e-retail, service management and hospitality. They hold out a lot of promise and are increasingly attractive to venture capitalists and angel investors, yet there is no public information, no traded share price and no observable beta.

The biggest challenge in valuing a private enterprise is arriving at the **cost of capital**, which in turn depends on its **beta**.

7.2 Why beta needs adjustment

Most publicly listed companies have **leveraged** capital structures, whereas privately owned firms may have **little or no debt**. Yet a strategic investor taking a stake would like to grow the firm further on **leveraged funds** — raising money through corporate bonds and debt instruments. As on the valuation date, however, the beta must reflect the **un-leveraged** position; hence we use the **unlevered beta** rather than a levered one.

The same problem arises for an **existing listed company** entering a **brand new line of business**. It should not use its own WACC to evaluate the project; instead it should assess the WACC **appropriate to the risk level of the new business**. For this it needs the **Asset Beta** (ungeared beta) of the business, which is then **re-geared** according to its own gearing level.

> **Key idea.** *Asset beta* measures only the **business risk** of the underlying project or asset; it contains no financial risk. *Equity beta* adds financial risk from leverage; therefore equity beta ≥ asset beta, and the two are equal only for a debt-free company.

7.3 The formulas

7.3.1 Asset beta from equity beta (with debt beta)

$$\beta_a = \beta_e \times [E \div (E + D(1 - t))] + \beta_d \times [D(1 - t) \div (E + D(1 - t))]$$

Symbol	Meaning
β_a	Ungeared or asset beta

Symbol	Meaning
β_e	Geared or equity beta
β_d	Debt beta
E	Equity (market value)
D	Debt (market value)
t	Tax rate

This is the WACC logic applied to betas: the asset beta is the weighted average of equity beta and debt beta, with weights E and D(1 − t).

7.3.2 The usual simplification: debt beta = 0

Generally it is assumed that the debt beta tends to be zero, because bond returns are not linked to the volatility of the market portfolio. Then:

$$\beta_a = \beta_e \times [E \div (E + D(1 - t))]$$

7.3.3 Re-gearing (equity beta from asset beta)

$$\beta_e = \beta_a \times [(E + D(1 - t)) \div E] = \beta_a \times [1 + (1 - t) \times D/E]$$

> **Pitfall.** Use the debt-to-equity ratio at *market* values for the peers when un-levering and the target's *target* debt-to-equity ratio when re-levering. D/E here is ratio of debt to equity, not debt to total capital: a "40:60 debt : equity" means D/E = 40/60.

7.4 The five-step method for the equity beta of a new business or an unlisted company

Step	Action
1	Identify **pure-play firms** (proxy companies engaged entirely in the same business) and their **equity betas**, to surrogate the equity beta of the new project or business
2	**De-gear** each proxy's beta to compute its **asset beta**, since different companies have different gearing levels
3	If there is only one proxy, continue with its asset beta; if there are several, take the **average** of asset betas (or choose the most appropriate company's asset beta)
4	**Re-gear** the asset beta to the capital structure of the appraising company, to reflect its financial risk: $\beta_e = \beta_a \times [E + D(1 - t)]/E$
5	Insert the computed β_e into **CAPM** to compute the required rate of return for the project or the value of the business

Worked example 7.1: Steps 1–4

Three listed pure-play firms in retail:

Peer	Equity beta	D/E (market)	Tax	Asset beta (debt beta 0)
P	1.20	50%	30%	1.20 ÷ (1 + 0.7 × 0.5) = 1.20 ÷ 1.35 = **0.889**
Q	1.00	25%	30%	1.00 ÷ 1.175 = **0.851**
R	1.50	80%	30%	1.50 ÷ 1.56 = **0.962**
Average				**0.901**

The target plans to run at D/E = 40% with tax 30%: β_e = 0.901 × (1 + 0.7 × 0.40) = 0.901 × 1.28 = **1.153**.

Step 5: with Rf = 6% and Rm = 12%: Ke = 6% + 1.153 × 6% = **12.92%**.

7.5 The source chapter's Illustration 1 — X Pvt. Ltd.

A privately held company X Pvt. Ltd. operating in the retail space is scouting for angel investors.

Data

- The company has achieved break-even this year; **EBITDA = 90**, which includes an extraordinary gain of 10 for the year, and a potential write-off of preliminary sales promotion costs of 20 is still pending.
- The **unlevered beta** based on the industry in which it operates is **1.8**; the average debt to equity ratio is **40:60**.
- The rate of return provided by risk-free liquid bonds is **5%**; internal assessment of market return for the industry is **11%**.
- EV is to be taken at a **multiple of 5 on EBITDA**.
- Future cash flows for the next three years: **Y1 100, Y2 120, Y3 150**.
- Pre-tax cost of debt **12%**; tax **30%**.

Question. What is the potential value to be placed on X Pvt. Ltd.?

Solution

1. **Levered beta** = 1.8 × [1 + (1 − 0.3) × 40/60] = 1.8 × 1.4667 = **2.64**.
2. **Adjusted EBITDA** = 90 − 10 (extraordinary gain) − 20 (pending write-off) = **60**.
3. **EV by the multiple** = 5 × 60 = **300**.
4. **Cost of equity** (CAPM) = 5% + 2.64 × (11% − 5%) = 5% + 15.84% = **20.84%**.
5. **WACC** = 20.84% × 60/100 + 12% × (1 − 0.30) × 40/100 = 12.504% + 3.36% = **15.864%**.
6. **Discounting** the three-year cash flows at 15.864%:

	Y1	Y2	Y3
Future cash flows	100	120	150
Discount factor	0.863	0.745	0.643
PV	86.30	89.40	96.45

Value of the firm = 272.15 (the source chapter's answer).

Commentary on Illustration 1

- The EBITDA-multiple value (300) and the DCF value (272.15) are two estimates; the DCF over three years carries no terminal value, so it understates a going concern's value; the multiple-based figure is higher.
- A reasoned *range* would be **₹272–₹300**, with the final price negotiated — the lesson of Chapter 1.
- The key technical points: (i) levered beta uses *D/E = 40/60*; (ii) weights in WACC use *debt = 40%, equity = 60%*; (iii) adjust EBITDA for one-offs *before* applying the multiple.

7.6 Beta and cash

If a peer holds large cash balances, its operating-asset beta is *higher* than the unlevered beta suggests, because cash has a beta near zero:
β_operating = β_unlevered ÷ (1 − Cash/Firm value).
Relevant for cash-rich sectors such as technology.

7.7 Valuing an unlisted company: a complete procedure

1. **Normalise** earnings and balance sheet (Chapter 19).
2. **Choose** a primary method: DCF (if forecasts reliable), capitalisation/PE (if stable), asset value (if asset-heavy).
3. **Compute Ke** using bottom-up beta (this chapter), re-levered at the target structure.
4. **Compute WACC** with a cost of debt reflecting the firm's own credit standing (not a peer's).
5. **Forecast** FCFF; compute TV.
6. **Adjust** for lack of marketability and control, if the standard of value requires.
7. **Cross-check** with peer multiples (adjusted for size and liquidity).

7.8 Common mistakes

1. Using book values of debt and equity for peers.
2. Ignoring the tax rate in de-levering/re-levering.
3. Forgetting to re-lever to the target's structure.
4. Using the target's *current* accidental capital structure when it will change.
5. Mixing D/E and D/V in the formulas.

Chapter 7 summary

- Unlisted firms and new businesses need an **asset beta** from pure-play peers.
- $\beta_a = \beta_e \times E/[E + D(1 - t)]$ (debt beta 0); $\beta_e = \beta_a \times [E + D(1 - t)]/E$.
- Five steps: find proxies → de-gear → average → re-gear → CAPM.
- Illustration 1: levered β 2.64, Ke 20.84%, WACC 15.864%, value 272.15.

Check your understanding

1. Why can't a listed company use its own WACC to appraise a new business?
2. Un-lever: $\beta_e = 1.4$, D/E = 60%, t = 25%.
3. Re-lever the result to D/E = 30%.
4. Compute the weight of equity in WACC if D:E is 40:60.
5. Why is EBITDA adjusted before applying a multiple?

Answers. (1) Risk of the new business differs; use asset beta re-geared. (2) 1.4 ÷ (1 + 0.75 × 0.6) = 1.4 ÷ 1.45 = **0.966**. (3) 0.966 × (1 + 0.75 × 0.3) = 0.966 × 1.225 = **1.183**. (4) 60/100 = **60%** (debt 40%). (5) To remove non-recurring items so the multiple is applied to maintainable earnings.

> **Exam focus.** Questions often give the unlevered (asset) beta and ask for re-levered beta, Ke, WACC and value (as in Illustration 1). Mind D/E vs D/V and the (1 − t) factor.

Part IV — Relative Valuation and Other Approaches

Chapter 8 — Relative Valuation

8.1 Intrinsic versus relative

The three approaches of Part II (asset, earnings, cash flow) arrive at the **intrinsic value** of an enterprise. **Relative valuation** arrives at a *relative* value using a **comparative analysis** with peers or similar enterprises. Increasingly, contemporary analysts use relative valuation in conjunction with the other approaches to **validate** the intrinsic value arrived at earlier.

The concept. One way to look at the practical implementation of fair value is to identify assets **similar to the ones held** by the acquiree so that values can be compared. This is a significant departure from intrinsic value. Trying to get a value nearest to the **market price** means that the valuation of a particular portfolio, or a divestiture in an entity, would happen at an agreeable price that fits into the **normal distribution**.

In one sense relative valuation is already used in a limited way whenever we speak of expected market returns or an index-based comparative: the more the pricing of an asset is correlated with similar assets in the market, the more inclusive it gets. When comparing bonds, the closer a bond's YTM to the government index of return, the more credible the pricing.

Relative valuation is also called **valuation by multiples**: it uses financial ratios to derive a desired metric (the *multiple*) and then compares it with that of **comparable firms** — those with similar asset and risk dispositions assumed to continue so over the comparison period. In the process there may be extrapolations to the desired range.

8.2 The four steps

1. Find the **drivers** that best represent the multiple.
2. Determine the results on the chosen driver(s) through **financial ratios**.
3. Find the **comparable firms** and carry out the comparative analysis.
4. **Iterate** the value of the firm obtained to smoothen out the deviations.

Step 1 — Choose the driver (and so the type of multiple)

Family	Multiples	Best used for
Enterprise-value based	**EV/EBITDA**, **EV/Invested capital**, **EV/Sales**	EV/EBITDA is the most popular and uses critical inputs; EV/Invested capital suits **capital-intensive** enterprises; EV/Sales suits companies that are **cash-rich, have a huge order book** and forecast organic growth through own capital
Equity-value based	**P/E ratio**, **PEG ratio**	P/E is the celebrated equity multiple; **PEG = P/E ÷ expected growth rate** is more suitable for relative valuation of **high-growth or sunrise industries**

Step 2 — Choose the right financial ratio

A **factor-based** approach helps. A firm earning most revenue from **exports** will be heavily influenced by foreign-exchange fluctuations; a pure P/E ratio may not reflect that (for instance, the impact that Brexit triggered on currency values). EV/Invested capital would be a **misfit** for a company that is light on core assets or has significant investment properties.

Step 3 — Choose the right comparables

Perhaps the most challenging step: **no two entities can be the same**, even if operating within the same risk and opportunity perimeter. Software company X may seem comparable to similar-sized company Y — same capital structure, operating environment, head count — yet X may derive revenue from dedicated service contracts with **Full-Time Equivalent (FTE) pricing**, whereas Y earns through a **Unit Transfer Pricing (UTP)** model. This changes the risk structure dramatically — "take benchmarks with a pinch of salt".

For a **niche market** where comparables are scarce, one may look beyond the current operating market to similarly structured companies in other industries. Alternatively, take closely relevant firms and perform a **regression** to arrive at the comparable metrics.

Step 4 — Iterate / extrapolate

Smooth out deviations and arrive at an estimate. Conclude: relative valuation is a **comparative-driven** approach that assumes the value of similar firms is a good indicator of the value of the tested firm.

8.3 Assumptions inherent in relative valuation

1. The **market is efficient** (it prices peers correctly on average).
2. The function between **fundamentals and multiples is linear**.
3. Comparable firms are **similar in structure, risk and growth pattern**.

Violations: bubbles and panics (market not efficient); non-linear value drivers (e.g., growth options); peers with different risk.

8.4 Enterprise value (EV): two ways

EV = Market capitalisation + Debt (+ preference + minorities) − Cash and cash equivalents.

The source chapter lists two approaches:
(a) take **entity value as the base** and adjust for debt values to arrive at EV; or
(b) take a **balance-sheet approach** and arrive at EV.

8.5 Illustration 2 — A Ltd. (per-share earning value and equity value)

A Ltd. made a gross profit of ₹10,00,000 and incurred indirect expenses of ₹4,00,000. Issued equity shares 1,00,000. Debt ₹3,00,000; surplus funds ₹5,00,000. Risk-free rate 4.5%; market rate 12%; beta 0.9.

Determine (a) per-share **earning value** and (b) **equity value** if the applicable EBITDA multiple is 5.

Solution

(a) Capitalisation rate by CAPM = 4.5% + 0.9 × (12% − 4.5%) = 4.5% + 6.75% = **11.25%**.

(₹ thousand)	
Gross profit	1,000
Less: indirect expenses	(400)
EBITDA	**600**
Earning value of the company = 600 ÷ 0.1125	5,333.33
Number of shares	1,00,000
Earning value per share	**₹53.33**

(b) Equity value using the EBITDA multiple of 5

(₹ thousand)	
EBITDA	600
EBITDA multiple	5
Capitalised value (EV)	3,000
Less: debt	(300)
Add: surplus funds	500
Equity value	**3,200**

Commentary

- Method (a) values 600 at a CAPM-based capitalisation rate ⇒ an implied multiple of 1/0.1125 = 8.9×, hence a higher value (5,333) than the market multiple of 5× (3,000).
- In (b), the bridge from EV to equity subtracts debt and adds *surplus* funds — the standard treatment.
- The two estimates (₹53.33 per share by earning value; ₹32.00 per share by multiple) differ widely; a valuer would examine whether the 5× peer multiple fairly reflects this company's growth and risk, or whether 11.25% understates the required return.

8.6 Illustration 3 — H K Ltd. (Enterprise value from a balance sheet)

The balance sheet of H K Ltd. (₹ thousand):

Assets		Liabilities	
Non-current assets	1,000	Shareholders' funds	800
Current assets: trade receivables	500	Long-term debt	200
Cash and cash equivalents	500	Current liabilities and provisions	1,000
Total	**2,000**	**Total**	**2,000**

Shares are actively traded; CMP ₹12 per share. Shareholders' funds represent 70,000 shares of ₹10 each and the rest is retained earnings. Calculate the EV.

Solution

Shares outstanding	70,000
CMP	₹12
Market capitalisation	**₹8,40,000**
Add: debt	₹2,00,000
Less: cash and cash equivalents	(₹5,00,000)
Enterprise value	**₹5,40,000**

Note: current liabilities and provisions (₹10,00,000) are *operating* liabilities (payables, provisions) and are not treated as debt. Cash is netted because a buyer of the enterprise would also get the cash.

8.7 Choosing and adjusting multiples in practice

1. **Same definition across firms** — e.g., EBITDA after or before lease rent (Ind AS 116 effects), same treatment of exceptional items.
2. **Calendarise** to the same period (trailing 12 months or forward).
3. **Use a median** of several peers; examine outliers.
4. **Adjust** for differences: growth, margin, size, leverage, liquidity.
5. **Apply** to the subject's *normalised* metric.
6. **Move from EV to equity** with the bridge.

8.7.1 What drives a multiple?

A multiple is a compressed DCF. Higher growth, higher return on capital and lower risk all justify higher multiples. For a stable firm:

- P/E ≈ payout × (1 + g) ÷ (Ke − g);
- EV/EBITDA rises with margin and growth and falls with capital intensity and risk.

So before applying a peer's multiple, ask whether the peer's growth/risk justify it *for your company*.

8.8 Limitations

- Peers never perfectly comparable; "pinch of salt".
- Inherits market mispricing: if the sector is in a bubble, relative valuation reproduces it.
- Accounting differences distort ratios.
- Hides assumptions behind a single number.
- Not helpful for unique businesses or start-ups without comparables.

8.9 Relative valuation vs intrinsic: using both

A sound practice: do a DCF and a multiples analysis; compare the *implied multiple* from the DCF with the peer multiples (e.g., DCF EV ÷ EBITDA = 8.6× vs peers 6–7×). If they diverge, find out why — different growth? lower margin? A **reverse DCF** (solving for the growth or margin that the market price implies) is a neat way to see what the market believes.

Chapter 8 summary

- Relative valuation = comparative analysis with similar firms using multiples; used to validate intrinsic value.
- Four steps: drivers → ratios → comparables → iterate.
- EV multiples (EV/EBITDA, EV/Invested capital, EV/Sales) and equity multiples (P/E, PEG).
- Assumptions: efficient markets, linearity, comparable structure and risk.
- Illustration 2: earning value ₹53.33/share, equity value ₹3,200 thousand; Illustration 3: EV ₹5,40,000.

Check your understanding

1. Name the EV-based and equity-based multiples and when each is suitable.
2. Why might two apparently similar software companies deserve different multiples?
3. Compute EV: market cap 900, debt 250, cash 120, minority interest 30.
4. What is PEG and for whom is it suitable?
5. State the three assumptions of relative valuation.

Answers. (1) EV/EBITDA (general), EV/Invested capital (capital-intensive), EV/Sales (cash-rich, large order book); P/E and PEG (growth). (2) Different revenue models (FTE vs UTP) imply different risk. (3) 900 + 250 + 30 − 120 = **₹1,060**. (4) P/E ÷ growth; high-growth/sunrise industries. (5) Efficient market; linear relationship; comparable structure, risk and growth.

> **Exam focus.** Know the four steps and the multiple families; be able to compute EV from a balance sheet (Illustration 3) and equity value from EBITDA multiple (Illustration 2).

Chapter 9 — Contemporary Approaches: PER, Goodwill and Leveraged Buy-Outs

9.1 Why "other approaches" exist

As business became technology-driven, and management teams began to position themselves as "value creators" through inorganic growth and conglomerate-style structures, the concept of value and valuation underwent a paradigm shift. The "MNC" tag and inorganic growth gave rise to complex structures and tiers of management styles. Yet the game-plan stays the same: *separate the seed from the chaff* using the time-honoured asset-based or income-based approaches, or the more mature cash-flow approach, or a combination of all three. But in bull markets analysts liberally use terms like "PEs", "exit multiples", "LBOs" and "brand value".

Some of these concepts arose from the peculiarities of certain industries. An **internet company** has virtually zero fixed assets but a robust online presence and huge brand-recall value. That gave rise to metrics such as **price per page visited** or, for an online play store, **price per subscriber**. Yet discerning analysts still ask for the cash-to-sales ratio and apply a DCF before they "put the money in the pot".

9.2 Goodwill-based approach

Another contemporary approach is the **goodwill-based** valuation. A retail giant desperate to acquire a traditional mom-and-pop store in a particular hotspot (so as to secure the location) can reasonably adopt this method:

1. First take an **asset-based valuation** (net assets).
2. Then value the **goodwill separately** by linking a multiple to the store's **annual sales** or its **footfall**.

Worked example 9.1. A store with net assets of ₹2 crore, annual sales ₹6 crore, in a high-footfall location. The acquirer pays goodwill at 0.5× annual sales = ₹3 crore. Total price = 2 + 3 = **₹5 crore**. The goodwill represents the value of location and customer base, which the asset approach cannot capture. The key judgement is the multiple, which should be supported by comparable transactions or by the profit the location earns above a normal return.

9.3 Price Earning Ratio (PER)

The price-earning ratio **equates the EPS to the price prevailing on the stock market** — the logic being that the market prices the stock on its fundamentals, so you need not look beyond it to value the stock.

Example from the source chapter: EPS ₹40; average share price over the last quarter ₹50; PER = 50 ÷ 40 = **1.25**.

Note the source's observation that PER is a **relative figure**; comparison *across industries in the same sector* can give a more **median PER** that may be acceptable for valuation. (Conventionally P/E is price ÷ EPS and is typically a larger number, e.g. 10–30; the example is simply to illustrate the arithmetic.)

Where PER is used: to estimate value from earnings (price = EPS × PER), to compare companies, and to infer the market's growth expectations.

9.4 Leveraged Buy-Outs (LBOs)

The increasingly complex nature of commerce gave rise to a new class of "strategic investors": **Private Equity (PE) firms**. They scout for enterprises "in the rough", acquire them using a clever mix of **debt and equity** (typically **70:30 debt to equity**), and then target to **sell within a medium-term period, say 3 to 5 years**. In the process they leverage the debt and create value (both perceived and real) and then either spin off the management control to another entity for a price or go for an outright sale.

9.4.1 The mechanics

1. **Sources and uses.** Purchase price (+ fees) funded by debt (senior and subordinated) and sponsor equity.
2. **Operating forecast** — revenue, EBITDA, capex, working capital.
3. **Debt schedule** — interest and principal; cash flow after debt service used to repay debt.
4. **Exit.** Exit EV = EBITDA (or similar) × exit multiple; equity = EV − remaining debt.
5. **Returns** — sponsor's IRR and multiple of money.

9.4.2 The source chapter's LBO example

X is a small software company providing a niche data-control and testing service, with 60 employees and some steady contracts, generating an EBITDA of ₹100 lakh per year.

- A **venture capitalist (VC)** convinces the managing director to sell the **majority stake** to him, valued at a premium of **100% per share over the**

book value, plus a **one-time goodwill payout of ₹50 lakh**, using an income-based valuation approach. Total consideration: **₹250 lakh**.

- The VC then ropes in a **banker to pump in ₹200 lakh** for the acquisition-cum-expansion as well as for brand marketing. The remaining **₹50 lakh** gap is the VC's contribution as promoter equity towards securities premium.
- Since the core operations team is not dismantled, the company easily achieves approximately **20% average growth** in each of the next 3 years.
- At the end of the third year, the VC puts the company on the "sale block" and attracts a leading MNC. The exit multiple is **7× EBDAT**.

The hypothetical entity value (₹ lakh):

	Y_0	Y_1	Y_2	Y_3
EBIDTA	100.00	120.00	144.00	178.00
Less: interest (on debt ₹200 lakh at 18%, repaid linearly over 6 years)	36.00	30.00	24.00	18.00
EBDTA	64.00	90.00	120.00	160.00
Less: taxes @ 30%	19.20	27.00	36.00	48.00
EBDAT	**44.80**	**63.00**	**84.00**	**112.00**
Multiple				7
Capitalised value at end of Y_3				**784**
Less: debt				(100)
Equity value				**684**

Check. Debt outstanding at the end of year 3 is $200 - 3 \times 33.33 = 100$; interest falls as debt is repaid (18% of 200, 166.7, 133.3, 100 = 36, 30, 24, 18) ✔. *Note on Y_3 EBITDA:* 20% growth on 144 would give 172.8; the source table shows 178 and the downstream numbers (160, 48, 112, 784, 684) are built on 178. With 172.8, EBDAT would be $(172.8 - 18) \times 0.7 = 108.4$ and equity value $7 \times 108.4 - 100 =$ **₹658.5 lakh**. The conclusion is unchanged.

Return to the VC. The VC put in ₹50 lakh of equity (if we take the 250 consideration funded by ₹200 lakh bank and ₹50 lakh VC equity) and receives an equity value of ₹684 lakh in 3 years — a multiple of 13.7× and an IRR of about 139% a year ($13.68^{1/3} - 1$). Even allowing for sharing with management, the leverage effect is dramatic. (The book does not compute the IRR; the figures above are the natural implication.)

9.4.3 Where the value comes from

1. **Growth** in EBITDA (20% a year).
2. **Debt pay-down** from operating cash flow.
3. **Multiple**: here exit at 7× — what the entry multiple was matters (entry EV/EBITDA = 250/100 = 2.5×!). A jump from 2.5× to 7× is an enormous

multiple expansion, the biggest part of the return; this is why the example should be read as an illustration of mechanics, not as an expected outcome.

9.4.4 The risk of an LBO

"One of the prime casualties in an LBO model is that the **future cannot be predicted with exactitude**. Thus, if at the end of the third year the industry is caught in a **cyclical slowdown**, the VC will find itself saddled with a huge loan and burgeoning interest costs that are difficult to recycle." High leverage magnifies both gains and losses.

Risk factors: cyclical downturn; interest-rate rise; covenant breach; failure to deliver growth; exit-market closure; key-person departure.

9.5 LBO as a valuation tool (ceiling price)

By reversing the model, a valuer can compute the **maximum price a financial buyer could pay** for a target IRR (say 20–25%). This gives a ceiling for a financial buyer; strategic buyers with synergies may pay more.

Worked example 9.2. Target EBITDA 200; sponsor wants 25% IRR over 5 years; exit at 7× EBITDA; debt capacity 4× EBITDA = 800 at 10%, with 60% of free cash flow after interest swept to debt reduction. If year-5 EBITDA is 300 and remaining debt 400, exit equity at 7× = 2,100 − 400 = 1,700. Required entry equity = $1{,}700 \div 1.25^5 = 1{,}700 \div 3.0518 =$ **557**. Maximum entry EV = equity 557 + debt 800 = **1,357** → 6.8× EBITDA. Paying more than ~6.8× reduces the sponsor's IRR below target.

9.6 Choosing among approaches

Situation	Contemporary tool	Caveat
Internet/e-commerce	Price per user/page/subscriber	Check cash-to-sales and DCF
Retail hotspot	Asset + goodwill by sales/footfall	Support the multiple
Listed comparables	PER	Relative, not absolute
PE acquisition	LBO	Sensitive to leverage and exit

Chapter 9 summary

- Contemporary approaches arise from industry peculiarities; they supplement, not replace, asset/income/cash-flow methods.
- Goodwill approach: asset value plus goodwill by sales/footfall multiple.
- PER equates EPS to price; relative figure.
- LBO: ~70:30 debt-equity, 3–5 year horizon; returns from growth, debt pay-down and exit multiple; major risk is cyclical downturn.

Check your understanding

1. Why do internet companies lead to metrics like "price per subscriber"?
2. Outline the goodwill-based approach.
3. What debt-equity mix is typical of an LBO and what is the horizon?
4. In the source LBO example, why is the "equity value" 684 and not 784?
5. Why should an LBO analysis be treated cautiously?

Answers. (1) Few fixed assets; value lies in user base and brand. (2) Net assets plus goodwill valued by multiple of annual sales or footfall. (3) About 70:30; 3–5 years. (4) 784 is capitalised enterprise value; deduct debt of 100. (5) Heavily dependent on forecast and exit; cyclical slowdown can leave a heavy debt burden.

> **Exam focus.** Reproduce the LBO table, the 70:30 mix and the sentence on cyclical-slowdown risk.

Chapter 10 — The Chop-Shop (Break-Up Value) Method

10.1 The idea

The chop-shop method is named after the practice of buying a car and selling it in parts, because the parts together fetch more than the whole. In business: the approach attempts to identify **multi-industry companies that are undervalued** and that would have more value if **separated from each other**. In other words, an attempt is made to **buy assets below their replacement value**. It is also called the **break-up value** or **sum-of-the-parts** approach.

It suits conglomerates whose segments belong to different industries, where the market applies a blended multiple that fits none of the segments well — the **conglomerate discount**.

10.2 The three steps

Step	Task
1	Identify the firm's various **business segments** and calculate the **average capitalisation ratios** for firms (pure-play) in those industries
2	Calculate a **"theoretical" market value** based upon each of the average capitalisation ratios
3	**Average** the "theoretical" market values to determine the **"chop-shop" value** of the firm

Capitalisation ratios are ratios of market capitalisation (or value) to a financial metric: **capitalisation/sales**, **capitalisation/assets**, **capitalisation/operating income**. Each ratio is taken from *pure-play* firms in the segment's industry.

10.3 Illustration 4 — Cornett GmbH

*Using the chop-shop (or break-up value) approach, assign a value for Cornett GmbH, whose stock is currently trading at a total market price of **€4 million**. For Cornett, the accounting data are set out in three business segments: consumer wholesaling, specialty services and assorted centres.*

Data for the firm's segments

Business segment	Segment sales	Segment assets	Segment income
Consumer wholesaling	€1,500,000	€750,000	€100,000
Specialty services	€800,000	€700,000	€150,000
Assorted centres	€2,000,000	€3,000,000	€600,000

Industry data for pure-play firms

Business segment	Capitalisation/sales	Capitalisation/assets	Capitalisation/operating income
Consumer wholesaling	0.75	0.60	10.00
Specialty services	1.10	0.90	7.00
Assorted centres	1.00	0.60	6.00

Solution

(1) Using the capitalisation/sales ratio

Segment	Ratio	Segment sales	Theoretical value
Consumer wholesaling	0.75	€1,500,000	€1,125,000
Specialty services	1.10	€800,000	€880,000
Assorted centres	1.00	€2,000,000	€2,000,000
Total			**€4,005,000**

(2) Using the capitalisation/assets ratio

Segment	Ratio	Segment assets	Theoretical value
Consumer wholesaling	0.60	€750,000	€450,000
Specialty services	0.90	€700,000	€630,000
Assorted centres	0.60	€3,000,000	€1,800,000
Total			**€2,880,000**

(3) Using the capitalisation/operating income ratio

Segment	Ratio	Segment income	Theoretical value
Consumer wholesaling	10.00	€100,000	€1,000,000
Specialty services	7.00	€150,000	€1,050,000
Assorted centres	6.00	€600,000	€3,600,000
Total			**€5,650,000**

Average theoretical value = (4,005,000 + 2,880,000 + 5,650,000) ÷ 3 = 12,535,000 ÷ 3 = **€4,178,333 ≈ €4,178,000**.

Interpretation

The chop-shop value (€4.178 million) is slightly above the current market price (€4 million): the firm is **modestly undervalued** (by about 4.5%) on a break-up basis. The wide dispersion of the three estimates (€2.88 m on assets, €4.0 m on sales, €5.65 m on income) is itself informative: the **asset-based** figure is low because the assorted centres use a lot of assets; the **income-based** figure is high because they earn high income relative to sales. Averaging blends these views.

10.4 Variations and refinements

1. **Weighted average.** If one ratio is more reliable for a segment (e.g., income for stable businesses), weight it more.
2. **Use EV multiples** (EV/EBITDA) instead of equity-level ratios, then bridge to equity.
3. **Deduct unallocated corporate costs** (head-office) capitalised at an appropriate multiple.
4. **Adjust for taxes** on the sale of segments and for transaction costs.
5. **Adjust for intra-group transactions** and shared assets.
6. **Add non-core assets** (investments, surplus land) at market value, and **deduct net debt**.
7. **Use segment-specific DCFs** rather than ratios where cash flows are forecastable.

10.5 Worked example: sum-of-the-parts for a holding group

Bhavani Industries has three businesses (₹ crore):

Segment	EBITDA	Peer EV/EBITDA	Segment EV
Cement	300	8.0×	2,400
Packaging	120	10.0×	1,200
Hotels	80	12.0×	960
Total	**500**		**4,560**

- Less: capitalised unallocated corporate costs (₹20 crore at 8×): (160)
- **Group EV (SOTP)** = 4,400
- Less: net debt (900) → **Equity value = 3,500**.

If the market capitalisation is ₹2,800 crore, the market is applying a **20% conglomerate discount** (1 − 2,800/3,500). A buyer who could unlock value by separating (or by improving capital allocation) sees the opportunity; but the discount may reflect governance and tax leakage costs that a break-up would not necessarily remove.

10.6 When does chop-shop make sense?

Favourable	Unfavourable
Diversified group with distinct segments	Integrated business with shared assets and synergies
Good segment disclosure (Ind AS 108)	Poor segment data
Market values the group at a discount	Group trades at a premium (market sees synergy)
Buyer plans to sell parts	Buyer plans to run as one
Clear pure-play comparables	Segments with no peers

10.7 Link to asset-based valuation

The chop-shop approach complements the asset approach (Chapter 3): both consider the entity as separable parts. Where the asset approach values the *assets* separately, chop-shop values the *businesses* separately on market ratios. Replaceable value (Chapter 3) and chop-shop both aim at "buying assets below their replacement value".

10.8 Limitations

1. **Peers**: finding pure-play peers for each segment.
2. **Segment data**: allocation of common costs and assets is arbitrary.
3. **Synergies lost**: break-up may destroy synergy and incur separation costs, taxes and dis-synergies.
4. **Averaging** of ratios blends methods with different reliability.
5. **Not a prediction**: the break-up value says what the parts are worth *if* separated, not that separation will occur.

Chapter 10 summary

- Chop-shop = break-up/sum-of-the-parts value: segment ratios from pure-play firms applied to segment metrics and then averaged.
- Cornett GmbH: values €4.005 m (sales), €2.88 m (assets), €5.65 m (income); average **€4.178 m** vs market €4 m.
- Useful for conglomerates trading at a discount; limited by comparability and separation costs.

Check your understanding

1. List the three steps of the chop-shop method.
2. Why average the values from three different ratios?
3. A segment has sales 600, assets 450, income 90; ratios: cap/sales 0.9, cap/assets 0.8, cap/income 7. Compute the average theoretical value.
4. What does a gap between SOTP value and market value indicate?
5. Name two refinements to the basic method.

Answers. (1) Identify segments and average capitalisation ratios; compute theoretical values; average them. (2) Each ratio has biases; averaging reduces dependence on one. (3) 540, 360, 630 → average **510**. (4) A conglomerate discount (or a governance/tax cost). (5) Deduct unallocated corporate costs; use EV multiples and net debt; weight ratios.

Exam focus. Practice a Cornett-style computation: three tables, a final average. Show all three theoretical values clearly.

Chapter 11 — Economic Value Added (EVA) and Market Value Added (MVA)

11.1 The gap EVA fills

Up to now we have seen performance metrics such as ROI and ROCE, and approaches based on the asset base, earnings or free cash flows, to find the "worth" of the entity. **Why EVA?** What gap is it trying to fill that the others couldn't?

The answer lies in the way EVA looks at the **performance of the management** of a company. All approaches seen so far were largely a function of *number-crunching*. EVA tries to make management **more accountable for individual decisions** and the **impact of decisions on the path to progress** of the company.

Illustration of the idea. Suppose management has two dissimilar but equal-risk opportunities. It would probably choose the project that **breaks even earlier**, cutting down the risk of future losses. But if both projects had been undertaken, each would have generated a **positive IRR**, though the second would have had a longer pay-back period. This impact of strategic decision-making comes out evidently in EVA computations, whereas under earlier techniques this performance-driven aspect would never have been highlighted. EVA evaluates whether **returns cover the cost of capital**.

EVA is therefore not only a valuation technique but also a **holistic method of evaluating financial performance**: it measures the economic contribution of a company. Its core concept is that a company generates **value only if there is a creation of wealth in terms of returns in excess of the cost of capital invested**. EVA insists on **separation of the firm's operation from its financing**. If EVA is negative the company is not generating value from the funds invested; if positive, it is.

11.2 The formula

EVA = NOPAT − (Invested capital × WACC) or **EVA = NOPAT − Capital charge**

- **NOPAT** (Net Operating Profit After Tax) = **EBIT − tax expense** (i.e., EBIT × (1 − t)). The logic: find the cash returns that business operations would make after tax payments. Depreciation is left untouched here, being an operational expense for the limited purposes of EVA.
- **Capital charge** = Invested capital × WACC.

11.3 Steps to compute EVA

1. **Compute NOPAT.** Start with EBIT, deduct tax on operating profit, and adjust for **non-cash expenses** (e.g., the **provision for bad and doubtful debts**, which is just a book entry) to arrive at the "actual" cash earnings.
2. **Compute invested capital.** From published financials: **total assets less non-interest-bearing current liabilities** (sundry creditors, billing in advance, tax liabilities, etc.); equivalently, **equity plus long-term debt**, generally at the *start* of the year. Make adjustments for non-cash elements (e.g., provision for bad debts) in both invested capital and NOPAT.
3. **Compute WACC** (using market or book weights as the question directs).
4. **Capital charge** = Invested capital × WACC.
5. **EVA** = NOPAT − Capital charge.

11.4 Illustration 5 — A Ltd.

Compute the EVA of A Ltd. from the following (all figures in ₹ lakh).

Profit and loss statement		Balance sheet	
Revenue	1,000	PPE	1,000
Direct costs	(390)	Current assets	300
SG&A	(200)	**Total**	**1,300**
EBIT	**410**	Equity	700
Interest	(10)	Reserves	100
EBT	400	Non-current borrowings	100
Tax expense	(120)	Current liabilities and provisions	400
EAT	**280**	**Total**	**1,300**

Assume that a bad-debt provision of ₹20 lakh is included in SG&A and the same amount is reduced from trade receivables in current assets. Pre-tax cost of debt 12%; tax rate 30%; cost of equity 8.45%. The current liabilities also include a tax liability of 100.

Solution

Step I — NOPAT

EBIT	410
Less: taxes (30% × 410)	(123)
Add: non-cash expenses (bad-debt provision)	20
NOPAT	**307**

Step II — Invested capital

Total assets	1,300

Less: non-interest-bearing liabilities (current liabilities and provisions)	(400)
	900
Add: non-cash adjustment (provision)	20
Invested capital	**920**

(Check: equity 700 + reserves 100 + non-current borrowings 100 = 900 before the adjustment.)

Step III — WACC (weights from equity 800 and debt 100 on 900):

WACC = (800/900 × 8.45%) + (100/900 × 12% × (1 − 0.30)) = 7.511% + 0.933% = **8.44%**.

Step IV — Capital charge = 920 × 8.44% = **77.65**.

Step V — EVA = 307 − 77.65 = **₹229.35 lakh**.

Interpretation

A positive EVA of ₹229 lakh means A Ltd. earned ₹229 lakh above the return required by capital providers. Equivalently NOPAT/IC = 307/920 = 33.4% vs WACC 8.44%: a spread of nearly 25 points.

11.5 Why NOPAT and not net income?

EVA separates operations from financing. Net income is *after* interest on debt but *before* any charge for equity capital; it flatters firms that rely on equity (since equity appears "free" in the income statement). NOPAT is *before* all financing costs, so EVA charges *both* debt and equity through the WACC. A company can report a positive net income yet have negative EVA.

11.6 Market Value Added (MVA)

MVA = Current market value of the firm − Invested capital

If the invested capital is ₹920 lakh and the market value of the firm is ₹1,000 lakh, then **MVA = 1,000 − 920 = ₹80 lakh**.

MVA attempts to resolve some of the issues involved in EVA (for example, EVA ignores **value drivers** and **book values**). It does not give a basis for *share valuation* but is an alternative way to gauge performance efficiencies of an enterprise from a **market-capitalisation point of view**: the logic is that the **market will discount the efforts taken by the management fairly**. Hence the MVA of 80 is the true value added that is **perceived by the market**; in contrast EVA is a **derived** value added meant for the more discerning investor.

11.6.1 Relationship between EVA and MVA

In theory, **MVA = PV of all future EVAs** (discounted at WACC). A company that is expected to keep generating positive EVA has positive MVA. Hence EVA is the *flow* and MVA the *stock*.

Illustration. If expected EVA is ₹12 lakh next year, growing at 5% for ever, with WACC 12%: MVA ≈ 12 ÷ (0.12 − 0.05) = ₹171 lakh. Firm value = invested capital + MVA.

11.6.2 Behavioural note

"Since MVA represents market views regarding the company's future value generation, companies with a higher MVA will naturally become the darlings of the share market and would eventually become 'pricey' from a pure pricing perspective. In such cases, the **EVA may also sometimes have a slightly negative correlation as compared to MVA**. But this will be a short-term phenomenon as eventually the gap will get closed by investors themselves. A stock going ex-dividend will exhibit such propensities."

The main objective of EVA is thus to show **management efficiency in generating returns over and above the hurdle rate of invested capital**.

11.7 EVA vs MVA — summary

Feature	EVA	MVA
Formula	NOPAT − (Invested capital × WACC)	Market value of firm − Invested capital
Nature	Periodic **flow** (a year's performance)	Cumulative **stock** (market's view of all future value)
Basis	Accounting (adjusted), derived	Market, observed
Who uses	Managers, discerning investors	Shareholders, analysts
Strength	Management accountability; charges for equity	Reflects market perception
Weakness	Historical, accounting adjustments	Not available for unlisted firms; reflects sentiment
Link	MVA = PV of future EVAs	

(This is the content for the standard theory question "Differentiate between EVA and MVA".)

11.8 Adjustments in practice

Stern Stewart's original EVA made 160+ adjustments; common ones:

1. **Capitalise R&D** and marketing/brand spend and amortise them.
2. Add back **provisions** (bad debts, warranties) as non-cash.
3. Treat **operating leases** as debt.
4. Remove **non-recurring** items.

5. Adjust for **deferred taxes** (use cash taxes).
6. Include **unrecorded assets** (patents) in invested capital — the cases of Herbal Gyan and Constant Engineering (Chapter 21, Q15–Q16) illustrate this.

11.9 Using EVA for decisions

- **Performance measurement** and **incentive compensation** (EVA bonuses).
- **Capital allocation**: projects are accepted if they earn above WACC.
- **Divisional performance**: EVA by division with division-specific WACC.
- **Valuation**: Value = invested capital + PV of future EVA (Chapter 19 compares with DCF).

11.10 Limitations

1. Historical (the base year) — may not reflect future value creation.
2. Depends on the accuracy of invested capital (book values, depreciation policy).
3. WACC estimation uncertainty.
4. Short-term focus unless multi-year.
5. Does not by itself yield a share price.

Chapter 11 summary

- EVA = NOPAT − Invested capital × WACC; positive EVA ⇒ value creation.
- NOPAT = EBIT − tax (adjusted for non-cash items); invested capital = total assets − non-interest-bearing liabilities (± adjustments).
- Illustration 5: NOPAT 307, IC 920, WACC 8.44%, capital charge 77.65, **EVA 229.35**.
- MVA = market value − invested capital; MVA = PV of future EVAs.
- EVA measures management efficiency; MVA reflects market perception.

Check your understanding

1. Define EVA and say why NOPAT rather than net income is used.
2. Compute EVA: EBIT 500, tax 30%, invested capital 2,000, WACC 11%.
3. Compute MVA if market value is 3,200 and invested capital 2,000.
4. What is the relationship between EVA and MVA?
5. Why add back provision for doubtful debts in NOPAT?

Answers. (1) Excess of NOPAT over capital charge; NOPAT excludes financing so both debt and equity are charged. (2) NOPAT 350; charge 220; EVA **130**. (3) **1,200**. (4) MVA is PV of future EVAs. (5) It is a non-cash book entry; EVA seeks cash-based operating returns.

> **Exam focus.** "Differentiate between EVA and MVA" is a repeated question; also computational EVA questions (Chapter 21: Q9–Q16). Present NOPAT, invested capital, WACC, capital charge, EVA in five labelled steps.

Chapter 12 — Shareholder Value Analysis (SVA)

12.1 Why SVA?

We saw that **EVA** is the residual that remains if the capital charge is subtracted from NOPAT. A positive residual simply states that profits earned are adequate to cover the cost of capital. But is NOPAT the only factor that affects shareholders' wealth? Not strictly — but it is definitely *inadequate*, because it **does not take future earnings and cash flows into account**. NOPAT is a historical figure; it cannot fully represent the future potential of the entity. More importantly, it **does not capture future investment opportunities** (or the opportunity cost, whichever way you look at it).

SVA plugs this gap by tweaking the value analysis to take into its fold certain **drivers** that can expand the horizon of value creation. The key drivers considered are:

1. **earnings potential in terms of sales**,
2. **investment opportunities**, and
3. **cost of incremental capital**.

SVA (associated with Alfred Rappaport's shareholder-value approach) is, in effect, a **DCF valuation structured around value drivers**.

12.2 The seven value drivers (Rappaport)

Though the syllabus names three groups, the classical SVA framework has seven drivers: **sales growth rate, operating profit margin, income-tax rate, incremental fixed-capital investment, incremental working-capital investment, cost of capital, and the length of the value-growth (forecast) period**. A change in any of these changes value; management can focus on the drivers with the highest leverage.

12.3 Steps in SVA

Step	Task
(a)	Arrive at the **future cash flows (FCFs)** using a judicious mix of the **value drivers**
(b)	**Discount** these FCFs using the **WACC**
(c)	Add the **terminal value** to the present values computed in step (b)
(d)	Add the **current market value of non-core assets and marketable investments**
(e)	**Reduce the value of debt** from the result in step (d) to arrive at the **value of equity**

12.4 The progressive case study ($ million)

Step a.1 — Operating cash flows from value drivers

	Y1	Y2	Y3	Y4 onwards
EBIT (growing at 5% yearly)	100.00	105.00	110.25	115.76
Interest cost	(5.00)	(6.00)	(7.00)	(8.00)
EBT	95.00	99.00	103.25	107.76
Taxes @ 33%	(31.35)	(32.67)	(34.07)	(35.56)
EAT	**63.65**	**66.33**	**69.18**	**72.20**
Add back: depreciation	5.00	5.00	6.00	7.00
Add back: one-time write-offs	1.00	–	–	–
Operating cash flow	**69.65**	**71.33**	**75.18**	**79.20**

Step a.2 — Free cash flows

	Y1	Y2	Y3	Y4 onwards
Operating cash flow	69.65	71.33	75.18	79.20
Less: forecast incremental capital investment	–	(12.00)	(6.00)	(9.00)
Less: forecast increase in net working capital	(5.00)	(5.00)	(6.00)	(7.00)
Free cash flow	**64.65**	**54.33**	**63.18**	**63.20**

Step b — Discount at WACC 12%

	Y1	Y2	Y3	Y4 onwards
FCF	64.65	54.33	63.18	63.20
Discount factor	0.89	0.80	0.71	0.64
PV	**57.54**	**43.46**	**44.86**	**40.45**

Step c — Terminal value

From year 4 onwards the free cash flow is level at 63.20 — a **perpetuity**. The source chapter capitalises it by multiplying the present value of the year-4 flow (63.20 × 0.64 = 40.45) by the perpetuity multiplier 1 ÷ 0.12 = 8.33:

PV of the perpetuity = 40.45 × 8.33 = 336.95.

(The exact value of a level perpetuity beginning at the end of year 4 is worked out in §12.4.1.)

Steps d and e — Equity value

	$ million
PV of FCFs, Y1–Y3 (57.54 + 43.46 + 44.86)	145.86
PV of the perpetuity from Y4	336.95
Total PVs	**482.81**
Add: investment property (at fair value)	35.00
Less: carrying cost of debt	(19.00)

	$ million
Value of equity	**498.81**

"Thus, we observe that SVA brings out a **futuristic sense of value for shareholders**. In fact, this can be a good benchmark for shareholders from a cash return on investment perspective too."

12.4.1 A note on the perpetuity

Multiplying the *present value of the year-4 flow* by 1/WACC values a perpetuity whose first flow is one year later than year 4, i.e. it discounts one year too many. The exact value of a level perpetuity of 63.20 starting at the end of year 4 is the capitalised value at the end of year 3, discounted three years:

- Value at end of year 3 = 63.20 ÷ 0.12 = **526.67**
- PV today = 526.67 × 0.7118 (= $1/1.12^3$) = **374.9**

On that basis the total PV would be 145.86 + 374.9 = 520.8 and the value of equity 520.8 + 35 − 19 = **$536.8 million**, about 8% above the source's $498.81 million. An examiner following the source expects **498.81**; the *method* being tested — level cash flow valued as a perpetuity, non-core assets added, debt subtracted — is the same.

> **Pitfall.** When writing your own valuation, state at which date the perpetuity value is measured and discount it from *that* date.

12.5 SVA versus EVA versus DCF

	EVA	SVA	DCF (general)
Orientation	Historical residual profit	Forward-looking value-driver cash flows	Forward-looking cash flows
Value drivers explicit?	No	Yes (sales growth, margin, investments, cost of capital, period)	Optional
Output	Annual surplus over capital charge	Value of equity	Enterprise/equity value
Use	Performance measurement	Strategy and valuation	Valuation

SVA is *DCF with the drivers made explicit* — which makes it a management tool: you can ask "if sales growth rises by 1 point, what happens to shareholder value?" and rank levers.

12.6 Using SVA for strategy

1. **Sensitivity of value to each driver.** In the case study, a change in EBIT growth or in investment requirements changes FCF; running each driver ±10% reveals the **value leverage**.

2. **Threshold margin**: the minimum operating margin at which value creation is zero for given growth and investment.
3. **Business-unit analysis**: SVA by unit identifies units destroying value.
4. **Capital allocation**: invest only where incremental value > 0.

Worked example 12.1 (threshold). Suppose incremental sales of ₹100 crore need ₹30 crore of incremental capital; tax 30%; WACC 12%. The minimum incremental EBIT margin m* at which the incremental investment earns WACC solves m* × 100 × 0.7 = 0.12 × 30 → m* = 3.6 ÷ 70 = **5.14%**. Below 5.14% margin on new sales, growth destroys value.

12.7 Limitations

1. Needs credible forecasts of all drivers.
2. Complexity; many inputs.
3. Like DCF, sensitive to WACC and horizon.
4. Does not by itself separate operating from financing effects unless FCFF is used (the source case includes interest in the cash flow build).

Chapter 12 summary

- SVA is a value-driver-based DCF that supplements EVA by including future earnings potential, investment opportunities and cost of incremental capital.
- Five steps: FCFs from drivers → discount at WACC → add TV → add non-core assets/market investments → subtract debt.
- Case study: total PV 482.81; with investment property 35 and debt 19, equity value $498.81 m (as in the source; exact discounting of the perpetuity gives about $536.8 m).
- SVA helps management see which drivers create most value.

Check your understanding

1. Why does SVA improve on EVA?
2. List the five SVA steps.
3. Compute equity value: PV of FCFs 300; PV of TV 600; non-core assets 40; debt 150.
4. Name the key value drivers considered.
5. What does the threshold margin tell management?

Answers. (1) It brings in future earnings, investment opportunities and incremental capital cost. (2) See §12.3. (3) 300 + 600 + 40 − 150 = **₹790**. (4) Sales/earnings growth, margin, tax, fixed and working-capital investment, cost

of capital, forecast period. (5) The minimum profitability at which growth adds value.

> **Exam focus.** Be able to reproduce steps (a)–(e) and the final table; the source's perpetuity convention gives 498.81 (exact discounting would give 536.8).

Part V — Fair Value and Special Entities

Chapter 13 — Arriving at Fair Value; Going-Concern and Non-Going-Concern Valuation

13.1 What is "fair value" for an acquirer?

The ultimate purpose of a potential acquirer of a **controlling stake** or of the **takeover** of a company is that "he would purchase the same at the **fair price** — no less, no more". The approaches to valuation in this book, together with the various methods of value-added analysis (EVA, MVA, SVA), aim to identify entities that are **attractive in terms of true value** to a potential investor.

The same phrase means different things to different people:

Perspective	What "fair value" means
Chartered accountant	A transaction measured at **arm's length**
Financial analyst	The **present value of an entity in cash terms**
Speculative investor	The **arbitrage opportunities** that open up among similar entities that carry dissimilar values
Financial reporting	Exit price between market participants at the measurement date (Ind AS 113)

In an upward boom, valuations defy fair value. The dot-com boom saw companies valued at astronomical sums; when the downturn arrived, some vanished and others were just able to stand their ground. Fair value is therefore anchored in *fundamentals*, not in momentum.

13.2 No single method is "correct"

"Though the methods have their own pros and cons, it depends on the **vision of the ultimate decision maker** which method is suitable for his/her purpose. Further, it can be said that there is **no single answer to the method of valuation as the correct one** and it will be better if a **range of values**, i.e., **minimum acceptable by seller and maximum payable by the buyer**, could be determined. Ultimately the final deal would depend on the **negotiation** among the parties."

13.2.1 Building the range

Method	Typically indicates
Net realisable (liquidation) value	**Floor** — a seller will not accept less
Net asset value (book)	Lower reference
Capitalisation of earnings / P/E	**Minimum acceptable** to the target's shareholders (for P/E on a peer basis)

Method	Typically indicates
DCF	Intrinsic value for a going concern
Replaceable value	**Maximum an acquirer would pay** for equivalent assets
DCF with synergies	The buyer's *walk-away* price

The **seller's minimum** is the maximum of (liquidation value, standalone going-concern value, next-best alternative). The **buyer's maximum** is the minimum of (replacement cost, standalone value + synergies it can realise and keep, its own financial capacity). A deal is possible only if the buyer's maximum exceeds the seller's minimum; the gap is the **bargaining zone**.

13.3 The answer-writing approach for valuation questions

The source chapter gives a three-point approach to questions involving *evaluation and synthesis*:

1. **Unless specified otherwise, calculate valuation by as many methods as possible with the available data.**
2. **Give comments on the valuation by each of these methods.**
3. **Supplement your conclusion with any additional information if available.**

The merger question on AB Ltd. and XY Ltd. (Chapter 21, Q5), where the merchant bankers recommend averaging the DCF value and the net-asset value, is a typical application.

13.4 The going-concern assumption

One of the basic accounting assumptions is that an enterprise is a **going concern** and will continue in operation for the foreseeable future. It is assumed that the enterprise has **neither the intention nor the need to liquidate or curtail materially the scale of its operations**; if such an intention or need exists, financial statements may have to be prepared on a different basis and, if so, the basis used needs to be disclosed.

The valuation of the assets of a business entity depends on this assumption. Traditionally **historical costing** is followed in the majority of cases.

13.5 Non-going-concern (liquidation) valuation

Non-going-concern valuation is also known as **liquidation valuation**, because it is the **net value realised after disposing of all assets and discharging all liabilities**. An on-going firm could continue to earn profits, which contribute to its value in addition to its liquidation value; the **going-concern value** is therefore known as the **total value**.

13.5.1 Why going-concern value is generally higher

1. When a business is acquired as a going concern, the buyer pays for the value of its assets **and** for the value of its **future profitability, intangible assets and goodwill**, so the acquired firm can charge a premium.
2. **Liquidation** not only implies laying off employees, but also creates a **feeling of bad reputation** among potential investors; it destroys organisation value and relationships.
3. Assets sold piecemeal or in a forced sale fetch less than their value in use.

> **Rule.** Valuation based on non-going-concern should be applied **only when investors are of the view that the firm has no longer value as a going concern.**

13.5.2 The relationship

Going-concern value (total value) = Liquidation value + Value of going-concern premium (goodwill, intangibles, future profitability)

Worked example 13.1. For Sunrise Textiles (Chapter 19): liquidation (NRV) value ₹680 crore; going-concern DCF value ₹761 crore; going-concern premium = 761 − 680 = **₹81 crore**. If the DCF were *lower* than NRV (say ₹600 crore) the rational action would be to liquidate — the business is worth more dead than alive.

13.5.3 When is liquidation value the right basis?

Situation	Basis
Business continuing with positive cash flows	Going concern
Business that will earn less than the return on its liquidation proceeds	Liquidation (consider selling)
Legal winding-up, insolvency liquidation	Liquidation
Finite-life project at end of life	Salvage/liquidation
Distressed but recoverable business	Both: probability-weighted (Chapter 14)
Accounting for impairment (Ind AS 36)	Higher of fair value less costs of disposal and value in use

13.6 Liquidation: orderly versus forced

	Orderly liquidation	Forced (distress) liquidation
Time	Sufficient time to market assets	Immediate; auction
Prices	Close to market	Heavy discounts
Costs	Moderate	High
Use	Planned wind-up; lender security assessment	Insolvency

Valuation reports must state which premise applies.

13.7 A worked comparison

Kaveri Plastics Ltd (₹ lakh): Estimated going-concern DCF value of the enterprise: 900. Debt 400. Realisable value of assets (orderly): 780; liquidation expenses 40; liquidation (forced) discounts reduce assets to 600.

	Orderly	Forced
Realisable assets	780	600
Less: liquidation costs	(40)	(40)
Net realisable value of the enterprise	740	560
Less: debt	(400)	(400)
Equity — liquidation	**340**	**160**
Equity — going concern (900 − 400)	500	500
Going-concern premium over liquidation	160	340

The going-concern value of equity (500) is above liquidation value under both premises; the continuing business is preferable to closure, and the equity holders should not accept a price below ₹340–₹500 lakh depending on the strength of their alternatives.

13.8 Fair value and the bargaining zone: a numerical illustration

A seller values its business at ₹800 crore standalone (DCF), with a liquidation floor of ₹620 crore. A buyer's DCF with synergies is ₹930 crore; replacement cost for similar capacity is ₹1,050 crore; the buyer's financial capacity is ₹950 crore.

- **Seller's minimum** = max(620, 800) = **₹800 crore**.
- **Buyer's maximum** = min(1,050, 930, 950) = **₹930 crore**.
- **Bargaining zone** = 800 to 930 = ₹130 crore.

A price of ₹865 crore (the midpoint) would split the synergy gain equally. Negotiating strength (competition, alternatives, information) shifts the price within the zone.

13.9 Valuation date and information

Fair value is measured at a **specific date**; only information known or knowable at that date should be used. Subsequent events matter only to the extent they provide evidence about conditions at the valuation date.

Chapter 13 summary

- Fair value is the price a buyer should pay — "no less, no more" — and its meaning depends on the viewpoint (arm's length, PV, arbitrage).
- No single method is correct; determine a range from seller's minimum to buyer's maximum.
- For synthesis questions: use as many methods as data allow, comment on each, add supplementary information.
- Going-concern value > liquidation value (future profitability, intangibles, goodwill; no reputational damage). Use liquidation value only when no going-concern value remains.

Check your understanding

1. Give three perspectives on "fair value".
2. State the three-point approach to evaluation and synthesis questions.
3. Why is going-concern value generally higher than liquidation value?
4. When should non-going-concern valuation be used?
5. A business has DCF value 450 and liquidation value 520. What is the rational conclusion?

Answers. (1) Arm's length transaction; PV in cash terms; arbitrage opportunity. (2) Value by many methods; comment on each; supplement with additional information. (3) Includes future profitability, intangibles and goodwill; liquidation hurts reputation and organisation. (4) When investors believe the firm has no value as a going concern. (5) It is worth more liquidated; consider selling the assets (net of costs and taxes).

> **Exam focus.** "Going concern vs non-going concern" is a standard short note; remember the *total value* terminology and the "only when no longer value as a going concern" rule.

Chapter 14 — Valuation of Distressed Companies

14.1 What is distress?

Some firms are clearly exposed to possible distress, though the source of the distress varies:

- For some firms it is **too much debt**, which creates the potential for failure to make debt payments and its consequences (**bankruptcy, liquidation and reorganisation**).
- For others, distress arises from the **inability to meet operating expenses**.

A company is said to be in distress when it is **unable to meet, or has difficulty paying off, its financial obligations to its creditors**, typically due to **high fixed costs, illiquid assets, or revenues being sensitive to economic downturns**. Such distress can lead to **operational distress** as increasing costs of borrowing take a toll on operations.

Distressed companies are businesses that are likely to, or already have, **defaulted on their debts**. Although a company may not be making payments on some or all of its debt obligations, **there may still be some value remaining** on the instruments they hold. Just because a company cannot make payments on its debt **does not mean the company is entirely worthless**.

14.2 Why conventional methods fail

Conventional method	Why it fails for a distressed firm
DCF with terminal value	TV is based on an **infinite life and ever-growing cash flows**; the assumption of perpetuity of cash flows may not hold because of **negative cash flows**
Cash-flow estimation	A distressed firm usually has **negative and declining revenues**, expects to lose money for some more time, and has a **high risk of bankruptcy**. For firms expected to fail, DCF values the firm as a going concern. Even if the firm is expected to survive, projections must run until cash flows turn positive, otherwise **DCF yields a negative value** for equity or firm
Discount rates	Conventional rates reflect companies that are **operationally and financially sound**. They have to be **adjusted for the probability of failure** for distressed companies

The reasons "warrant adjustments or amendments and modifications to be made to conventional methods". The syllabus lists four approaches.

14.3 Method 1 — Modified discounted cash flow valuation

This method requires coming up with **probability distributions for the cash flows** (across all possible outcomes) to estimate the **expected cash flow** in each period. While computing, the **likelihood of default** should be adjusted

for. Along with the cash-flow estimates, discount rates are also estimated:

- use **updated debt-to-equity ratios and unlevered beta** to estimate the cost of equity;
- use **updated measures of the default risk** of the firm to estimate the cost of debt.

If the entire distribution cannot be estimated, the **probability of distress** is estimated for each period and used for the expected cash flow:

Expected cash flow (t) = Cash flow (t) × (1 − Probability of distress (t))

Worked example 14.1

Cash flows if the firm survives: Y1 100, Y2 140, Y3 180; terminal value at Y3 1,500 (if survival). Probability of the firm being in distress *by* each year (cumulative): Y1 10%, Y2 20%, Y3 30%. Discount rate 12%.

	Y1	Y2	Y3
Cash flow if surviving	100	140	180 + 1,500
× (1 − P)	0.90	0.80	0.70
Expected cash flow	90	112	1,176
Discount factor (12%)	0.8929	0.7972	0.7118
PV	80.4	89.3	837.1

Expected value = ₹1,006.8. If survival were certain, the value would be 100 × 0.8929 + 140 × 0.7972 + 1,680 × 0.7118 = 89.3 + 111.6 + 1,195.8 = **₹1,396.7**. The distress probabilities cut the value by about **28%**.

Note on double-counting. If the cash flows are already probability-weighted for distress, **do not also add a distress premium to the discount rate**; discount at a rate reflecting only systematic risk.

14.4 Method 2 — DCF valuation plus distress value

A DCF values a business as a going concern. DCF valuations will **understate** the value of the firm if there is a possibility that it will fail before it reaches stable growth, *and* the assets will be sold for a value **less than the present value of the expected cash flows** (a **distress sale value**). Conversely, a plain going-concern DCF **overstates** the value of a firm that may fail.

Steps to value a distressed firm (source chapter):

Step	Task
(i)	Value the business as a **going concern** by looking at the expected cash flows it will have if it follows the **path back to financial health**
(ii)	Determine the **probability of distress** over the lifetime of the DCF analysis
(iii)	Estimate the **distress sale value** as a **percentage of book value** or as a percentage of the DCF value of equity estimated as a going concern

Formula:

Value of equity = DCF value of equity × (1 − Probability of distress) + Distress sale value of equity × Probability of distress

Worked example 14.2

Going-concern DCF value of equity: ₹800 crore. Probability of distress: 35%. Distress sale value of equity: ₹100 crore (assets sold below debt, equity gets a residual).

Value of equity = 800 × 0.65 + 100 × 0.35 = 520 + 35 = **₹555 crore**.

If the probability rises to 50%: 400 + 50 = ₹450 crore. *Sensitivity*: each 10 points of distress probability removes about ₹70 crore.

Distress sale value as % of book value. If book value of assets is ₹1,000 crore and the distress sale fetches 55% = ₹550 crore, debt ₹900 crore: equity gets nothing (550 − 900 < 0) — distress sale value of equity = 0. For enterprise value the figure of 550 matters; allocate by seniority.

14.5 Method 3 — Adjusted present value (APV)

APV is based on the logic of **separating the investment decision from the financing decision**. First, the value of the firm is computed **without debt (the unlevered firm)**; then the effect of debt on firm value is adjusted:

Firm value = Unlevered firm value + (Tax benefits of debt − Expected bankruptcy cost from the debt)

- The first part is computed by discounting the free cash flows to the firm at the **unlevered cost of equity**.
- The second part reflects the present value of the **expected tax benefits** from the use of debt.
- **Expected bankruptcy cost** can be estimated as the difference between the unlevered firm value and the **distress sale value**, multiplied by the probability of distress:

Expected bankruptcy costs = (Unlevered firm value − Distress sale value) × Probability of distress

Worked example 14.3

Unlevered firm value ₹2,000 crore; debt ₹800 crore at 10%, tax 30% (perpetual debt: tax benefit = t × D = 0.3 × 800 = ₹240 crore); distress sale value ₹1,100 crore; probability of distress 20%.

- Expected bankruptcy cost = (2,000 − 1,100) × 0.20 = **₹180 crore**.
- Firm value (APV) = 2,000 + 240 − 180 = **₹2,060 crore**.

- Equity = 2,060 − 800 = **₹1,260 crore**.

Compare: without distress costs, 2,240; the possible distress consumes three-quarters of the tax benefit. If P rises to 40%: expected cost 360, APV = 1,880 — **below the unlevered value**: at that probability debt destroys value.

14.6 Method 4 — Relative valuation

Relative valuation multiples such as **revenue and EBITDA multiples** are used more popularly for distressed firms than for healthy ones, because multiples such as **price-earnings or price-to-book** often **cannot even be used** for a distressed firm (negative earnings, eroded book value). Analysts aware of the possibility of distress often consider it **subjectively** when they compare the multiple for the firm they are analysing to the industry average.

Example from the source chapter: Assume the average telecom firm trades at **2 times revenues**. Adjust this multiple down to **1.25 times revenues** for a distressed telecom firm. (A 37.5% discount.)

Worked example 14.4. Distressed telecom with revenue ₹1,600 crore: value = 1.25 × 1,600 = ₹2,000 crore (EV), against ₹3,200 crore at the healthy multiple. If net debt is ₹1,900 crore, equity value = ₹100 crore — a thin slice, consistent with the fact that leverage magnifies the effect of distress on equity.

14.7 Comparison of the four methods

Method	Core idea	Best when	Weakness
Modified DCF	Probability-weighted cash flows	Distress probabilities estimable by period	Needs probability estimates; risk of double-counting
DCF + distress value	Blend of going-concern and distress sale values	Binary outcome (recover vs fail)	Single probability and a single distress value
APV	Unlevered value + tax shield − bankruptcy costs	Capital structure issues central; firm with significant debt	Needs cost-of-bankruptcy and probability
Relative	Adjusted peer multiples	Quick cross-check; loss-making firms	Subjective discount; peers not distressed

14.8 Liquidation value as a floor

For a distressed firm the liquidation value is a **floor** for the enterprise value to creditors; if the going-concern value is below liquidation value, creditors may prefer liquidation. The valuer should compute both (Chapter 13) and state which premise applies. Equity is an **option** on the firm: even when the enterprise value is below debt, equity has some value if there is a chance of recovery before the debt matures — one reason distressed-company equity can trade above zero.

14.9 Practical issues

1. **Management forecasts are optimistic**; stress-test.
2. **Time to resolution** — legal delays erode value.
3. **Contingent liabilities** and litigation dominate.
4. **Hierarchy of claims** — allocate enterprise value by priority (secured, statutory, unsecured, preference, equity).
5. **Cash trapped** in subsidiaries or in escrow.
6. **Tax losses** may be restricted on change in control.
7. **Indian context:** the Insolvency and Bankruptcy Code process uses *fair value* and *liquidation value* determined by registered valuers for the committee of creditors; check current regulations.

Chapter 14 summary

- Distress = inability to meet obligations due to excess debt or operating weakness; the firm is not necessarily worthless.
- Conventional DCF fails: infinite-life TV, negative cash flows, discount rates of healthy firms.
- Four methods: modified DCF (expected CF × (1 − P)); DCF + distress value (equity = DCF × (1 − P) + distress value × P); APV (unlevered + tax benefits − expected bankruptcy cost); relative valuation (e.g., 2× revenue → 1.25×).
- Liquidation value is a floor; equity behaves like an option.

Check your understanding

1. Why do conventional DCF methods fail for distressed firms?
2. Compute equity value: DCF equity 600, distress equity 80, probability 25%.
3. APV: unlevered 1,500, tax benefit 180, distress sale 900, probability 15%. Compute firm value.
4. Why are revenue and EBITDA multiples preferred for distressed firms?
5. How do you avoid double-counting distress risk?

Answers. (1) Infinite-life TV, negative cash flows, going-concern assumption, discount rates of healthy firms. (2) 450 + 20 = **470**. (3) Expected bankruptcy cost = (1,500 − 900) × 0.15 = 90; value = 1,500 + 180 − 90 = **1,590**. (4) P/E and P/B are meaningless with losses and eroded book value. (5) Either probability-weight cash flows or raise the discount rate, not both.

> **Exam focus.** Write the four methods with their formulas; know the "2× → 1.25×" telecom example.

Chapter 15 — Valuation of Start-ups

15.1 Why start-ups are different

The three most common globally accepted methods of valuing a business are:

1. **Earning / cash-flow approach** — estimated cash flows for the foreseeable future are discounted to present value.
2. **Asset approach** — used when the business is not a going concern (liquidation, untimely losses); assets and liabilities are valued at current realisable value.
3. **Market approach** — values a business based on comparable companies in the same or similar industries, adjusted for their specific parameters.

One common feature of these approaches is that they **presuppose an established business that generates cash flows using its assets**. A start-up is not "established" in any sense, and its cash flows (if not already spent on marketing) cannot be assumed constant. **Profitability** seems to be a cursed word in start-up investor circles.

Start-up valuation is often required to bring in investment (equity or debt). The most significant differentiating factor is that there is **no historical data** on which future projections can be drawn. The value rests entirely on **future growth potential**, which in many cases depends on an **untested idea** not based on adequate sampling of consumer behaviour, and on the **competence, drive and self-belief** of the (often highly qualified) managers and their capacity to convert a promising idea into commercial success.

The major roadblock is the **absence of past performance indicators**: "founders walking in the dark and making the investors believe that they are wearing night-vision goggles." This is why valuation becomes critical: a professional helps investors navigate the dark **using facts, rather than fairy tales**.

15.2 Why traditional methods cannot be applied

Method	Why it fails for a new-age start-up
Income approach	A vast majority of start-ups operate on the assumption of **not generating positive cash flows in the foreseeable future** (a model now accepted by investors). With no or minimal positive cash flows, it is not easy to value the business correctly
Asset approach	(i) Start-ups have **negligible assets**; a large chunk of value is in **intellectual property and other intangibles**, which are hard to value and harder to agree with investors. (ii) Start-ups are new but usually operate under the **going-concern assumption**, so value should not be limited to the realisable value of assets today
Market approach	New-age start-ups are **disruptors**; they function in markets without established competitors, so numbers may be skewed and **not comparable** enough to form a base. However, elements of the market approach are used in **advanced funding rounds**

15.3 Value drivers for start-ups

Every start-up is different, but the following drivers recur. Most are **highly subjective**, which is why standard methods (next section) are needed.

Driver	Impact on valuation
Product	Uniqueness and readiness of the product/service; a functional prototype attracts higher value than an "idea"; market testing and customer response are sub-drivers
Management	More than half of Indian unicorn founders come from IIT/IIM; credentials and **balance of the team** (engineers + finance professionals + MBAs) matter; the profile of owners plays a crucial role
Traction	**Quantifiable evidence** that the product works and there is demand
Revenue	More revenue streams ⇒ more value; revenues are not mandatory but a better indicator than traction alone
Industry attractiveness	Logistics, distribution channels, customer base; a tourism start-up is worth less if lockdowns are expected
Demand–supply	If the industry is attractive there will be more investor demand, raising individual company value
Competitiveness	Fewer competitors ⇒ more value; first-mover advantage; but an existing proven model (e.g., Uber for Ola) makes it easier to convince investors, though it adds the burden of differentiation

15.4 Methods for valuing start-ups

15.4.1 Berkus approach

Created by American venture capitalist and angel investor **Dave Berkus**, this approach values a start-up on a detailed assessment of **five key success factors**:

1. **Basic value** (soundness of the idea)
2. **Technology** (prototype)
3. **Execution** (quality of management)
4. **Strategic relationships** in its core market
5. **Production and consequent sales**

Each factor is assigned a monetary value up to a cap, and the sum is the valuation. The method **caps pre-revenue valuations at $2 million and post-revenue valuations at $2.5 million**. Although it ignores other market factors, its limited scope is useful for businesses looking for an uncomplicated tool.

Worked example 15.1. Using a five-factor scale with up to $0.4 m per factor (total $2 m pre-revenue): idea 0.35; prototype 0.30; management 0.25; relationships 0.10; early sales 0.00 → total **$1.0 million**.

15.4.2 Cost-to-duplicate approach

Takes into account **all costs and expenses** of the start-up and its product development, including purchase of physical assets; the sum is the fair market value. Criticised for **not focusing on future revenue projections or the assets** of the start-up. It gives a floor.

Example. Product development ₹2.4 crore + equipment ₹0.6 crore + legal/IP ₹0.2 crore + marketing and brand ₹0.8 crore = **₹4.0 crore**.

15.4.3 Comparable transactions method

Using the **market approach**, this is lucrative for investors because it is built on **precedent**: "How much were similar start-ups valued at?"

Example from the source chapter: **XYZ Ltd.**, a logistics start-up, was acquired for **₹560 crore**. It had **24 crore active users** — about ₹23 per user. **ABC Ltd.**, another logistics start-up, has **1.75 crore users**; at ₹23 per user, ABC's valuation ≈ **₹40 crore**.

Computation: 560 ÷ 24 = ₹23.33 per user; × 1.75 crore = ₹40.8 crore ≈ ₹40 crore.

"With any comparison model, one needs to factor in ratios or multipliers for anything that is a **differentiating factor**" — proprietary technology, intangibles, industry penetration, locational advantages. Depending on these, the multiplier may be adjusted (e.g., ABC's users are in lower-ARPU cities ⇒ 0.8× adjustment ⇒ ₹32.6 crore).

15.4.4 Scorecard valuation method

Another option for **pre-revenue** businesses; it compares the start-up with others already funded, with added criteria.

1. Find the **average pre-money valuation of comparable companies**.
2. Consider how the business stacks up on **qualities**, each with a weight:

Quality	Weight
Strength of the team	0–30%
Size of the opportunity	0–25%
Product or service	0–15%
Competitive environment	0–10%
Marketing, sales channels and partnerships	0–10%
Need for additional investment	0–5%
Others	0–5%

3. Assign each quality a **comparison percentage**: on par (100%), below average (<100%) or above average (>100%) versus competitors/industry.

E.g., the marketing team has a 150% score (well trained, tested customer base): multiply its 10% weight by 150% to get a factor of **0.15**.

4. **Sum** the factors; multiply by the **average valuation in the sector** to get the pre-revenue valuation.

Worked example 15.2. Average pre-money valuation of comparable seed-stage companies: ₹10 crore.

Quality	Weight	Comparison	Factor
Team	30%	125%	0.375
Opportunity	25%	110%	0.275
Product	15%	100%	0.150
Competition	10%	80%	0.080
Marketing/sales	10%	150%	0.150
Need for additional investment	5%	90%	0.045
Others	5%	100%	0.050
Sum	100%		**1.125**

Pre-revenue valuation = 1.125 × ₹10 crore = ₹11.25 crore.

15.4.5 First Chicago method

Combines a **discounted cash flow approach** and a **market approach**: valuation is done for **three scenarios — worst case, normal case and best case** — each valued and multiplied by a **probability factor** to arrive at a **weighted average value**.

Worked example 15.3. Exit value in year 5 and probabilities; discount rate 25% (factor 0.3277):

Scenario	Probability	Exit value (₹ crore)	PV	Weighted
Best	20%	1,000	327.7	65.5
Normal	50%	300	98.3	49.2
Worst	30%	10	3.3	1.0
Value today				**₹115.7 crore**

15.4.6 Venture capital method

Venture capital firms made this method famous. Investors seek a return equal to some **multiple of their initial investment** or strive to achieve a specific **IRR** according to the perceived risk. The method uses the relevant time frame in **discounting a future value** attributable to the firm. The **post-money value** is calculated by discounting at the investor's required rate of return; the investor may seek 10×, 20×, 30× their original investment at exit.

Post-money value = Expected exit value ÷ Required multiple (or ÷ (1 + IRR)^n)
Pre-money value = Post-money value − Investment
Investor's required ownership = Investment ÷ Post-money value

Worked example 15.4. Expected exit value in 5 years ₹300 crore. The investor puts in ₹10 crore and wants **10×**.

- Post-money = 300 ÷ 10 = **₹30 crore**; pre-money = 30 − 10 = **₹20 crore**.
- Ownership = 10 ÷ 30 = **33.3%**.
- Equivalent IRR = $10^{1/5} - 1$ = **58.5%** a year.
  If the investor expects 25% dilution by the founders' later rounds, it needs 33.3% ÷ 0.75 = **44.4%** today.

> **Pitfall.** A 10× or 58% "required return" is not a cost of capital: it embeds the high failure rate of start-ups. Mixing it with probability-weighted outcomes (as in the First Chicago method) double-counts risk.

15.5 Choosing a method

Stage	Preferred methods
Idea / pre-revenue	Berkus, scorecard, cost-to-duplicate, comparable transactions
Early revenue	Scorecard, comparable transactions, VC method
Growth	VC method, First Chicago, DCF with scenarios, multiples (EV/revenue)
Late stage	DCF, multiples, precedent rounds

Use more than one method and present a **range**.

15.6 The role of the professional

"While the traditional methods fall short, there is no shortage of new innovative methods used to value start-ups based on their value drivers. However, the valuation of a start-up is much more than the application of ways — it is about understanding the **story of the future trajectory** and communicating that narrative using **substantial numbers**." The valuer's job is to test the story against facts: market size, unit economics, team, traction, burn rate, runway.

15.7 Red flags

1. Revenue projections growing 10× with no cost of acquisition.
2. Market size defined top-down only ("1% of a huge market").
3. No evidence of traction.
4. Team without relevant experience.
5. Valuation requested based solely on the last round's price.

6. Cap-table complexity (liquidation preferences) ignored.

Chapter 15 summary

- Traditional approaches presuppose established, cash-generating businesses; start-ups have no history, little assets and no comparables.
- Value drivers: product, management, traction, revenue, industry attractiveness, demand–supply, competitiveness.
- Methods: Berkus ($2m/$2.5m caps), cost-to-duplicate, comparable transactions (₹23 per user example), scorecard (weights 30/25/15/10/10/5/5), First Chicago (three scenarios), venture capital (post-money = exit ÷ multiple).

Check your understanding

1. Why does the asset approach fail for new-age start-ups?
2. List the scorecard factors and their maximum weights.
3. Compute pre-money: exit ₹200 crore, investor wants 8× on ₹10 crore.
4. How does First Chicago handle uncertainty?
5. Value a start-up with 3 crore users if a comparable sold at ₹30 per user.

Answers. (1) Negligible tangible assets, intangibles hard to value, and going-concern value exceeds asset value. (2) Team 30, opportunity 25, product 15, competition 10, marketing 10, funding need 5, others 5. (3) Post-money = 200 ÷ 8 = 25; pre-money = **₹15 crore**. (4) Three scenarios with probabilities. (5) 3 × 30 = **₹90 crore**.

> **Exam focus.** Memorise the three traditional methods and why each fails; the Berkus caps; scorecard weights; the XYZ/ABC comparable transaction example; First Chicago's three scenarios.

Chapter 16 — Valuation of Digital Platforms

16.1 What is a digital platform?

A **digital platform** is a **software-based online infrastructure that facilitates interactions and transactions between users**. Platforms are principally built to facilitate **many-to-many interactions**. Based on the kind of services provided, the source chapter classifies them as follows:

Category	Description	Examples
Marketplace	Multiple buyers are matched to multiple suppliers	Booking.com (guests–hotels), Uber (travellers–drivers), Amazon (sellers–buyers)
Search engine	Multiple people looking for information are matched to multiple information sources; a search request triggers the system to seek the desired information	Google, Bing, Baidu
Repository	Multiple suppliers "deposit" materials into a kind of library, to be retrieved by users later	Spotify, YouTube, GitHub
Digital communication	Multiple users send messages/documents or interact in real time by voice/video	WhatsApp, Microsoft Teams, Telegram, Slack
Digital community	People who want to remain virtually connected for a long time find each other and interact	Facebook, LinkedIn
Payments platform	Matching between those owing money and those wanting to be paid	Paytm, GPay

The *principles* of valuation for digital platforms are largely like those for other types of companies, with certain nuances peculiar to the industry. Platforms are valued using the **income**, **market** and **cost** approaches.

16.2 Why platforms are tricky to value

1. **Network effects** — value to each user rises with the number of users; growth can be non-linear.
2. **Negative early cash flows** — platforms burn cash to acquire users ("penetrative strategies": cash burn and lower margins to create market share and popularise the platform).
3. **Capital-light** — low book value of assets; the main asset is code and data.
4. **Scarce comparables** — few listed peers; each platform's business model differs.
5. **Winner-takes-most dynamics** — outcomes are binary-ish.
6. **Dependence on management** — survival depends on the quality of promoters, investors and their capability to execute and adapt.

16.3 Income approach

Valuation methods under the income approach lay emphasis on **projected financial performance**, taking into account **future revenues and costs** using company-specific revenue and cost drivers and applicable capital expenditure and working-capital cycles.

16.3.1 Top-down approach

Backward working is required under the **top-down approach**, which starts with an analysis of the **total potential market** for the platform on a global or domestic scale, often called the **Total Addressable Market (TAM)**. The next step is to estimate the **share** the company expects to gain in the target market and the time to reach such share — referred to as the **Serviceable Addressable Market (SAM)** and **Serviceable Obtainable Market (SOM)**. Then the company estimates its business plan to accomplish its objectives and strategy: how it will gain market share, increase revenues while optimising or using cash.

The financial forecast must reflect the type and features of the platform's business model: a repository allowing streaming may earn revenue based on **subscribers**; a payments platform may earn on the **number of transactions**. Direct operating costs are unique to each type of platform.

Worked example 16.1 (top-down). TAM (online grocery in the country) ₹3,00,000 crore; SAM (cities where the platform will operate) = 25% → ₹75,000 crore; SOM (share obtainable by year 5) = 2% of SAM → **₹1,500 crore** of gross merchandise value. With a take-rate of 12%, revenue = ₹180 crore; at a 15% steady EBITDA margin, EBITDA = ₹27 crore in year 5. The path to that point includes years of negative cash flow; the DCF must fund them.

16.3.2 Bottom-up approach

The top-down approach can be **ambitious** for a nascent company, since estimating market size and market share poses practical challenges. Under the **bottom-up approach**, the platform estimates its earnings from the **limited resources it has**: a young platform estimates its revenue and costs given its financial constraints and deploys strategies to **target high-margin sales and cost-cutting** to generate more cash. This is more in line with efficient **capital-budgeting decisions**, which ultimately help forecast earnings and cash flows.

Worked example 16.2 (bottom-up). 20 lakh active users; average order frequency 1.5 a month; average order value ₹600; take-rate 10%; churn 20% a year; marketing budget funds 8 lakh new users a year. Revenue per user per

year = 1.5 × 12 × 600 × 10% = ₹1,080. Year-1 revenue ≈ 20 lakh × ₹1,080 = ₹216 crore; subsequent growth depends on net new users (new − churned) = 8 − 4 = +4 lakh a year → +20%.

Under both scenarios, **the value of a digital platform depends on the quality of the financial forecasts** — and the growth and survival of an entity depend on its promoters, investors and stakeholders creating products or services that fill a market need, and on their capability to execute and adapt to unexpected circumstances.

16.3.3 Discount rate (11.1)

The discount rate should be based on the **type of cash flows being discounted**:

- **Free cash flow to the firm (FCFF)** → discount at the **Weighted Average Cost of Capital (WACC)**.
- **Free cash flow to equity (FCFE)** → discount at the **cost of equity (Ke)**.

CAPM: **R = Rf + β × (Rm − Rf)** (Chapter 6).

16.3.4 Specific considerations (11.1.2)

(a) **Beta** measures the sensitivity of a stock or company to the market. Practically, a company's beta is estimated from the sensitivity of its share price (or its comparables' or industry's) to the market. Given the unique nature of each platform and the **scarcity of listed traded comparables**, estimating beta is challenging: one might need to compare the general diversified sector, the industry driving the revenue, or international comparables.
(b) The **survival** of a platform depends upon the **quality of management, ability to adapt to change quickly and foresee opportunity**.

Thus there are specific risks that **cannot be estimated using CAPM with only the industry or general sector beta**. A **Company-Specific Risk Premium (CSRP)** or **alpha** needs to be estimated and added to determine the appropriate cost of equity used to discount the estimated cash flows. The CSRP for **nascent companies would be higher** than for mature platforms with large operations and a large customer base.

Worked example 16.3. Rf 7%, sector beta 1.2, market premium 6%, CSRP 4% for a nascent platform: Ke = 7% + 1.2 × 6% + 4% = **18.2%**; for a mature platform with CSRP 1%: **15.2%**.

> **Caution.** Where possible, reflect specific risks in *probability-weighted cash flows* rather than inflating the discount rate; the CSRP is a practical shortcut when scenarios are unavailable.

16.4 Market approach

The **market approach** values a company by comparing it with similar valued companies using multiples such as **P/E**, **EV/EBITDA**, **Price to Book Value** and **Price to Revenue/Sales**. Selection of comparables is vital; parameters such as **market capitalisation, revenue, profit margins and capital structure** are used.

For digital platforms the comparison becomes difficult because:

- **listed comparables are scarce** and even absent for many platforms;
- the underlying value — specifically **profit and EBITDA — may be negative**;
- platforms are **capital-light**, making their **book value very low**.

Hence the market approach for digital platforms **emphasises revenue** — and, more precisely, **the primary driver of revenue** for that kind of platform.

16.4.1 Revenue drivers by category

Category	Drivers of revenue
Marketplace (supply and demand)	Number of bookings made, number of registered users, volume of transactions
Payment (billing and payments)	Number of active subscribers, number of merchants registered on the platform, compatibility and speed of the operating system, security, ease of use
Community (network of contacts)	Number of users, subscription fees, platform for professionals
Communication (network for messaging)	Number of users, sponsored links, advertising revenue
Repository (supply library)	Number of readers and contributors, authenticity of data, duration of use, quality and variety of data
Search (machine queries and information)	Number of users, relevant search results, time taken per search

Two search engines can be compared on **total active users** and **average time taken to show relevant results**; the one with more relevant results in shorter time is valued at a premium. For a **repository** platform drawing subscription/advertising revenue per view and duration of visit, comparison can be drawn on the **number of users, average number of views per user and average revenue per user**.

Example (source). A search-engine company valued at **₹100 crore** with **50 million users** and content of 1,00,000 hours can be used to compare a similar platform with fewer users but the same or similar revenue parameters.

Worked example 16.4. Platform A: ₹100 crore ÷ 5 crore users = **₹20 per user**. Platform B: 3 crore users, with 25% higher average revenue per user than A. Value of B = 3 crore × ₹20 × 1.25 = **₹75 crore** (versus ₹60 crore if ARPU were

equal). Then adjust for differences in growth, engagement, and margin.

16.4.2 Regression-based comparison

With many comparables, regress EV/Revenue on growth and margin to estimate a "fair" multiple for the subject given its growth and margin — more defensible than a simple average.

16.5 Cost approach

The **cost approach** estimates value from the **sum total of the cost to build the same platform or a similar platform with the same utility**. Since the asset behind a digital platform is the **code**, the **hours spent by developers** to write it are the primary cost. However, this approach may not be most appropriate because it **fails to take into account the revenue-generating capacity of the platform**, which may create significantly higher value for shareholders than the cost of developing it.

Worked example 16.5. 40 developers × 18 months × ₹2.5 lakh a month = ₹18 crore; add design and testing ₹3 crore; add servers and tools ₹2 crore → **₹23 crore** cost to rebuild. If the platform has 2 crore users generating revenue, the income and market approaches would give far more; the cost approach provides a *floor* and is relevant for purchase price allocation of the technology asset.

16.6 Reconciliation

Approach	Strength	Weakness for platforms
Income (DCF)	Captures growth, burn, scale economics	Highly sensitive to forecasts; long loss period
Market (revenue multiples)	Reflects market view	Scarce comparables; sentiment
Cost	Objective floor	Ignores revenue capacity

Conclusion (source): "The valuation of digital platforms can be tricky based on the peculiarities mentioned above. However, the **fundamentals of valuation remain the same**. The understanding of the business, the revenue model, the quality of management, and the risk-reward parameters determine the value of the digital platform."

16.7 Useful metrics for analysis

Metric	Meaning
CAC	Customer acquisition cost
LTV	Lifetime value of a customer (contribution margin × expected lifetime)
LTV/CAC	Healthy ≥ 3×
Churn / retention	Customers lost / retained per period

Metric	Meaning
ARPU	Average revenue per user
Take rate	Platform revenue ÷ GMV
GMV	Gross merchandise value
Burn rate; runway	Cash consumed per month; months of cash left

Chapter 16 summary

- Platforms: marketplace, search, repository, communication, community, payments.
- Income approach: top-down (TAM → SAM → SOM) or bottom-up; discount FCFF at WACC, FCFE at Ke; add CSRP for nascent firms.
- Market approach: multiples are hard; use revenue drivers specific to the platform type.
- Cost approach: rebuild cost of code — a floor.
- Fundamentals of valuation remain unchanged.

Check your understanding

1. Name three categories of digital platform with their revenue drivers.
2. Explain TAM, SAM and SOM.
3. Why is beta hard to estimate for a platform, and what is the remedy?
4. Platform A is valued at ₹150 crore with 3 crore users. Value a platform with 2 crore users at the same value per user.
5. Why is the cost approach not enough?

Answers. (1) E.g., marketplace (bookings, users, transaction volume); payments (active subscribers, merchants); search (users, relevance, time per search). (2) Total market; the part that can be served; the part realistically obtained. (3) Few listed comparables; use sector/international betas and add a CSRP. (4) ₹50 per user → **₹100 crore**. (5) Ignores revenue generating capacity.

> **Exam focus.** Memorise the six categories and their revenue drivers; know top-down vs bottom-up; and the role of CSRP/alpha.

Chapter 17 — Valuation of Professional and Consultancy Firms

17.1 The nature of the business

Professional services firms can be defined as firms that provide **customised, knowledge-based services to clients** — chartered accountants, advocates, management consultancy firms, architects, doctors' practices, design agencies. Even within the industry, firms vary significantly because of the different nature of the services each provides.

Their economic characteristics make valuation special:

Feature	Implication for valuation
Assets are people — fee-earners, partners, relationships	Little balance-sheet value; the asset approach is of limited use
Key-person dependence	Value depends on whether clients and staff stay after a change in ownership
Owner-managers	Partners' remuneration mixes salary and profit; normalisation is critical
Revenue is repeat or project-based	Recurring work (audit, retainers) is worth more than one-off assignments
Low capital intensity	High return on capital; working capital in the form of receivables and unbilled work
Cyclicality/regulation	Demand for consulting is cyclical; regulated professions have rules on ownership

17.2 The information used

As with any business valuation, understanding the **present and projected industry trends** plays a significant role in determining an accurate valuation amount. But experts generally look at the firm's **historical data** and compare it with **industry Key Performance Indicators (KPIs) and benchmarks**. Valuation experts also compare the firm against its **competitors**. The main sources of information are **audited annual statements and income-tax returns** (and, for partnership firms, partners' capital and drawings accounts).

17.3 Projections and risk

When using the **income approach**, although historical data is important, **projected growth (terminal value) also impacts the overall value**. Valuation experts plan for future growth and compare it with projected trends after conversations with management, but there is an **inherent risk associated with using future earnings potential**, since results may or may not materialise. Hence this **risk should be factored into the overall calculation** — through a higher discount rate or a lower projection.

17.4 Normalisation of income and cash flows

In addition to analysis of financial statements and comparison with industry standards, **normalisation of net income and cash flows** is another important aspect. This step allows **comparison of firms on an equal footing**. It involves **adding back non-cash items and specific items that might not apply to a new firm**; the normalised cash flows are then applied to the chosen valuation method and used in calculating the overall value.

Typical normalisation adjustments

Adjustment	Reason
Replace partners' actual remuneration by **market-rate** pay for equivalent professionals	The buyer will have to hire or pay equivalent professionals
Add back personal, non-business expenses (vehicles, travel, club fees)	Not recurring for a new owner
Remove one-time fees (a large non-recurring assignment)	Not maintainable
Remove non-operating income (interest, rent from surplus property)	Valued separately
Convert cash-basis accounting to accrual (receivables, WIP)	Comparability
Adjust rent to market rates if premises are owned by a partner	Related-party
Provide for unrecorded liabilities (retirement dues, leave encashment)	Real obligations

17.5 KPIs and benchmarking

"One commonly used method is to analyse the extent to which a firm meets expectations in comparison to **current industry benchmarks and KPIs**. Since professional services include several different types of firms, KPIs can vary greatly, and hence it is equally important to look at **specific indicators which align with the acquirer firm's goals**."

KPI	Why it matters
Revenue per partner / per professional	Productivity
Utilisation (billable hours ÷ available hours)	Efficiency
Realisation rate (fees collected ÷ standard fees)	Pricing power
Leverage (professionals per partner)	Profit per partner
Client concentration (% of fees from top 5 clients)	Risk
Recurring-revenue share	Stability
Staff attrition	Continuity
Receivable days; WIP days	Cash conversion
Profit margin before partner pay	Operating profitability

17.5.1 The valuation methods

1. **Income approach — capitalisation or DCF** of normalised earnings (preferred).

2. **Market approach — multiples of revenue (fees) or of normalised EBITDA/earnings** from transactions of similar firms. Revenue multiples are common in practice (e.g., 0.8× to 1.5× annual recurring fees, depending on sector, recurring share and growth), but should be sanity-checked against earnings.
3. **Asset approach** — net tangible assets plus goodwill (excess earnings).
4. **Excess-earnings method** (capitalised "super profit"): value = net tangible assets + goodwill, where goodwill = (Normalised earnings − return on net tangible assets) ÷ capitalisation rate.

17.6 Worked example 17.1 — A chartered-accountancy firm

Mehta & Associates (₹ crore):

	Reported	Normalised
Gross fees	12.0	11.4 (removing a one-time ₹0.6 crore special assignment)
Operating costs excluding partner pay (staff, rent, etc.)	(6.0)	(6.0)
Partners' pay (4 partners × ₹1.0 crore)	(4.0)	(3.2) market rate ₹0.8 crore × 4
Profit before tax	**2.0**	**2.2**

Calculation of the normalised figure: 11.4 − 6.0 − 3.2 = **2.2**. Tax at 25% → after-tax maintainable earnings = 1.65.

Capitalisation: Because of key-person risk and a high concentration of fees (top five clients 40%), the valuer selects a required return of 25%:
Value of the practice = 1.65 ÷ 0.25 = **₹6.6 crore** (equity).

Cross-check by revenue multiple: 0.6× normalised fees = 6.8 (at the lower end of the typical 0.6–1.0× range for a practice with high concentration) ✔.

Excess-earnings: Net tangible assets (receivables, WIP, equipment) ₹1.5 crore; normal return 12% → 0.18; excess earnings = 1.65 − 0.18 = 1.47; capitalised at 25% → goodwill 5.88; total value = 1.5 + 5.88 = **₹7.4 crore**.

Range: ₹6.6–7.4 crore. With a retention agreement (earn-out over 3 years) for the two partners who control most clients, the buyer might pay **70% upfront** and 30% contingent on retention of 90% of fees.

17.7 Adjusting for key-person risk

- **Scenario analysis:** if two rainmaker partners leave, 30% of fees go over two years → value falls by about 25–35%.
- **Transition arrangements:** non-compete, non-solicitation, earn-outs, deferred consideration.

- **Discount rate:** a higher required return (e.g., 25% vs 18%) for key-person risk — or reflect the risk in cash-flow scenarios (preferred).

17.8 Special points for partnership firms in India

- **Partners' capital and drawings** and goodwill on admission/retirement (accounting) are different from market-based valuation but may be a starting point.
- **Profit-sharing ratios** determine who bears the value.
- **Regulation**: restrictions on ownership of professional firms (e.g., by non-professionals) limit the buyer universe and therefore value.
- **Tax:** personal taxation of partners; conversion to LLP; goodwill treatment.

17.9 Common pitfalls

1. Capitalising reported profit before replacing partners' pay with market salaries.
2. Ignoring client concentration and key-person dependence.
3. Applying a revenue multiple from a firm with a different recurring share.
4. Ignoring work-in-progress and receivable quality.
5. Using historical growth for a practice built on one-off assignments.

Chapter 17 summary

- Professional firms are people businesses; value rests on client relationships and recurring revenue.
- Compare historical data with industry KPIs and benchmarks; sources are audited statements and tax returns.
- Normalise income and cash flows (partner pay, non-recurring, personal items) for comparison.
- Factor risk of future earnings into the discount rate or projections.
- Use KPIs aligned with the acquirer's goals.

Check your understanding

1. Why is normalisation of partners' remuneration critical?
2. Name five KPIs for professional firms.
3. Fees 8.0, operating costs excluding partner pay 3.5, partners (3) paid 0.9 each but market rate 0.7. Compute normalised PBT.
4. Why might revenue multiples differ between two firms?
5. How can a buyer protect against client attrition?

Answers. (1) Owner pay includes profit; a buyer must pay market rates. (2) Revenue per professional, utilisation, realisation, leverage, client concentration, attrition. (3) $8.0 - 3.5 - 2.1 = \mathbf{2.4}$. (4) Different recurring share, concentration, growth and profitability. (5) Earn-outs, deferred consideration, non-compete, retention agreements.

> **Exam focus.** Remember: sources (audited statements, ITRs), KPIs and benchmarks, normalisation of net income and cash flows, factoring in risk of projections.

Chapter 18 — Impact of ESG on Valuation

18.1 What is ESG?

Environmental, Social and Governance (ESG) is a framework designed to be embedded into an organisation's strategy that considers the **needs and ways in which to generate value for all organisational stakeholders** — employees, customers, suppliers and financiers. An illustrative list of what the three factors include:

Environmental	Social	Governance
Climate change	Employee development	Board independence
Water	Diversity and inclusion	Board diversity
Waste generation	Community development	Anti-corruption and bribery
Emissions	Health and safety	Tax transparency
Biodiversity	Customer	Ethical conduct

ESG is on the radar of several investors today. Focusing on ESG issues can bring out **risks and opportunities** for the company's ability to create sustainable value. The key environmental aspects are **climate change and natural-resource scarcity**; social issues include **diversity and inclusivity, labour practices, health and safety, and cyber security**; the governance aspect emphasises **board diversity and independence, executive pay and tax transparency**.

18.2 The momentum

The source chapter lists the key developments:

- **Investment pace in ESG funds:** ESG funds tapped in excess of **$50 billion in 2020** and total assets with ESG focus crossed more than **$35 trillion** in the same period.
- **Green bonds:** the green-bonds market crossed a milestone of **$1 trillion in 2020**.
- **Sustainability taxonomy on the rise:** key regions (for example the **European Union**) have already defined sustainability taxonomies; others are in the process.
- **Convergence of ESG frameworks:** the IFRS Foundation launched work to develop a **single global reporting standard** on ESG.
- **SEBI:** in **February 2023**, the Securities Exchange Board of India proposed a **regulatory framework on ESG disclosures by listed entities**.

ESG performance and linked ratings have begun to play an influencing role for companies going to market to raise funds. The high ESG focus of investors, lenders and financial institutions has reached a *tipping point* and has started to impact financing options: companies with high ESG focus stand to benefit through **preferential/lower cost of debt** and access to specialised products like **Green, Social and Sustainability-linked Bonds**.

18.3 From "good to have" to "must have"

Traditionally ESG was viewed as "good to have" in the area of business ethics, sustainability, diversity and community. With heightened interest from stakeholder groups, directors realise it is moving into **"must-have" territory**. The business case for ESG generally begins with **operational efficiency and risk reduction** and extends to longer-term **operational and organisational resiliency and sustainability**. Boards recognise the strong and direct link between building a profitable business and focusing on environmental and social considerations; they also know that this requires **robust governance practices** that fortify the company's portfolio as a strong contender with investors and shareholders.

18.4 How do ESG factors enter valuation?

The most popular technique of valuing any business is **discounting future cash flows**. Accordingly, the impact of ESG risks can be incorporated in **either the discount rate or the expected cash flows**.

18.4.1 Option A — adjust the discount rate

Management and investors are generally more interested in adjusting the discount rate by including a **risk premium**. This is more *practical* but the impact of ESG factors may **not be explicit**: a reader of the valuation cannot see *what* risk the premium represents, *when* it hits, or *how large* it is.

18.4.2 Option B — adjust the expected cash flows (more explicit)

Adjustment of ESG factors in the **cash flows** would be more explicit. The source chapter shows how each factor can be incorporated:

Factor	How to incorporate in expected cash flows
E — Environment	Carry out **2-degree scenario analysis** (e.g., what happens if the temperature of the plant increases by 2 degrees); adjust cash flows by considering **carbon points** (carbon price/credits)
S — Social	Adjust the impact of **social-measures cost on revenue**, such as better labour working conditions, **CSR** and other welfare measures for the various stakeholders
G — Governance	Adjust the impact of **poor governance on revenue** in the form of **penalty, fines, taxes**

18.5 Worked example 18.1: carbon cost — cash-flow vs discount-rate approach

Vindhya Cement Ltd. Expected FCFF next year ₹300 crore; growth 3%; WACC 11%. Baseline enterprise value = 300 ÷ (0.11 − 0.03) = **₹3,750 crore**.

The plant emits 1 million tonnes of CO_2 a year. A carbon price of ₹500 per tonne is expected to apply (full pass-through not possible): cost = 10 lakh × ₹500 = ₹50 crore a year before tax; after tax (25%) = **₹37.5 crore**, growing with the business at 3%.

Option B — cash flows: adjusted FCFF = 300 − 37.5 = ₹262.5 crore; EV = 262.5 ÷ 0.08 = **₹3,281 crore** → a decline of **12.5%** (₹469 crore).

Option A — discount rate: if the valuer instead adds a 0.5% ESG risk premium to WACC (11.5%): EV = 300 ÷ 0.085 = **₹3,529 crore** → a decline of 5.9%.

The two approaches give very different answers (12.5% vs 5.9%), and only the cash-flow approach tells the reader *why*. The cash-flow approach also lets the valuer test **mitigation**: for instance, an abatement investment of ₹150 crore that halves emissions would reduce the carbon cost to ₹18.75 crore after tax; EV with abatement = (300 − 18.75) ÷ 0.08 − 150 = 3,515.6 − 150 = **₹3,365.6 crore**, i.e., better than doing nothing (3,281) — the investment is **worth ₹84.4 crore**.

18.6 Worked example 18.2: governance penalty

A company expects a regulatory penalty of ₹40 crore with 25% probability in each of the next two years, and an increase in its annual compliance cost of ₹5 crore afterwards. Expected penalty cost = 0.25 × 40 = ₹10 crore a year for two years; compliance cost ₹5 crore a year in perpetuity after year 2 (after tax 3.75, PV at 11% as a perpetuity from year 3: 3.75 ÷ 0.11 × 0.8116 = 27.7 crore).

PV of penalties (after-tax 7.5 each year): 7.5 × (0.9009 + 0.8116) = 12.8. Total negative effect ≈ ₹40.5 crore — about 1.1% of a ₹3,750 crore value. A valuer who ignores it overvalues; one who adds a large "governance discount" without quantifying risks hiding the numbers.

18.7 ESG and the cost of capital

Evidence of a systematic ESG effect on the cost of capital is mixed, but several channels exist:

1. **Cost of debt:** sustainability-linked loans and green bonds can carry lower coupons; poor ESG can raise spreads or restrict lenders.

2. **Cost of equity:** high ESG investors may accept lower returns; poor governance may raise the required return (e.g., through concerns about minority treatment).
3. **Access to capital:** some funds exclude companies with poor ESG, narrowing the investor base.

In valuation, adjust the discount rate *only* for risks that are **systematic** and not already captured in cash flows; avoid double-counting.

18.8 ESG across the valuation approaches

Approach	ESG channels
Income/DCF	Cash-flow adjustments (carbon cost, capex for transition, fines, welfare costs, green revenue); discount rate; terminal growth (durability)
Market/relative	Peers' ESG-driven multiple premia/discounts; adjust peer set; ESG-related de-rating
Asset-based	Impairment of stranded assets; environmental liabilities (remediation, decommissioning); revaluation of green assets
Distress	Environmental claims raise probability of distress

18.9 Data and frameworks

- Company-level disclosures: Business Responsibility and Sustainability Reporting (BRSR) for large listed Indian companies; global frameworks (ISSB/IFRS sustainability standards, GRI, TCFD).
- **ESG ratings** differ widely across providers; use primary data (emissions, accident rates, board composition) over scores where possible.
- **Materiality**: focus on issues that affect financial performance in the particular sector (e.g., emissions for cement; data security for IT; safety for mining).

18.10 Practical guidance

1. **Identify material ESG issues** for the sector.
2. **Quantify** exposure (emissions, water use, incident rates, governance flags).
3. **Build scenarios** (regulation, carbon price, demand shifts, physical risk).
4. **Insert** costs and revenues in cash flows with timing.
5. **Document** discount-rate choices; avoid double-counting.
6. Include **opportunities** (green products, lower financing cost, resilience) as well as risks.

18.11 Limits

ESG data are inconsistent; regulation is evolving; time horizons of ESG risks are long, beyond typical explicit forecast periods; scenario analysis requires judgement. A valuer should disclose the ESG assumptions and their sensitivity.

Chapter 18 summary

- ESG = environmental, social, governance; now "must-have", with momentum in funds, green bonds, taxonomies, a single global reporting standard and SEBI's 2023 proposal.
- High-ESG companies may get lower cost of debt and access to green/sustainability-linked financing.
- Incorporate ESG in valuation via the discount rate (practical but opaque) or via expected cash flows (more explicit): E — 2-degree scenario and carbon points; S — cost of social measures; G — penalties and fines.

Check your understanding

1. Give two examples each of E, S and G factors.
2. Why are cash-flow adjustments for ESG more explicit than discount-rate adjustments?
3. How can the "E" risk be incorporated in cash flows?
4. State two channels through which ESG performance affects financing.
5. EV baseline = 2,000; ESG cost reduces FCFF by 8% in perpetuity. What is the revised EV (same WACC and growth)?

Answers. (1) E: climate change, water; S: health and safety, diversity; G: board independence, anti-corruption. (2) They show what the risk is, when, and how large; premium hides this. (3) 2-degree scenario analysis; carbon points/price. (4) Lower cost of debt; access to green/sustainability-linked bonds; investor base. (5) 2,000 × 0.92 = **1,840**.

> **Exam focus.** Know the E/S/G lists, the developments (including SEBI Feb 2023), and the three-line method of adjusting cash flows for E, S and G.

Part VI — Cases, Practice and Integration

Chapter 19 — Normalisation and an Integrated Case: Valuing One Company by Every Method

The previous chapters introduced the methods one at a time. In practice a valuer applies several to the *same* company and must reconcile the answers into a defensible range. This chapter first covers **normalisation** — the clean-up of the numbers on which every method depends — and then values a single fictional company, **Sunrise Textiles Ltd**, by the asset, income, DCF, EVA and market approaches.

19.1 Normalisation: cleaning the numbers

Whatever the method, reported figures are rarely a perfect basis. Normalisation adjusts them to represent the **maintainable** level of earnings and the **true** level of assets and liabilities. It also makes firms comparable on equal footing (the point made for professional firms in Chapter 17).

19.1.1 Income-statement adjustments

Adjustment	Treatment
Extraordinary/exceptional items (gain on sale of asset, litigation settlement)	Remove (after tax); consider if recurring
Non-operating income (interest on surplus funds, rent on idle property)	Remove; value the asset in the bridge
Prior-period items	Reallocate or remove
Related-party transactions at non-market terms	Restate to market
Owner/promoter remuneration above/below market	Replace with market rate
Accounting policy differences (depreciation, inventory)	Align to a common basis
Non-cash charges (provisions, ESOP)	Treat as cost if real; add back if purely book entry
Full-year effect of acquisitions/closures	Pro-forma
Cyclical peaks and troughs	Use mid-cycle margin

19.1.2 Balance-sheet adjustments

Remove fictitious assets (accumulated losses, preliminary expenses), add unrecorded assets (brands, licences), revalue land and plant, recognise unrecorded liabilities (contingent liabilities, deferred consideration), and separate operating from non-operating assets.

19.1.3 The tax effect

Every adjustment to profit changes taxable profit; apply the tax rate to each. An adjustment of ₹10 crore pre-tax is ₹7.5 crore after tax at 25%.

19.1.4 Example

Reported PBT ₹120 crore includes: gain on sale of a flat ₹15 crore; insurance receipt for fire ₹5 crore; one-time legal cost ₹8 crore; promoter's salary ₹3 crore above market.

Normalised PBT = 120 − 15 − 5 + 8 + 3 = **₹111 crore**. At 25% tax: PAT = ₹83.25 crore.

19.2 Sunrise Textiles Ltd: the facts

Sunrise Textiles Ltd is a fictional integrated textile manufacturer. All amounts in ₹ crore; 10 crore equity shares; valuation date: end of the latest financial year (FY0). Purpose: a strategic investor is considering acquiring 100% of the equity.

Profit and loss (FY0, normalised):

Revenue	1,200
EBITDA	190
Depreciation	(50)
EBIT	**140**
Interest on debt of 300 at 9%	(27)
PBT	113
Tax @ 25%	(28.25)
PAT	**84.75**
NOPAT = EBIT × 0.75	105

Profit after tax history (normalised): FY−3: 62; FY−2: 70; FY−1: 78; FY0: 85.

Balance sheet (book values):

Assets		Liabilities and equity	
Fixed assets (net)	600	Long-term debt	300
Net current assets (operating)	250	Equity (net worth)	590
Non-operating investments	40		
Total	**890**	**Total**	**890**

Other information.

- *Realisable values:* fixed assets 780 (land is worth much more than book); net current assets 215 (debtors and old stock have to be written down); investments 40; liquidation expenses 15.
- *Replacement cost:* fixed assets 840; net current assets 250; investments 40.
- *Operating invested capital:* 600 + 250 = **850**.
- *Peer listed company:* P/E 12; EV/EBITDA 5–6× for textile manufacturers.

- *Cost of capital:* risk-free 7%, market premium 6%, asset beta for textile pure-plays 0.85 (equity beta at the company's low D/E ≈ 0.85 used); cost of debt 9% pre-tax; target debt-to-total-capital of 25% → WACC taken as **12%**.
- *Growth:* NOPAT grows 8%, 8%, 7%, 6%, 6% over five years; incremental ROIC 14%; terminal growth 5% with ROIC 13%.

19.3 Asset-based approach

Basis	Calculation	Value of equity	Per share
Net asset value (book)	600 + 250 + 40 − 300	**590**	₹59.0
Net realisable value	780 + 215 + 40 − 15 (costs) − 300	**720**	₹72.0
Replaceable value	840 + 250 + 40 − 300	**830**	₹83.0

Comment. Book value (590) is the lower limit; NRV is higher because the land is undervalued; replacement cost (830) is the **maximum price** an acquirer would pay for equivalent assets. The asset approach gives a **range of ₹590–₹830 crore** — but says nothing about earning power.

19.4 Income-based approach

19.4.1 Capitalisation of earnings

Weighted-average PAT (weights 1, 2, 3, 4): (62 + 140 + 234 + 340) ÷ 10 = **77.6**. This looks backward; the latest year's PAT is 85. Using 77.6 is conservative; it adds an economy-of-scale adjustment of +4 → maintainable PAT **81.6**.

Capitalisation rate	Value	Per share
13%	627.7	₹62.8
14%	**582.9**	₹58.3
15%	544.0	₹54.4

(CAPM gives Ke = 7% + 0.85 × 6% = 12.1% for equity; a 14% rate allows for the fact that the firm will not grow forever and is not listed. This method implicitly assumes **no growth**, so it understates a growing firm.)

19.4.2 P/E method

Peer P/E 12; **20% reduction** for non-listing → 9.6. Maintainable EPS = 81.6 ÷ 10 = ₹8.16.
Price = 8.16 × 9.6 = **₹78.3** → equity **₹783 crore**.

19.4.3 Goodwill (super-profit)

Net assets (book) 590; normal return 12% → 70.8; maintainable PAT 81.6 → super profit 10.8; capitalised at 14% → goodwill 77.1; value = 590 + 77.1 = **₹667 crore**.

19.5 Cash-flow approach (DCF)

Forecast of FCFF (₹ crore). NOPAT grows 8%, 8%, 7%, 6%, 6% from 105. The reinvestment each year funds the *next* year's growth at an incremental ROIC of 14% (reinvestment = NOPAT × next-year growth ÷ 0.14).

	Y1	Y2	Y3	Y4	Y5
NOPAT	113.4	122.5	131.0	138.9	147.2
Net reinvestment	(64.8)	(61.2)	(56.2)	(59.5)	(52.6)
FCFF	**48.6**	**61.2**	**74.9**	**79.4**	**94.7**
PVF @ 12%	0.8929	0.7972	0.7118	0.6355	0.5674
PV	43.4	48.8	53.3	50.5	53.7

PV of explicit FCFF = **249.7**.

Terminal value. Growth 5%; ROIC 13% ⇒ reinvestment rate 5/13 = 38.5%. $FCFF_6$ = 147.2 × 1.05 × (1 − 0.385) = **95.1**. TV at year 5 = 95.1 ÷ (0.12 − 0.05) = **1,359.2**; PV = 1,359.2 × 0.5674 = **771.2**.

	₹ crore
PV of explicit FCFF	249.7
PV of terminal value	771.2
Enterprise value	**1,020.9**
Less: debt	(300.0)
Add: non-operating investments	40.0
Equity value	**760.9**
Per share	**₹76.1**

TV share of EV = 75.5%.

Sensitivity (₹ per share)

WACC \ g	4%	5%	6%
11%	89.5	93.8	99.4
12%	74.2	**76.1**	78.4
13%	62.2	62.8	63.5

If terminal ROIC equals WACC (12%) — no excess returns — value per share falls to ₹72.1.

19.6 EVA and MVA

Invested capital rises from 850 with the reinvestments: 914.8, 976.0, 1,032.2, 1,091.7, 1,144.3.

	FY0	Y1	Y2	Y3	Y4	Y5
NOPAT	105.0	113.4	122.5	131.0	138.9	147.2
Opening invested capital	850.0	850.0	914.8	976.0	1,032.2	1,091.7
Capital charge @ 12%	102.0	102.0	109.8	117.1	123.9	131.0
EVA	**3.0**	**11.4**	**12.7**	**13.9**	**15.0**	**16.2**

- **FY0 EVA = 105 − 0.12 × 850 = ₹3.0 crore** — the firm barely covers its cost of capital today (ROIC 12.35% vs WACC 12%).
- EVA rises as the firm grows at an ROIC above WACC.
- **Value by EVA** = Opening invested capital + PV of future EVA (including terminal) ≈ 850 + 189 = **≈ ₹1,039 crore**, within 2% of the DCF enterprise value (the small gap arises because the terminal EVA assumes a different reinvestment pattern). Equity ≈ 1,039 − 300 + 40 = ₹779 crore (₹77.9 per share).
- **MVA** (at the DCF value) = 1,020.9 − 850 = **₹170.9 crore**: the value the market (or the acquirer) is expected to put on the future EVAs.

Interpretation. The business today earns just its cost of capital; **the entire premium over invested capital (₹171 crore) depends on the forecast improvement** — a message the DCF alone does not make as visible.

19.7 Relative valuation

EV/EBITDA of peers 5–6×; EBITDA 190.

Multiple	EV	Less debt	Add investments	Equity	Per share
5×	950	(300)	40	690	₹69.0
6×	1,140	(300)	40	880	₹88.0

The DCF-implied EV/EBITDA is 1,020.9 ÷ 190 = **5.4×**, inside the peer range — a comfort check.

19.8 Putting the methods together

Method	Equity value (₹ crore)	Per share (₹)	Comment
Net asset value (book)	590	59	Lower limit
Net realisable value	720	72	Floor for a seller
Replaceable value	830	83	Upper limit for a buyer of assets

Method	Equity value (₹ crore)	Per share (₹)	Comment
Capitalisation of earnings (13–15%)	544–628	54–63	No growth assumed
P/E (adjusted)	783	78	Peer-based
Super-profit (goodwill)	667	67	Blend of asset and income
DCF (WACC 11–13%, g 4–6%)	622–994	62–99	**Central ₹76**
EVA-based	779	78	Consistent with DCF
EV/EBITDA 5–6×	690–880	69–88	Market cross-check

19.8.1 Reading the range

- The **asset-based** floor is ₹59–72 per share; no rational seller would accept less than the NRV of ₹72 *unless* the business loses money (it does not).
- The **market-based** (P/E and EV/EBITDA) and **DCF/EVA** estimates cluster around ₹70–₹80 per share.
- The **capitalisation** method is low because it ignores growth.
- The **replacement cost** (₹83) is the ceiling an asset-buyer would pay.

19.8.2 Conclusion (following the three-point approach of Chapter 13)

1. **Valued by many methods:** asset (book, NRV, replacement), income (capitalisation, P/E, goodwill), DCF, EVA, multiples.
2. **Commented on each:** the capitalisation method understates a growing business; the asset values give floor and ceiling; the DCF depends heavily on terminal assumptions (75% of value) and on a WACC of 12%.
3. **Supplementary information:** the firm earns only its cost of capital today (EVA ₹3 crore); the buyer's synergy potential is not included; the land has development potential not captured in the operating forecast.

Concluded range for 100% of equity: ₹700-₹800 crore (₹70–₹80 per share); point estimate ₹750 crore (₹75 per share). The seller's minimum is the NRV of ₹720 crore; the buyer's maximum (before synergies) is the DCF of ₹761 crore and the replacement cost of ₹830 crore. The **bargaining zone** is roughly ₹720–₹800 crore.

19.9 What the case teaches

1. **No single number**: methods differ by up to 50% (₹54–₹99); the *range* and its reasons matter.
2. **Triangulation** gives confidence: three independent families (asset, income, market) overlap at ₹70–₹80.
3. **DCF and EVA tell the same story** from different angles; EVA reveals the base-year performance.

4. **Normalisation** matters: the base-year PAT (85) and the weighted history (77.6) lead to different capitalised values.
5. **Terminal value and WACC** are the key judgements.

Chapter 19 summary

- Normalise income and balance sheet (and apply tax) before any method.
- Sunrise Textiles: NAV 590, NRV 720, replacement 830, P/E 783, DCF 761, EVA ~779, EV/EBITDA 690–880.
- A reasoned range (₹700–800 crore) arises from the overlap of the families.
- Always state each method's comments and add supplementary information.

Check your understanding

1. Normalise: reported PBT 90, includes gain 12, one-time cost 5, promoter excess pay 2. Compute PAT at 30% tax.
2. Why is the capitalisation value lower than the DCF value for Sunrise?
3. What does an EVA of ₹3 crore in FY0 tell us?
4. Compute MVA given EV 1,000 and invested capital 850.
5. How would you determine the seller's minimum and buyer's maximum here?

Answers. (1) 90 − 12 + 5 + 2 = 85; PAT = 85 × 0.7 = **59.5**. (2) It assumes no growth. (3) The firm barely earns its cost of capital. (4) **150**. (5) Minimum = max(NRV, standalone); maximum = min(replacement cost, standalone + synergies, financial capacity).

> **Exam focus.** This case is the template for "value the company by various methods and comment" questions: lay out each method, give a one-line comment, and finish with a range.

Chapter 20 — Case Studies

Cases show how valuation behaves in the real world, where data are incomplete, methods disagree and the market reacts. This chapter analyses the two real-life case studies in the syllabus — the **Vodafone–Idea merger** and **Facebook's acquisition of WhatsApp** — and adds three constructed teaching cases.

20.1 Case study 1 — Vodafone and Idea Cellular: valuation in a merger

20.1.1 The facts (as in the source chapter)

Valuation was applied in the context of the **merger of Vodafone's Indian operations with Idea Cellular Ltd.** The valuation methods used by the appointed chartered-accountant firms were:

#	Method	What was done
(a)	**Market value method**	The share price observed on the NSE over a suitable time frame was considered
(b)	**Comparable companies' market multiple method**	Stock-market valuations of comparable companies on the BSE and NSE were considered
(c)	**NAV method**	The asset-based approach, to arrive at the net asset value of the merging entities as of **31 December 2016**

Surprisingly, the DCF method was not used. The reason stated: the managements of both Vodafone and Idea had **not provided the projected (future) cash flows** and other parameters necessary for DCF. The final valuation using methods (a)–(c) provided a basis for a merger on a **share-exchange** basis. (Information from the valuation report filed by Idea Cellular with NSE.)

20.1.2 The market's reaction

A business newspaper reported that **Idea Cellular slumped 9.6%** as traders said the implied deal price in the planned merger **under-valued the company's shares**. Although traders had initially reacted positively, doubts about Idea's valuation after the merger sent shares downward. **Idea fell as much as 14.57%**, reversing earlier gains of **14.25%**, after announcing it would merge with Vodafone's Indian operations.

20.1.3 Lessons

1. **Valuation methods, though technically correct, may not elicit a positive reaction.** There is a **"perceived value"** that is not quantifiable and depends on analyst interpretation, majority opinion and other factors.

2. **Method choice is constrained by data.** Without management forecasts, DCF could not be done; the valuers relied on market, comparable and asset approaches. Listed targets with liquid shares lend themselves to market-based methods.
3. **For a share-exchange merger the ratio matters more than the absolute value.** What matters is each company's value *relative* to the other, measured consistently.
4. **Information asymmetry and signalling:** initial positive reaction followed by a drop shows that the market took time to digest the exchange terms.

20.1.4 How an exchange ratio is derived (illustration)

Suppose the valuers' conclusions (hypothetical numbers) per share are:

Method	Company A (acquirer)	Company B (target)	Weight
Market (volume-weighted price)	₹ 100	₹ 40	50%
Comparable multiples	₹ 110	₹ 46	25%
NAV	₹ 60	₹ 30	25%
Weighted value	₹ 92.5	₹ 39.0	

Exchange ratio = 39.0 ÷ 92.5 = **0.42** shares of A per share of B. Compare this with the market-only ratio (40 ÷ 100 = 0.40): the weighting toward multiples and NAV favours B's shareholders. If B's shareholders and the market believe B is worth more, the ratio will be challenged — as in the case.

20.2 Case study 2 — Facebook's acquisition of WhatsApp

20.2.1 The facts (as in the source chapter)

Facebook announced the takeover of **WhatsApp for a staggering USD 21.8 billion** (2015 in the source). The key characteristics of WhatsApp that influenced the deal were:

(a) A **free text-messaging service** with a **$1 per year service fee**; it had **450 million users** worldwide close to the valuation date.
(b) **70%** of the users were **active users**.
(c) An aggressive rate of user account increase of **1 million users a day** would lead to a pipeline of **1 billion users** within a year's range.

The **gross per-user value** would come to an average of **USD 55**, which included a **USD 4 billion payout as a sweetener for retaining WhatsApp employees** post takeover. The payback for Facebook will eventually be to **monetise this huge user base** with recalibrated charges in the **international messaging** arena. Facebook believes that the future lies in **international, cross-platform communications**. (Information from the official website of CNBC.)

20.2.2 Analysis of the numbers

Measure	Calculation	Result
Price per registered user	21.8 bn ÷ 450 m	**$48.4**
Price per active user (70% active = 315 m)	21.8 bn ÷ 315 m	**$69.2**
Price per user excluding the $4 bn retention payout	17.8 bn ÷ 450 m	**$39.6**
Price per user on the "pipeline" of 1 bn users	21.8 bn ÷ 1,000 m	**$21.8**
Revenue yield: $1 a year × 450 m users (if all paid)	≈ $450 m a year	Price ÷ revenue ≈ **48×**

The source's "$55 per user" is a rounded, deal-specific figure (the exact arithmetic depends on the user count assumed); what matters is the order of magnitude: **tens of dollars per user** for a business earning about a dollar per user.

20.2.3 Which valuation approach does this reflect?

- **Market (user-based) approach:** price per user/active user — comparables from other communications platforms.
- **Income approach with a long runway:** the value rests on monetising a very large user base in the future (Chapter 16: top-down TAM/SAM/SOM).
- **Strategic (investment) value** to Facebook: protecting its own communication franchise from a rival, plus synergy with its ad platform — the value to *this* buyer exceeds fair value to a financial buyer.
- **The $4 billion retention payout** is a *cost of retaining human capital*; it is not paid for assets; in accounting terms such payments (if contingent on continued service) may be treated as compensation rather than purchase consideration.

20.2.4 Lessons

1. For a platform, **users are the asset**; the multiple-of-users approach is the only *market* measure when revenue is tiny.
2. **Growth rate and runway** (1 million new users a day) drove the valuation.
3. **Strategic value** can exceed intrinsic standalone value.
4. **Sanity check** by backing out the implied revenue and margin required: to justify $21.8 bn at a 10% discount rate in perpetuity, WhatsApp would need annual free cash flow of about $2.2 bn (21.8 × 0.10) — i.e., roughly **$2–3 of free cash per user** on a billion users, compared with $1 of revenue per user then. The valuation embeds a belief in a 3–5× rise in monetisation per user plus massive user growth.

20.3 Case study 3 — A distressed hospital chain

Lifeline Hospitals Ltd (₹ crore): facing high leverage after a debt-funded expansion and a downturn in occupancy.

Item	Value
Going-concern enterprise value (if recovery plan succeeds)	1,800
Debt (face value)	1,500
Distress-sale enterprise value (if the plan fails)	900
Probability that the plan fails	40%

Step 1 — Equity by state.

- Recovery state: 1,800 − 1,500 = **300**.
- Distress state: 900 − 1,500 < 0 → equity worth **0** (limited liability).

Step 2 — Value of equity = 0.60 × 300 + 0.40 × 0 = **₹180 crore**.

Step 3 — Value of debt = 0.60 × 1,500 + 0.40 × 900 = **₹1,260 crore** (84% of face value).

Check: equity + debt = 180 + 1,260 = 1,440 = 0.6 × 1,800 + 0.4 × 900 ✔.

Lesson. A naive "blended EV minus debt" = 1,440 − 1,500 = −60 would suggest zero equity; the **state-by-state** approach recognises that equity has value in the recovery state and floors at zero in distress — equity is an option on the firm. Raise the failure probability to 60% and equity falls to 0.4 × 300 = 120.

20.4 Case study 4 — A fintech start-up

PayNest Ltd: a payments start-up with **12 lakh active users**, revenue ₹6 crore (growing 150%), losses ₹9 crore a year, 24 months of cash. An investor proposes to invest **₹15 crore**.

Method 1 — comparable transactions. A recent acquisition of a similar payments app was at **₹300 per active user**. PayNest value = 12 lakh × ₹300 = **₹36 crore**; adjust +10% for faster growth = ₹39.6 crore.

Method 2 — scorecard. Average pre-money of seed/Series-A payments start-ups ₹40 crore; factors: team 30% × 110%, opportunity 25% × 120%, product 15% × 100%, competition 10% × 70%, marketing 10% × 100%, funding need 5% × 80%, other 5% × 100% → 0.33 + 0.30 + 0.15 + 0.07 + 0.10 + 0.04 + 0.05 = **1.04** → ₹41.6 crore.

Method 3 — venture capital method. Expected exit value in 5 years ₹400 crore; investor wants **6×** → post-money = 400 ÷ 6 = **₹66.7 crore**; pre-money = 66.7 − 15 = **₹51.7 crore**; ownership = 15 ÷ 66.7 = **22.5%**.

Method 4 — First Chicago. Success (25%): exit ₹700 crore; moderate (40%): ₹200 crore; failure (35%): ₹10 crore; discount 30% for 5 years (factor 0.2693): 0.25 × 188.5 + 0.40 × 53.9 + 0.35 × 2.7 = 47.1 + 21.5 + 0.9 = **₹69.6 crore** post-money-equivalent → pre-money ₹54.6 crore.

Summary of pre-money values (₹ crore):

Method	Pre-money
Comparable transactions	36–40
Scorecard	41.6
VC method (6×)	51.7
First Chicago	54.6

Conclusion. Pre-money of **₹40–₹52 crore**, with negotiation around ₹45 crore (investor gets 15 ÷ 60 = 25%). The *low-end* methods (market, scorecard) rely on precedent; the *high-end* (VC, First Chicago) on the exit story. The gap highlights that start-up valuation is a negotiation anchored by method.

20.5 Case study 5 — A cyclical manufacturer at the top of the cycle

Rajat Steels Ltd: EBITDA margins over the last eight years: 6%, 14%, 22%, 17%, 4%, −3%, 8%, 18% (average 10.75%). Revenue ₹2,000 crore; the latest year (18%) is near a peak.

- **Naive approach:** capitalise last year's EBITDA of ₹360 crore at 5× = ₹1,800 crore.
- **Normalised approach:** use the average margin 10.75% → EBITDA ₹215 crore × 5× = **₹1,075 crore**.
- **Difference:** the naive method overstates by **67%**.

Lesson. For cyclical businesses the base year must be normalised to a mid-cycle level (Chapter 19) before applying multiples or capitalising earnings. A DCF should model the cycle explicitly for some years and revert to mid-cycle in the terminal year.

20.6 What the five cases have in common

Theme	Case(s)
Method follows the data available	Vodafone–Idea (no cash-flow forecasts → market, comparables, NAV)
Perceived value vs computed value	Vodafone–Idea (market reaction)
Users as the asset; strategic value	WhatsApp
Equity as an option; state-by-state valuation	Lifeline Hospitals
Multiple methods, wide range, negotiation	PayNest
Normalise before valuing	Rajat Steels

Chapter 20 summary

- Vodafone–Idea: market value, comparable multiples and NAV were used; DCF was not (no projections); the market still reacted negatively — "perceived value".
- WhatsApp: $21.8 bn for 450 m users (70% active), growth 1 m users a day, ≈ $55 per user including a $4 bn retention payout; valuation rests on user growth and future monetisation.
- Constructed cases reinforce distress, start-up and cyclical valuation.

Check your understanding

1. Why was DCF not used in the Vodafone–Idea valuation?
2. What does the market reaction teach about valuation?
3. Compute the price per active user for WhatsApp.
4. In the Lifeline case, why is equity not simply "EV − debt"?
5. Why must a peak-year EBITDA not be capitalised?

Answers. (1) Managements did not provide projected cash flows and other parameters. (2) A technically correct valuation may still be rejected due to perceived value. (3) 21.8 bn ÷ 315 m ≈ **$69**. (4) Equity is limited-liability: it can't be negative; value by state. (5) It overstates sustainable earnings.

> **Exam focus.** Be able to name the three methods used in Vodafone–Idea, the reason DCF was not used, and the WhatsApp facts (450 m users, 70% active, 1 m a day, $55 per user, $4 bn sweetener).

Chapter 21 — Practical Questions from the Syllabus Chapter, Fully Solved

This chapter restates each of the sixteen practical questions in the source chapter's "Test Your Knowledge" section and solves them in full, with commentary on the technique tested. Attempt each question before reading the solution. Amounts are as in the questions.

(The two theoretical questions — "Differentiate between EVA and MVA" and "Relative valuation is the method to arrive at a 'relative' value using comparative analysis to peers or similar enterprises: elaborate" — are answered in Chapters 11 and 8; model answers are in Chapter 23.)

Question 1 — Acquisition of XYZ Ltd. by ABC Company

Data. XYZ has 1.5 crore shares; market price ₹400 per share. ABC's average cost of capital is 12%. XYZ's expected cash accruals for the next three years (₹ crore): Year 1: 250; Year 2: 300; Year 3: 400. PV factors at 12%: 0.893, 0.797, 0.712. **Calculate the range of valuation ABC has to consider.**

Technique. Market value (lower bound) versus DCF value (upper bound) — a range for negotiation.

Solution.

Valuation based on market price: ₹400 × 1.5 crore = **₹600 crore**.

Valuation based on DCF:
(250 × 0.893) + (300 × 0.797) + (400 × 0.712) = 223.25 + 239.10 + 284.80 = **₹747.15 crore**.
Per share = 747.15 ÷ 1.5 = **₹498.10**.

Range of valuation	Per share (₹)	Total (₹ crore)
Minimum (market)	400.00	600.00
Maximum (DCF)	498.10	747.15

Commentary. The market price (₹400) is the sellers' minimum — shareholders will not sell below the market price. The DCF value (₹498.10) is the maximum ABC should pay, given its 12% cost of capital. A control premium would be negotiated within the range. (A purist would also add a terminal value if cash accruals continue beyond year 3; the question gives only three years.)

Question 2 — Eagle Ltd.

Data. Eagle Ltd. reported a profit of ₹77 lakh after 30% tax for 2011-12. The income included **extraordinary items of ₹8 lakh** (income) and an **extraordinary loss of ₹10 lakh**. Existing operations (other than extraordinaries) will continue. A new product launch is expected to give: sales ₹70 lakh; material ₹20 lakh; labour ₹12 lakh; fixed costs ₹10 lakh. Required: (i) value of the business at a capitalisation rate of 14%; (ii) market price per equity share, given share capital of 1,00,000 13% preference shares of ₹100 each and 50,00,000 equity shares of ₹10 each and a P/E ratio of 10.

Technique. Normalise profit; add incremental profit; capitalise; deduct preference dividend; apply P/E.

Solution.

(i) Business value

	₹ lakh
Profit before tax = 77 ÷ (1 − 0.30)	110.00
Less: extraordinary income	(8.00)
Add: extraordinary loss	10.00
Maintainable PBT from existing operations	**112.00**
Profit from new product: 70 − (20 + 12 + 10)	28.00
Total expected PBT	**140.00**
Less: tax @ 30%	(42.00)
Future maintainable profit after tax	**98.00**
Capitalisation rate	14%
Value of the business = 98 ÷ 0.14	**₹700 lakh**

(ii) Market price per equity share

	₹
Future maintainable profit after tax	98,00,000
Less: preference dividend (1,00,000 × ₹100 × 13%)	(13,00,000)
Earnings available to equity shareholders	**85,00,000**
Number of equity shares	50,00,000
EPS	**₹1.70**
P/E ratio	10
Market price per share	**₹17**

Commentary. Notice (1) the profit is grossed up to PBT before adjusting the extraordinary items (which are stated pre-tax) and (2) the preference dividend is deducted before computing EPS. Forgetting either is the common error.

Question 3 — ABC Co.'s new sales strategy

Data. Last year: Sales ₹20,000; gross margin 20% (₹4,000); administration, selling and distribution 10% (₹2,000); PBT ₹2,000; tax 30% (₹600); PAT ₹1,400. Balance sheet: fixed assets ₹8,000; current assets ₹4,000; equity ₹12,000.
Under the new strategy sales grow **20% a year for three years**; from the 4th year cash flow stabilises. Gross margin, asset-turnover ratio, capital structure and tax rate remain unchanged. **Depreciation is 10% of net fixed assets at the beginning of the year.** The target rate of return is **15%**. **Determine the incremental value due to adoption of the strategy.**

Technique. Projected balance sheet and free cash flows; terminal (residual) value; compare with value without the strategy.

Solution.

Ratios. Fixed assets = 8,000/20,000 = **40% of sales**; current assets = 4,000/20,000 = **20% of sales**. PBT = 10% of sales; PAT = 70% of PBT.

Projected balance sheet

	Year 1	Year 2	Year 3	Year 4
Sales	24,000	28,800	34,560	34,560
Fixed assets (40%)	9,600	11,520	13,824	13,824
Current assets (20%)	4,800	5,760	6,912	6,912
Total assets / Equity	14,400	17,280	20,736	20,736

Projected cash flows (free cash flow to firm)

	Year 1	Year 2	Year 3	Year 4
Sales	24,000	28,800	34,560	34,560
PBT (10% of sales)	2,400	2,880	3,456	3,456
PAT (70%)	1,680	2,016	2,419.20	2,419.20
Add: depreciation (10% of opening net fixed assets)	800	960	1,152	1,382.4
Less: addition to fixed assets	(2,400)	(2,880)	(3,456)	(1,382.4)
Less: increase in current assets	(800)	(960)	(1,152)	–
FCF	**(720)**	**(864)**	**(1,036.80)**	**2,419.20**

Notes: Year 1 addition to fixed assets = 9,600 − 8,000 = 1,600 net, plus depreciation 800 = 2,400 gross. Year 4: sales constant, so gross investment equals depreciation (1,382.4) to maintain 13,824 of net fixed assets; no increase in current assets.

Present value (15%)

Year	FCF	PVF @ 15%	PV
1	(720)	0.870	(626.40)
2	(864)	0.756	(653.18)

Year	FCF	PVF @ 15%	PV
3	(1,036.80)	0.658	(682.21)
			(1,961.79)

Residual value (from year 4, a level perpetuity): 2,419.20 ÷ 0.15 = **16,128**. Present value = 16,128 ÷ $(1.15)^3$ = 16,128 ÷ 1.521 = **10,603.55**.

Value with the strategy = 10,603.55 − 1,961.79 = **₹8,641.76**.

Value without the strategy (PAT 1,400 perpetual) = 1,400 ÷ 0.15 = **₹9,333.33**.

Incremental value = 8,641.76 − 9,333.33 = **−₹691.57**.

Conclusion. The strategy is **not financially viable**: it destroys ₹691.57 of value. The reason: sales grow 20% but the business must invest heavily in fixed and current assets (about 60% of incremental sales), generating negative free cash flow for three years; the stable-year cash flow (2,419) is only 73% above the current PAT, not enough to compensate. Total assets must rise from ₹12,000 to ₹20,736 (an increase of ₹8,736) before the PAT of 1,400 grows to 2,419.

Question 4 — H Ltd. and B Ltd. (exchange ratio)

Data. H Ltd. buys the business of B Ltd. effective 1 April 2012. Balance sheets at 31 March 2012 (₹ crore):

Liabilities	H Ltd.	B Ltd.	Assets	H Ltd.	B Ltd.
Equity shares of ₹100 each	350.00	–	Net fixed assets	220.00	0.50
Equity shares of ₹10 each	–	6.50	Net current assets	1,020.00	29.00
Reserves and surplus	950.00	25.00	Deferred tax assets	60.00	2.00
Total	**1,300.00**	**31.50**	**Total**	**1,300.00**	**31.50**

Information: (1) weighted-average post-tax maintainable profits of the last four years: H ₹300 crore; B ₹10 crore. (2) Both envisage a capitalisation rate of 8%. (3) H has a **contingent liability of ₹300 crore** as at 31 March 2012. (4) H will issue ₹100 shares to B's shareholders in terms of the exchange ratio on a **fair value basis**, with **weights of 1 and 3** for the values on net-asset basis and earnings-capitalisation basis for both companies.
Required: value of shares under (i) net asset method, (ii) earnings capitalisation method, (iii) exchange ratio on a fair-value basis.

Solution.

Number of shares: H: 350 ÷ 100 = **3.50 crore**. B: 6.50 ÷ 10 = **0.65 crore**.

(i) Net asset value per share

	Calculation	₹ per share
H Ltd.	(1,300 − 300 contingent liability) ÷ 3.50	**285.71**
B Ltd.	31.50 ÷ 0.65	**48.46**

Assumption: the contingent liability of H will materialise at its full amount (stated in the answer).

(ii) Earnings-capitalisation value per share

	Calculation	₹ per share
H Ltd.	300 ÷ 0.08 = 3,750 ÷ 3.50	**1,071.43**
B Ltd.	10 ÷ 0.08 = 125 ÷ 0.65	**192.31**

(iii) Fair value (weights 1 : 3)

	Calculation	₹ per share
H Ltd.	(285.71 × 1 + 1,071.43 × 3) ÷ 4	**875.00**
B Ltd.	(48.46 × 1 + 192.31 × 3) ÷ 4	**156.35**

Exchange ratio = 156.35 ÷ 875 = **0.1787**. H Ltd. should issue **0.1787 share for each share of B Ltd.** (Each ₹100-face H share is valued at ₹875.)

Question 5 — AB Ltd. and XY Ltd.

Data. AB Ltd. plans to acquire and absorb the running business of XY Ltd., consideration in equity shares of AB. As at 31.3.2006 AB's paid-up capital: 80 lakh shares of ₹10; highest and lowest quotations in the last six months ₹570 and ₹430; the **price per share is the average of high and low**.

XY's balance sheet at 31.3.2006 (₹ lakh): Sources: Share capital — 20 lakh equity shares of ₹10 fully paid 200; 10 lakh equity shares of ₹10, ₹5 paid 50; Loans 100; total 350. Uses: Fixed assets (net) 150; net current assets 200; total 350.

Merchant bankers' cash-flow estimates (after-tax earnings for equity, ₹ lakh): year ended 31.3.07: 105; 31.3.08: 120; 31.3.09: 125; 31.3.10: 120; 31.3.11: 100, plus a **terminal value estimate of 200**. The business of XY may be valued on the **average of (i) aggregate discounted cash flows at 8% and (ii) net assets value**. PV factors at 8% for years 1–5: 0.93, 0.86, 0.79, 0.74, 0.68. Required: (i) total value; (ii) number of shares to be issued by AB; (iii) basis of allocation among XY's shareholders.

Solution.

AB's share price = (570 + 430) ÷ 2 = **₹500**.

Value of XY Ltd. based on future cash flows (₹ lakh)
105 × 0.93 = 97.65

Value of XY Ltd. based on future cash flows (₹ lakh)	
120 × 0.86	103.20
125 × 0.79	98.75
120 × 0.74	88.80
(100 + 200) × 0.68	204.00
Value on discounted cash flows	**592.40**
Value on net assets (350 − 100 loans = share capital paid-up 250)	**250.00**
Average value (592.40 + 250) ÷ 2	**421.20**

Number of shares to be issued = ₹4,21,20,000 ÷ ₹500 = **84,240 shares**.

Basis of allocation. Convert partly paid shares to fully-paid equivalents: 20 lakh + (10 lakh × 5/10) = **25 lakh** equivalent shares.

- Fully paid shareholders: 84,240 × 20/25 = **67,392 shares**.
- Partly paid shareholders: 84,240 − 67,392 = **16,848 shares**.

Commentary. Using the merchant bankers' recommended weighting (equal weight to DCF and NAV) the value is 421.2 lakh; allocation recognises that ₹5-paid shares carry half the economic interest of fully paid shares.

Question 6 — Hansel Ltd. (book vs market weights)

Data. Expected free cash flow ₹54 lakh next year; expected growth 9%; analyst's value ₹1,800 lakh, but he used **book values** of debt and equity. Cost of equity 20%; post-tax cost of debt 10%. The **value of equity is thrice its book value** and the **market value of debt is nine-tenths of its book value**. Compute the correct value.

Technique. Infer the WACC used, back out the book weights, convert to market weights, recompute.

Solution. $V_0 = FCFF_1 \div (Kc - g)$. 1,800 = 54 ÷ (Kc − 0.09) → Kc − 0.09 = 0.03 → **Kc = 12%** (the WACC the analyst used).

Let X be the weight of debt in *book* terms: 20% × (1 − X) + 10% × X = 12% → 20 − 10X = 12 → X = 0.80. Book weights: **debt 80%, equity 20%**. Take book value of capital = 100 ⇒ debt 80, equity 20.

Market values: equity = 3 × 20 = **60**; debt = 0.9 × 80 = **72**. Total = 132.
Market weights: equity 60/132 = 45.45%; debt 72/132 = 54.55%.

Correct WACC = 20% × 60/132 + 10% × 72/132 = 9.0909% + 5.4545% = **14.5455%**.

Correct value = 54 ÷ (0.145455 − 0.09) = 54 ÷ 0.055455 = **₹973.7 lakh** (the source rounds to ₹974.73 lakh using WACC 14.5455% → 54 ÷ 0.0554 = 974.7). The analyst overstated value by about ₹826 lakh because the book weights gave

too much weight to cheap debt.

Question 7 — XYZ Ltd. (two-stage DCF)

Data. XYZ Ltd. grows at a higher rate for 4 years, then stabilises. Base year: revenue ₹2,000 crore; EBIT ₹300 crore; capital expenditure ₹280 crore; depreciation ₹200 crore.

	High growth (4 years)	Stable growth
Growth in revenue and EBIT	20%	10%
Growth in capex and depreciation	20%	Capex offset by depreciation
Risk-free rate	10%	9%
Equity beta	1.15	1
Market risk premium	6%	5%
Pre-tax cost of debt	13%	12.86%
Debt : equity	1 : 1	2 : 3

Working capital is 25% of revenue throughout; corporate tax 30%. **What is the value of the firm?**

Solution.

Cost of capital

High growth: Ke = 10% + 1.15 × 6% = **16.9%**; Kd = 13% × 0.7 = **9.1%**; weights 50:50 → WACC = 0.5 × 16.9 + 0.5 × 9.1 = **13%**.
Stable growth: Ke = 9% + 1.0 × 5% = **14%**; Kd = 12.86% × 0.7 = **9.0%**; D/E 2:3 → debt 40%, equity 60% → WACC = 0.6 × 14 + 0.4 × 9 = **12%**.

Forecast FCFF (₹ crore)

	Yr 1	Yr 2	Yr 3	Yr 4	Terminal year
Revenue	2,400.00	2,880.00	3,456.00	4,147.20	4,561.92
EBIT	360.00	432.00	518.40	622.08	684.29
EAT (EBIT × 0.7)	252.00	302.40	362.88	435.46	479.00
Capital expenditure	336.00	403.20	483.84	580.61	456.19
Less: depreciation	(240.00)	(288.00)	(345.60)	(414.72)	(456.19)
Net capex	96.00	115.20	138.24	165.89	–
Δ Working capital (25% of Δ revenue)	100.00	120.00	144.00	172.80	103.68
FCF = EAT − net capex − ΔWC	**56.00**	**67.20**	**80.64**	**96.77**	**375.32**

(Capex and depreciation grow 20% a year from 280 and 200: 336/240 in year 1, etc. In the terminal year capex equals depreciation.)

Present value of explicit FCFs @ 13%

FCFF	PVF	PV
56.00	0.885	49.56

FCFF	PVF	PV
67.20	0.783	52.62
80.64	0.693	55.88
96.77	0.613	59.32
		217.38

Terminal value (stable growth 10%; WACC 12%): $TV_4 = 375.32 \div (0.12 - 0.10) =$ **18,766 crore**. PV = 18,766 × 0.613 = **₹11,503.56 crore**.

Value of the firm = 217.38 + 11,503.56 = **₹11,720.94 crore**.

Commentary. (1) The stable-growth WACC (12%) is *lower* than the high-growth WACC (13%) because beta falls to 1 and the debt ratio and cost of debt change. (2) The terminal value is **98%** of the total; WACC − g = 2% makes the result hypersensitive (a 1% rise in the stable WACC to 13% would cut TV to ₹12,510 crore × 0.613 ≈ 7,669 → total 7,886: a 33% fall). (3) Note the discounting of TV by the year-4 factor at the *high-growth* WACC, as the source does; in theory the stable-phase rate applies only from year 5, and the TV already embeds it.

Question 8 — WXY Ltd. (FCFF and WACC)

Data. WXY Ltd. grows at **20% a year for three years**, then 8% in perpetuity. Year ended 31 March 2014 (₹ crore): revenue 7,500; COGS 3,000; operating expenses 2,250; capital expenditure 750; depreciation (included in operating expenses) 600. During the high-growth period revenues and EBIT grow at 20% and **capex net of depreciation grows at 15%**; from year 4 revenues and EBIT grow at 8%, incremental capex is offset by depreciation; net working capital is **25% of revenue**; **WACC 15%**; tax 30%. PVIF at 15%: 0.8696, 0.7561, 0.6575. **Estimate the value using FCFF and WACC.**

Solution.

Base year: EBIT = 7,500 − 3,000 − 2,250 = **2,250** (depreciation is already in operating expenses). Operating expenses excluding depreciation = 2,250 − 600 = 1,650; depreciation 600; net capex = 750 − 600 = **150**.

₹ crore	Yr 1	Yr 2	Yr 3	Terminal year
Revenue	9,000.00	10,800.00	12,960.00	13,996.80
COGS	3,600.00	4,320.00	5,184.00	5,598.72
Operating expenses (excl. depreciation)	1,980.00	2,376.00	2,851.20	3,079.30
Depreciation	720.00	864.00	1,036.80	1,119.74
EBIT	**2,700.00**	**3,240.00**	**3,888.00**	**4,199.04**
Tax @ 30%	810.00	972.00	1,166.40	1,259.71
EAT	**1,890.00**	**2,268.00**	**2,721.60**	**2,939.33**

₹ crore	Yr 1	Yr 2	Yr 3	Terminal year
Net capex (capex − depreciation; +15% a year from 150)	172.50	198.38	228.13	–
Δ Working capital (25% of Δ revenue)	375.00	450.00	540.00	259.20
FCF	**1,342.50**	**1,619.62**	**1,953.47**	**2,680.13**

PV of explicit FCF @ 15%

FCF	PVIF	PV
1,342.50	0.8696	1,167.44
1,619.62	0.7561	1,224.59
1,953.47	0.6575	1,284.41
		3,676.44

Terminal value: 2,680.13 ÷ (0.15 − 0.08) = **38,287.57**; PV = 38,287.57 × 0.6575 = **25,174.08**.

Value of the firm = 3,676.44 + 25,174.08 = **₹28,850.52 crore.**

Commentary. In the terminal year, operating expenses (excluding depreciation) grow by 8% like revenue; EBIT = revenue growth × EBIT (8%). The terminal-year working-capital investment is 25% × (13,996.80 − 12,960) = 259.20. The terminal value is 87% of the total.

Question 9 — Jatayu Limited (EVA)

Data. Equity capital ₹160 lakh; reserves and surplus ₹140 lakh; 10% debentures ₹400 lakh. Cost of equity 14%; **financial leverage 1.5 times**; tax rate 30%. Compute EVA.

Solution.
Interest = 10% × 400 = **₹40 lakh**.
Financial leverage = PBIT ÷ PBT: 1.5 = PBIT ÷ (PBIT − 40) → 1.5 PBIT − 60 = PBIT → 0.5 PBIT = 60 → **PBIT = ₹120 lakh**.
NOPAT = 120 × (1 − 0.30) = **₹84 lakh**.
Capital employed = 160 + 140 + 400 = **₹700 lakh**.
WACC = 14% × (300/700) + 10% × (1 − 0.30) × (400/700) = 6.0% + 4.0% = **10%**.
EVA = 84 − 10% × 700 = **₹14 lakh**.

Question 10 — RST Ltd.

Data. Net income after tax ₹25,00,000; tax 30%. Debt (coupon 11%) ₹40 lakh; equity (share capital + reserves) ₹125 lakh; invested capital ₹165 lakh. Equity beta 1.36; risk-free 8.5%; market risk premium 9%. Required: (i) WACC; (ii)

EVA.

Solution.
Cost of equity = 8.5% + 1.36 × 9% = **20.74%**. *Post-tax Kd* = 11% × 0.7 = **7.70%**.
WACC = 20.74% × 125/165 + 7.70% × 40/165 = 15.71% + 1.87% = **17.58%**.
Taxable income = 25,00,000 ÷ 0.70 = **₹35,71,429**. *Interest* = 11% × 40,00,000 = **₹4,40,000**. *EBIT* = **₹40,11,429**.
NOPAT = 40,11,429 × 0.70 = **₹28,08,000**.
Capital charge = 17.58% × 1,65,00,000 = **₹29,00,700**.
EVA = 28,08,000 − 29,00,700 = **−₹92,700**.

Interpretation. Negative EVA: the company earns less than its cost of capital, even though it reports a profit of ₹25 lakh.

Question 11 — Tender Ltd.

Data. Net profit ₹15 lakh after 30% tax; interest ₹10 lakh; invested capital ₹95 lakh of which 55% is debt; WACC 13%. (a) Operating income; (b) EVA; (c) 6 lakh equity shares: how much dividend can be paid before the value of the entity starts declining?

Solution.
(a) Taxable income = 15 ÷ 0.70 = **₹21.43 lakh**; **operating income (EBIT)** = 21.43 + 10 = **₹31.43 lakh**.
(b) EVA = EBIT(1 − t) − WACC × invested capital = 31.43 × 0.70 − 13% × 95 = 22.00 − 12.35 = **₹9.65 lakh**.
(c) **EVA dividend** = EVA ÷ shares = ₹9,65,000 ÷ 6,00,000 = **₹1.6083 per share**. Dividends up to the EVA do not reduce value: beyond that the company would be distributing capital rather than surplus.

Question 12 — Orange, Grape and Apple

Data. Identical companies except capital structure; invested capital ₹1,00,000 each; tax 35%.

	Orange	Grape	Apple
Debt/assets	0.8	0.5	0.2
Shares outstanding	6,100	8,300	10,000
Pre-tax cost of debt	16%	13%	15%
Cost of equity	26%	22%	20%
EBIT	25,000	25,000	25,000

Required: (i) WACC; (ii) EVA; (iii) the best investment by EVA; (iv) price per share at industry P/E 11; (v) estimated market capitalisation.

Solution.

(i) WACC (post-tax Kd: 16% × 0.65 = 10.40%; 13% × 0.65 = 8.45%; 15% × 0.65 = 9.75%)

	Debt	Equity	WACC
Orange	80,000	20,000	10.4% × 0.8 + 26% × 0.2 = **13.52%**
Grape	50,000	50,000	8.45% × 0.5 + 22% × 0.5 = **15.225%**
Apple	20,000	80,000	9.75% × 0.2 + 20% × 0.8 = **17.95%**

(ii) EVA = EBIT(1 − t) − WACC × IC = 16,250 − WACC × 1,00,000:
Orange: 16,250 − 13,520 = **2,730**; Grape: 16,250 − 15,225 = **1,025**; Apple: 16,250 − 17,950 = **−1,700**.

(iii) Orange is the best investment: highest EVA and lowest WACC.

(iv) Price per share

	Orange	Grape	Apple
EBIT	25,000	25,000	25,000
Interest (debt × Kd)	12,800	6,500	3,000
Taxable income	12,200	18,500	22,000
Tax @ 35%	4,270	6,475	7,700
Net income	7,930	12,025	14,300
Shares	6,100	8,300	10,000
EPS	₹1.30	₹1.45	₹1.43
Price @ P/E 11	**₹14.30**	**₹15.95**	**₹15.73**

(v) Market capitalisation: Orange 14.30 × 6,100 = **₹87,230**; Grape 15.95 × 8,300 = **₹1,32,385**; Apple 15.73 × 10,000 = **₹1,57,300**.

Commentary. EVA ranks Orange first, yet the P/E-based price ranks it *last*. The P/E of 11 is applied uniformly although the three have different financial risk: Orange (80% debt) deserves a **lower** P/E. This is why P/E must be adjusted for leverage; EVA, which charges for the capital actually used, is not misled.

Question 13 — Delta Ltd.

Data. Net income ₹15,00,000; marginal tax rate 40%; interest expense ₹15,00,000; invested capital ₹1,00,00,000 of which 60% is debt; WACC 12.6%. Delta has 2,50,000 shares. (i) EBIT; (ii) EVA; (iii) dividend per share before the value starts to fall; what if no dividend is paid?

Solution.
(i) Taxable income = 15,00,000 ÷ (1 − 0.40) = **₹25,00,000**; EBIT = 25,00,000 + 15,00,000 = **₹40,00,000**.
(ii) EVA = 40,00,000 × 0.60 − 0.126 × 1,00,00,000 = 24,00,000 − 12,60,000 = **₹11,40,000**.
(iii) EVA dividend = 11,40,000 ÷ 2,50,000 = **₹4.56** per share. If Delta pays no dividend, the retained EVA is reinvested, growth and EBIT rise, and **the value of the firm is expected to increase**.

Question 14 — XYZ Inc. (software consultancy)

Data (31 Dec 2010, $ million). Income from consultancy 935; EBIT 180; interest 18; EBT 162; tax @ 35% 56.70; PAT 105.30. Balance sheet: equity stock (10 million shares at $10) 100; reserves 325; loans 180; current liabilities 180 (total 785); land and building 200; computers and software 295; current assets (debtors 150, bank 100, cash 40) 290. WACC 12%; share price $50. Compute (a) EVA and (b) MVA.

Solution.
NOPAT = 180 × (1 − 0.35) = **117.00**.
Capital employed = equity 100 + reserves 325 + loans 180 = **605**. *Capital charge* = 605 × 12% = **72.60**.
(a) EVA = 117.00 − 72.60 = $44.40 million.
Market value of equity = $50 × 10 million = **$500 million**; *equity funds (book)* = 100 + 325 = **425**.
(b) MVA = 500 − 425 = $75 million.

Question 15 — Herbal Gyan

Data. A beauty-cosmetics company; earnings average ₹12 lakh after tax, largely from a **patented** beauty cream. The patent has eight years to run and the company has been **offered ₹40 lakh** for it. Assets include ₹20 lakh working capital and ₹80 lakh property, plant and equipment; the patent is **not shown** in the books. Cost of capital 15%. **What is its EVA?**

Solution. Include the patent at its market value in the investment base:

Total investment	₹ lakh
Working capital	20
Property, plant and equipment	80
Patent rights	40
Total	**140**

EVA = income − (cost of capital × total investment) = 12 − (0.15 × 140) = 12 − 21 = **−₹9 lakh**.

Interpretation. Herbal Gyan has a negative EVA: it would earn more by selling the patent for ₹40 lakh and investing the proceeds at 15%. The *opportunity cost* of the unrecorded patent must be charged.

Question 16 — Constant Engineering Ltd.

Data. A high-tech product that reduces carbon emission has been patented; market value ₹100 crore, **not recorded** in the books. Net worth ₹200 crore; long-term debt ₹400 crore. The product generates **net operating profit after tax of ₹84 crore**. The 365-day government bond rate is 10%; the market portfolio generates 12%; the stock moves in tandem with the market (**β = 1**). Calculate EVA.

Solution.
Cost of equity (CAPM) = 10% + 1 × (12% − 10%) = **12%**. *Cost of debt:* the question treats debt cost as 10% (government bond/long-term rate) as in the model solution.
Invested capital = net worth 200 + long-term debt 400 + patent 100 = **₹700 crore**.
Weights: equity = (200 + 100)/700 = 300/700; debt = 400/700.
WACC = 12% × 300/700 + 10% × 400/700 = 5.14% + 5.71% = **10.85%**.
EVA = 84 − 10.85% × 700 = 84 − 75.95 = **₹8.05 crore**.

Cross-question patterns

Pattern	Questions
Market vs DCF range	1
Normalise profit, capitalise, P/E	2
Strategy evaluation with FCF and residual value	3
Merger exchange ratio: NAV + earnings weights	4, 5
WACC weights: book vs market	6
Two-stage FCFF	7, 8
EVA computation and EVA dividend	9–16
Include unrecorded patents in capital	15, 16

Chapter 22 — Additional Practice Problems with Solutions

These fifteen new problems mirror the style of the syllabus questions and cover every method in the book. Work each before reading the solution. Amounts are in ₹ crore unless stated.

Problems

P1 — Net asset value and NRV. A company has fixed assets (net) ₹900, net current assets ₹400 and long-term debt ₹500; 10 crore shares. Realisable values: fixed assets ₹1,000, net current assets ₹360; liquidation expenses ₹30. Compute value per share on book and on net-realisable bases.

P2 — Earnings capitalisation and P/E. Profits after tax of the last four years: 40, 46, 52, 58. Use weights 1, 2, 3, 4 to find maintainable profit. (a) Capitalise at 12.5%. (b) The company has 8 crore shares; a listed peer trades at P/E 15; apply a 25% discount for non-listing and compute the price per share.

P3 — Beta, Ke and WACC. The asset beta of a business is 0.9. The company plans a capital structure of D/E = 50% with a tax rate of 30%. Risk-free 6%; market return 12%; pre-tax cost of debt 10%. Compute the equity beta, cost of equity and WACC.

P4 — Three-year DCF. FCFF: 80, 90, 100 for years 1–3; WACC 10%; growth 4% after year 3. Debt 300; surplus cash 50; 10 crore shares. Compute EV, equity and value per share.

P5 — Enterprise value. 12 crore shares at ₹250; debt ₹800; cash ₹150; EBITDA ₹500. Compute EV and EV/EBITDA.

P6 — Chop-shop. Segment A: sales 800, assets 600, operating income 120; peer capitalisation ratios: sales 0.9, assets 0.7, income 7.5. Segment B: sales 500, assets 400, income 80; ratios 1.2, 1.0, 8.0. Compute the chop-shop value.

P7 — EVA and MVA. EBIT ₹650; tax 30%; invested capital ₹3,000 (equity 1,800 and debt 1,200); Ke 15%; pre-tax Kd 10%; market value of the firm ₹4,500. Compute NOPAT, WACC, EVA and MVA.

P8 — Exchange ratio. A: net asset value per share ₹120, earning-capitalisation value ₹200. B: ₹45 and ₹70. Use weights 1 (NAV) and 2 (earnings). Compute the exchange ratio.

P9 — Distressed valuation. (a) Going-concern equity ₹500, distress-sale equity ₹40, probability of distress 30%: value of equity. (b) Unlevered value ₹1,800; PV of tax benefits of debt ₹200; distress sale value ₹1,000; probability

of distress 25%: APV.

P10 — Venture capital method. Expected exit value ₹250 in 4 years; investor puts in ₹20 and wants 5× its money. Compute post-money, pre-money, ownership and the equivalent IRR.

P11 — ESG. FCFF next year ₹200; WACC 10%; growth 3%. A carbon cost of ₹25 (pre-tax; tax 25%) will apply every year, growing at 3%. Compute the change in EV (a) by cash-flow adjustment, (b) by adding 0.5% to the discount rate.

P12 — Going versus liquidation. A business has a DCF value of ₹450. Its assets could be sold for ₹560, with selling costs of ₹30 and taxes on gains of ₹10; liabilities ₹0 (ignore). What should the owners do?

P13 — Digital platform. Platform A: valued at ₹480 with 8 crore users. Platform B has 5 crore users but 20% lower ARPU. Value B on A's value per user, adjusted for ARPU.

P14 — Professional firm. Fees ₹9.0; operating costs (excluding partners' pay) ₹4.2; three partners each paid ₹0.9 but market pay is ₹0.7; tax 25%; required return 22%. Compute the value.

P15 — Two-stage FCFF. FCFF year 1 = ₹60, growing 15% a year for three years (60, 69, 79.35), then 5% for ever. WACC is 12% for years 1–3 and 10% in the stable phase. Compute the value of the firm.

Solutions

P1. *Book:* 900 + 400 − 500 = **₹800** → ₹80 per share. *NRV:* 1,000 + 360 − 30 − 500 = **₹830** → ₹83 per share. NRV exceeds book because the fixed assets are worth more than their book value (+100) by more than the shortfall on current assets (−40) and costs (−30).

P2. Maintainable profit = (40×1 + 46×2 + 52×3 + 58×4) ÷ 10 = (40 + 92 + 156 + 232) ÷ 10 = **52.0**.
(a) Capitalised value = 52 ÷ 0.125 = **₹416 crore**.
(b) EPS = 52 ÷ 8 = ₹6.50; adjusted P/E = 15 × 0.75 = 11.25; price = 6.50 × 11.25 = **₹73.13** → equity ₹585 crore. The P/E value is higher than the capitalisation value because the 12.5% rate implies a P/E of only 8, while the peer implies 11.25 — the market is pricing in growth.

P3. β_e = 0.9 × [1 + (1 − 0.30) × 0.50] = 0.9 × 1.35 = **1.215**. Ke = 6% + 1.215 × (12% − 6%) = **13.29%**. D/E = 0.5 ⇒ E/V = 2/3, D/V = 1/3: WACC = 13.29% × 2/3 + 10% × 0.7 × 1/3 = 8.86% + 2.33% = **11.19%**.

P4. PV of FCFF: 80/1.1 + 90/1.21 + 100/1.331 = 72.73 + 74.38 + 75.13 = **222.24**. TV (year 3) = 100 × 1.04 ÷ 0.06 = **1,733.33**; PV = 1,733.33 ÷ 1.331 = **1,302.28**. EV = **1,524.52**. Equity = 1,524.52 − 300 + 50 = **₹1,274.52**; per share = **₹127.45**. TV share = 85%.

P5. Market cap = 12 × 250 = ₹3,000. EV = 3,000 + 800 − 150 = **₹3,650**. EV/EBITDA = 3,650 ÷ 500 = **7.3×**.

P6.

Basis	Segment A	Segment B	Total
Sales (A: 800 × 0.9; B: 500 × 1.2)	720	600	**1,320**
Assets (600 × 0.7; 400 × 1.0)	420	400	**820**
Income (120 × 7.5; 80 × 8.0)	900	640	**1,540**

Average = (1,320 + 820 + 1,540) ÷ 3 = **₹1,226.7**.

P7. NOPAT = 650 × 0.70 = **455**. WACC = (1,800 × 15% + 1,200 × 10% × 0.70) ÷ 3,000 = (270 + 84) ÷ 3,000 = **11.8%**. Capital charge = 0.118 × 3,000 = 354. **EVA = 455 − 354 = ₹101. MVA = 4,500 − 3,000 = ₹1,500.**

P8. A: (120 × 1 + 200 × 2) ÷ 3 = **173.33**. B: (45 × 1 + 70 × 2) ÷ 3 = **61.67**. Exchange ratio = 61.67 ÷ 173.33 = **0.3558** shares of A for each share of B.

P9. (a) 500 × 0.70 + 40 × 0.30 = 350 + 12 = **₹362**. (b) Expected bankruptcy cost = (1,800 − 1,000) × 0.25 = 200. APV = 1,800 + 200 − 200 = **₹1,800**: the distress costs exactly offset the tax benefit; debt adds no value at that probability.

P10. Post-money = 250 ÷ 5 = **₹50**; pre-money = 50 − 20 = **₹30**; ownership = 20 ÷ 50 = **40%**; IRR = $5^{1/4} - 1$ = **49.5%**.

P11. Baseline EV = 200 ÷ (0.10 − 0.03) = **₹2,857**. (a) After-tax carbon cost = 25 × 0.75 = 18.75; adjusted FCFF = 181.25; EV = 181.25 ÷ 0.07 = **₹2,589** → −₹268 (**−9.4%**). (b) Discount rate 10.5%: EV = 200 ÷ 0.075 = **₹2,667** → −6.7%. The cash-flow approach shows a larger and more transparent effect.

P12. Net liquidation proceeds = 560 − 30 − 10 = **₹520** > DCF value 450. The business is worth more dead than alive: the owners should consider selling the assets (or restructuring to raise the going-concern value above 520). Going-concern value should be preferred only if it exceeds liquidation value.

P13. Value per user for A = 480 ÷ 8 = ₹60. B: 5 crore × ₹60 × 0.80 = **₹240 crore**.

P14. Normalised PBT = 9.0 − 4.2 − (3 × 0.7 = 2.1) = **2.7**. PAT = 2.7 × 0.75 = 2.025. Value = 2.025 ÷ 0.22 = **₹9.2 crore**. (Reported PBT with actual partner pay of 2.7: 9.0 − 4.2 − 2.7 = 2.1 — normalisation adds ₹0.6 crore.)

P15. PV of years 1–3 at 12%: 60/1.12 + 69/1.2544 + 79.35/1.4049 = 53.57 + 55.01 + 56.48 = **165.06**. $FCFF_4$ = 79.35 × 1.05 = 83.32. TV at year 3 = 83.32 ÷ (0.10 − 0.05) = **1,666.35**; PV = 1,666.35 ÷ 1.4049 = **1,186.08**. **Value = ₹1,351.13 crore.** (TV = 88% of the value.)

Self-assessment

Score (of 15)	Suggestion
13–15	Move on to timed practice of Chapter 21 questions
9–12	Revisit the chapters for the missed topics (NAV/NRV: Ch. 3; income: Ch. 4; DCF: Ch. 5; beta: Ch. 7; EVA: Ch. 11)
< 9	Re-read Parts I–III and redo the worked examples before attempting the problems

Chapter 23 — Answering Valuation Questions: Technique and Model Theory Answers

23.1 The anatomy of a valuation question

Almost every question on this chapter falls into one of three kinds:

1. **Computational** — compute value by a named method (NAV, capitalisation, DCF, EVA, exchange ratio).
2. **Evaluation and synthesis** — value by several methods, comment, conclude with a range.
3. **Theory / short notes** — explain a concept (relative valuation, EVA vs MVA, distressed firms, start-ups, ESG, going concern).

A reliable approach to each follows.

23.2 Computational questions — a seven-point checklist

1. **Read the data twice**: units (₹ lakh, ₹ crore, ₹ thousand), tax rate, dates, "per share" vs total.
2. **Write the formula** before the numbers — examiners award marks for method.
3. **Normalise** when profit includes extraordinary items; gross up to PBT if the items are pre-tax.
4. **Use the right base**: after-tax profit for earnings approaches; EBIT(1 − t) for NOPAT; FCFF for DCF.
5. **Label each step** (e.g., "Step III — WACC").
6. **Discount properly**: terminal value at the horizon must be discounted; use the PV factors given.
7. **State the answer with units** and a one-line interpretation.

23.2.1 Common computational traps

Trap	Example question	Correct approach
Forgetting preference dividend	Eagle Ltd. (Q2)	Deduct from earnings before dividing by equity shares
Not grossing up	Eagle Ltd.	PBT = PAT ÷ (1 − t) before removing extraordinary items
Using debt/total capital as D/E	Illustration 1	"40:60" is D/E = 40/60 for beta; weights 40% and 60% in WACC
Book weights in WACC	Hansel Ltd. (Q6)	Convert to market values
Omitting unrecorded intangibles	Herbal Gyan, Constant Engineering (Q15–Q16)	Include in invested capital

Trap	Example question	Correct approach
Wrong share count	Partly-paid shares (Q5)	Convert to fully-paid equivalents
Terminal value not discounted	Two-stage DCF (Q7–Q8)	Multiply by the cumulative factor
Ignoring contingent liability	H Ltd. (Q4)	State the assumption and deduct
Mixing EV and equity	EBITDA-multiple questions	Bridge: EV − debt + surplus cash

23.3 Evaluation and synthesis questions — the three-point approach

As the source chapter puts it:

1. Unless specified otherwise, calculate valuation by **as many methods as possible** with the available data.
2. **Give comments** on the valuation by each of these methods.
3. **Supplement your conclusion** with any additional information if available.

Template.

Section	Content
Method 1 (asset)	Calculation; comment: lower limit, ignores earning power
Method 2 (earnings)	Calculation; comment: depends on maintainable profit and rate
Method 3 (DCF)	Calculation; comment: sensitive to WACC and TV
Method 4 (market/relative)	Calculation; comment: peers' comparability
Summary	Table of values; range
Conclusion	Range; point estimate; seller's minimum, buyer's maximum
Additional information	Synergies, control premium, contingent liabilities, subsequent events

23.4 Model answers to theory questions

23.4.1 Differentiate between EVA and MVA *(syllabus Theoretical Question 1)*

Basis	Economic Value Added (EVA)	Market Value Added (MVA)
Meaning	Excess of NOPAT over the capital charge (invested capital × WACC) — the *economic* profit of a period	Excess of the market value of the firm over the invested capital — the *wealth* created for investors
Formula	NOPAT − (Invested capital × WACC)	Market value of the firm − Invested capital
Nature	A **flow**, measured for a period; derived from accounting data (with adjustments)	A **stock**, measured at a point in time; derived from market prices
Perspective	Internal — performance of management; "derived value added" for the discerning investor	External — the market's perception of value created
Link	MVA equals the present value of future EVAs	
Availability	Can be computed for any business unit or unlisted company	Needs a market value (listed companies)

Basis	Economic Value Added (EVA)	Market Value Added (MVA)
Strengths	Holds management accountable for capital; separates operations from financing	Reflects market's view of future prospects; includes value drivers beyond the accounts
Limitations	Historical; relies on accounting adjustments	Reflects sentiment; does not give a basis for share valuation; includes the effect of market volatility
Short-term relationship	Companies with high MVA may be "pricey" and show slightly negative correlation with EVA in the short term; the gap closes over time	

Illustration: NOPAT 307, IC 920, WACC 8.44% → EVA 229.35; market value 1,000 → MVA 80.

23.4.2 Relative valuation *(syllabus Theoretical Question 2)*

Relative valuation is the method to arrive at a "relative" value using a "comparative" analysis to its peers or similar enterprises. Elaborate.

Model answer.

- The asset-based, earnings-based and cash-flow-based approaches arrive at the **intrinsic** value of an enterprise. **Relative valuation**, in contrast, arrives at a **relative** value by comparing the enterprise with **similar enterprises (peers)**, and is increasingly used with the other approaches to **validate** the intrinsic value.
- It uses **multiples** — financial ratios — derived from comparable firms (those with similar asset and risk dispositions) and applied to the subject.
- **Steps:** (1) find the **drivers** that best represent the multiple; (2) determine the results through **financial ratios**; (3) find the **comparable firms** and perform the comparative analysis; (4) **iterate** the value to smoothen deviations.
- **Types of multiples:** *enterprise-value based* — EV/EBITDA, EV/Invested capital, EV/Sales; *equity-value based* — P/E and PEG (more suitable for high-growth or sunrise industries).
- **Choosing the right ratio** is vital: e.g., a firm with export revenue is exposed to currency fluctuations, so a pure P/E may not reflect that; EV/Invested capital is a misfit for asset-light firms.
- **Choosing comparables** is the most challenging step: no two entities are the same — e.g., FTE pricing versus Unit Transfer Pricing software firms have different risk structures; for niche markets, look at similarly structured firms in other industries or use regression.
- **Assumptions:** efficient markets; a linear relationship between fundamentals and multiples; comparable firms are similar in structure, risk and growth.
- **Illustration:** earning value per share and equity value from an EBITDA multiple (Illustration 2); enterprise value from a balance sheet (Illustration

3).

- **Conclusion:** relative valuation is a comparative-driven approach that assumes the value of similar firms is a good indicator of the value of the tested firm — benchmarks must be taken "with a pinch of salt".

23.4.3 Going concern vs non-going concern

- **Going concern:** the enterprise is assumed to continue for the foreseeable future with no intention or need to liquidate or curtail materially; assets are valued on that basis (historical cost in accounts); value includes future profitability, intangibles and goodwill — the **total value**.
- **Non-going concern (liquidation):** net value realised after disposing of all assets and discharging all liabilities.
- **Relation:** going-concern value is generally greater than liquidation value (premium for future profitability, goodwill; liquidation brings job losses and reputational damage).
- **When to use liquidation value:** only when investors believe the firm has no longer value as a going concern.

23.4.4 Valuation of distressed companies

- Define distress (high debt, inability to meet operating expenses; fixed costs, illiquid assets, cyclical revenue); not necessarily worthless.
- Why conventional methods fail (infinite-life TV, negative cash flows, discount rates, going-concern assumption).
- Four methods: modified DCF (expected CF = CF × (1 − P)); DCF + distress value (equity = DCF × (1 − P) + distress value × P); APV (unlevered value + tax benefits − expected bankruptcy costs); relative valuation (e.g., 2× → 1.25× revenue).
- Illustration; liquidation floor.

23.4.5 Valuation of start-ups

- Traditional methods presuppose established business; start-ups lack history, assets, comparables.
- Value drivers: product, management, traction, revenue, industry attractiveness, demand–supply, competitiveness.
- Methods: Berkus ($2m/$2.5m caps), cost-to-duplicate, comparable transactions, scorecard, First Chicago, venture capital method.

23.4.6 Valuation of digital platforms

- Types (marketplace, search, repository, communication, community, payments).

- Income approach: top-down (TAM/SAM/SOM) or bottom-up; discount FCFF at WACC / FCFE at Ke; CSRP for nascent platforms.
- Market approach: revenue-driver multiples because earnings and book values are low or negative.
- Cost approach: rebuild cost of code (a floor).

23.4.7 Valuation of professional firms

- People-based; normalise net income and cash flows; compare with industry KPIs; factor risk in projections; use KPIs aligned with the acquirer.

23.4.8 Impact of ESG on valuation

- Define ESG; list factors; momentum (ESG funds, green bonds, taxonomies, single reporting standard, SEBI's Feb 2023 proposal).
- Incorporate in discount rate (practical but not explicit) or cash flows (explicit): E — 2-degree scenario, carbon points; S — cost of social measures; G — penalties, fines, taxes.

23.4.9 CAPM versus APT

Table in Chapter 6.

23.4.10 LBO

- PE firms; 70:30 debt-equity; 3–5 year horizon; value created by leverage; exit by spin-off or sale; risk — cyclical slowdown and heavy interest burden.

23.4.11 Chop-shop method

Three steps; Cornett example; used for undervalued multi-industry firms; buying assets below replacement value.

23.5 Time management

For a 5–10 mark valuation question allow roughly one minute per mark. Attempt computational questions first if confident; do short notes in bullets with headings and one illustration.

23.6 Revision plan for the last week

Day	Focus
1	Chapters 3–5: approaches; redo Q1, Q2
2	Chapter 6–7: CAPM, APT, beta; redo Illustration 1; Q10
3	Chapter 8–10: relative, LBO, chop-shop; Illustrations 2–4
4	Chapter 11–12: EVA, MVA, SVA; Q9–Q16
5	Chapter 13–14: fair value, distressed; Q3, Q4, Q5
6	Chapter 15–18: start-ups, platforms, professional firms, ESG (theory)

Day	Focus
7	Timed paper: Chapter 22 problems; revise Appendix A formula sheet

23.7 A final checklist of formulas to memorise

(See Appendix A.)

- PV, NPV, IRR (decision rule), ROI on average investment; Gordon $P_0 = D_1/(Ke - g)$
- NAV, NRV, replaceable value
- Price = EPS × P/E; capitalised value = profit ÷ rate; rate = EPS/price = 1/PE
- FCFF; TV = CF × (1 + g)/(WACC − g); equity bridge
- CAPM; APT; β_a and β_e; WACC
- EV = market cap + debt − cash; chop-shop average
- NOPAT; invested capital; EVA; MVA
- Distressed: modified DCF; equity = DCF(1 − P) + distress × P; APV and expected bankruptcy cost
- Scorecard; VC method

Chapter 23 summary

- Three kinds of question: computational, synthesis, theory; each has a template.
- For synthesis: many methods, comment on each, add supplementary information.
- Know model answers for EVA vs MVA, relative valuation, going vs non-going concern, distressed, start-ups, platforms, professional firms, ESG.

Part VII — Extensions

The chapters in this part deepen topics that the syllabus chapter treats briefly or assumes from earlier study. They can be read in any order, but Chapter 24 (valuation of securities) is the natural prerequisite for the rest of the book and may be read first by students who need a refresher.

Chapter 24 — Background: Valuing Securities (Bonds, Preference Shares and Equity)

The syllabus notes that the valuation of **securities** — equity shares, preference shares, bonds — has been covered earlier, and that the focus now is the valuation of the **business**. Business valuation, however, rests on the same principles: every security is a claim on cash flows, valued by discounting those cash flows at a required return. This chapter revises the valuation of the three standard securities, because (i) the debt and preference components appear in the equity bridge, (ii) the cost of debt is derived from bond yields, and (iii) equity valuation by dividends is the model behind the Gordon formula.

24.1 The common principle

Value of any security = Σ (expected cash flow in period t) ÷ (1 + required return)^{t}

What changes between securities is (a) the **pattern of cash flows** (fixed coupons and principal; fixed preference dividends; uncertain, growing dividends) and (b) the **required return** (which reflects the risk of that claim: lowest for debt, higher for preference, highest for equity).

24.2 Valuing bonds (debentures)

A conventional bond pays a fixed **coupon** each year and repays the **face value** at maturity:

Price = $\Sigma_{t=1}^{n}$ [Coupon ÷ (1 + y)^{t}] + Face value ÷ (1 + y)^{n}

where y is the **yield to maturity (YTM)**, the market's required return on bonds of that risk and maturity.

24.2.1 Worked example 24.1

A 5-year debenture of face value ₹1,000 with a 9% annual coupon.

Required yield	Price	Relationship to face value
7%	**₹1,082.0**	Premium (coupon > yield)
9%	**₹1,000.0**	Par (coupon = yield)
11%	**₹926.1**	Discount (coupon < yield)

Calculation at 11%: PV of coupons = 90 × annuity factor (5 years, 11%) 3.6959 = 332.6; PV of face value = 1,000 × 0.5935 = 593.5; price = **926.1**.

Inverse relationship: when market yields rise, bond prices fall, and vice versa. If the debenture trades at ₹920, the YTM is about **11.17%**.

24.2.2 Why this matters in business valuation

1. **Cost of debt.** The pre-tax cost of debt in WACC is the YTM of the firm's bonds (or the current borrowing rate), not the old coupon.
2. **Market value of debt.** In the equity bridge and in WACC weights, debt should be at market value; it differs from face value when rates have moved or when default risk is high.
3. **Distressed debt.** The promised yield overstates the expected return; debt of a distressed firm trades at a discount that reflects the probability of default (Chapter 14).

24.3 Valuing preference shares

A preference share pays a **fixed dividend** ahead of equity.

- **Irredeemable preference share** (perpetual): **Value = Dividend ÷ Required return**.
- **Redeemable preference share: Value = Σ Dividend ÷ $(1 + k)^t$ + Redemption value ÷ $(1 + k)^n$** (like a bond).

24.3.1 Worked example 24.2

A ₹100 preference share paying 10% dividend, required return 12%.

- *Irredeemable:* 10 ÷ 0.12 = **₹83.33**.
- *Redeemable at par after 5 years:* 10 × 3.6048 + 100 × 0.5674 = 36.05 + 56.74 = **₹92.79**.

The redeemable share is worth more because the capital is returned at par (₹100) in five years.

Role in valuation. Preference capital is **deducted in the equity bridge** at its market or redemption value, and no tax shield applies to preference dividends (they are paid out of after-tax profit). In the capitalisation method the preference dividend is deducted before computing earnings for equity (Question 2 in Chapter 21).

24.4 Valuing equity shares

Equity holders receive uncertain, growing dividends (and ultimately the sale price). The **Dividend Discount Model (DDM)** values equity as the present value of expected dividends:

$P_0 = \Sigma D_t \div (1 + Ke)^t$

24.4.1 Constant growth (Gordon)

$P_0 = D_1 \div (Ke - g)$ — Chapter 2.

24.4.2 Two-stage growth

Dividends grow at g_1 for n years and then at g_2 for ever:

$$P_0 = \Sigma_{t=1}^{n} [D_0(1 + g_1)^t \div (1 + Ke)^t] + [D_{(n+1)} \div (Ke - g_2)] \div (1 + Ke)^n$$

Worked example 24.3. D_0 = ₹5; g_1 = 12% for 3 years; g_2 = 6%; Ke = 13%.

Year	Dividend	PV @ 13%
1	5.60	4.96
2	6.27	4.91
3	7.02	4.87
		14.74

$D_4 = 7.02 \times 1.06 = 7.45$; TV at year 3 = $7.45 \div (0.13 - 0.06)$ = **106.37**; PV = $106.37 \div 1.13^3$ = **73.72**.

P_0 = 14.74 + 73.72 = ₹88.46.

24.4.3 Link with growth and retention

Sustainable growth = Retention ratio × ROE. A company that retains 40% of earnings and earns ROE 15% grows at 6%; its dividend payout is 60%.

24.4.4 P/E from the Gordon model

$P_0 \div E_0$ = Payout × (1 + g) ÷ (Ke − g). With payout 60%, g 6%, Ke 13% → P/E = 0.6 × 1.06 ÷ 0.07 = **9.09×**. This links the P/E method (Chapter 4) to fundamentals: higher growth and lower required return raise the P/E.

24.5 From securities to the business

Claim	Valued by	Appears in the business valuation as
Debt	Discounting coupons and principal at YTM	Deduction in the bridge; weight and cost in WACC
Preference	Fixed dividend and redemption	Deduction in the bridge; sometimes a WACC component
Equity	Dividends/FCFE at Ke	The residual: EV minus other claims
Whole firm	FCFF at WACC	Enterprise value

The value of the firm is the **sum of the values of all claims**: V = D + P + E. The business-valuation approaches of this book value either the whole (DCF at WACC) or the residual (earnings, FCFE).

24.6 Convertible and other hybrid securities

A **convertible debenture** is a bond plus an option to convert into equity. Value = straight-bond value + conversion option value. For an investor, if the share price is far below the conversion price the instrument behaves like debt; if far above, like equity. In the equity bridge, treat it as debt if conversion is unlikely, as equity (additional shares) if very likely, and split into components in between.

24.7 Common mistakes

1. Using the coupon rate as the cost of debt rather than the current YTM.
2. Treating preference dividends as tax-deductible.
3. Using face value of debt in the bridge when the market value differs materially.
4. Applying the Gordon model with g ≥ Ke.
5. Ignoring that equity is a *residual* claim and therefore the riskiest.

Chapter 24 summary

- Value = PV of cash flows at the required return; risk differs by claim.
- Bond price rises as yield falls: ₹1,000 5-year 9% bond = ₹926 at 11%, ₹1,000 at 9%, ₹1,082 at 7%.
- Preference: irredeemable = D/k; redeemable like a bond.
- Equity: DDM; two-stage example ₹88.46; P/E = payout × (1 + g)/(Ke − g).
- Business value is the sum of the claims' values.

Check your understanding

1. Price a 4-year ₹100 bond with 8% coupon at a 10% yield.
2. Value an irredeemable preference share with ₹8 dividend at 10%.
3. Explain why bond prices fall when yields rise.
4. Compute P/E: payout 50%, g 5%, Ke 12%.
5. Why is the cost of debt based on YTM and not the coupon?

Answers. (1) 8 × 3.1699 + 100 × 0.6830 = 25.36 + 68.30 = **₹93.66**. (2) 8 ÷ 0.10 = **₹80**. (3) Fixed cash flows are discounted at a higher rate. (4) 0.5 × 1.05 ÷ 0.07 = **7.5×**. (5) YTM is the market's current required return; the coupon is historical.

> **Exam focus.** Expect bond/preference pricing as sub-parts in capital-structure and WACC questions.

Chapter 25 — Valuation in Mergers and Acquisitions: Synergies, Price and Exchange Ratios

The source chapter introduces valuation primarily for **mergers and acquisitions**, "the next topic for discussion". This chapter shows how the methods of the book are used in a deal: the standalone value of the target, the value of synergies, the maximum price a buyer should pay, the effect on the buyer's earnings per share, and the exchange ratio in a share-swap.

25.1 Three values in every deal

1. **Standalone value** of the target — what it is worth under its current owners and strategy (asset, income, DCF).
2. **Value to the acquirer** = Standalone value + PV of synergies − integration and transaction costs.
3. **Price** — the negotiated amount, somewhere between the seller's minimum and the buyer's maximum (Chapter 13).

Value created for the buyer = Value to the acquirer − Price paid.
Value gained by the seller = Price paid − Standalone value.

If the buyer pays the entire synergy value to the seller, it earns just its cost of capital on the deal.

25.2 Sources of synergy

Type	Examples	Reliability
Cost synergies	Overhead consolidation, procurement scale, plant rationalisation, logistics	High
Revenue synergies	Cross-selling, wider distribution, new products	Low
Financial synergies	Lower cost of debt, unused tax losses, higher debt capacity	Medium
Operating improvements	Better working-capital and capex management	Medium
Dis-synergies	Customer loss, talent attrition, culture clash	Often underestimated

In the source chapter's wording, "if any additional earning is there due to acquisition, the same should also be considered in valuation" (Chapter 4) — this is the synergy.

25.3 Valuing synergies

Treat synergy as a project: forecast the incremental after-tax cash flows, subtract costs to achieve them, and discount at a rate that reflects their risk (usually the target's WACC plus a premium for execution risk).

25.3.1 Worked example 25.1

Pioneer Ltd considers acquiring *Target Ltd*, whose **standalone equity value is ₹800 crore** (10 crore shares, ₹80 per share).

- Cost synergies: **₹40 crore a year pre-tax at run-rate**, reached 50% in year 1 and 100% from year 2.
- One-off integration cost ₹30 crore in year 1.
- Tax 25%; discount rate 12.5%; after year 5 the synergy grows 3% a year.

₹ crore	Y1	Y2	Y3	Y4	Y5
Pre-tax synergy (40 × ramp)	20	40	40	40	40
Integration cost	(30)	–	–	–	–
Net pre-tax	(10)	40	40	40	40
After tax (× 0.75)	(7.5)	30	30	30	30

PV of years 1–5 at 12.5% = **73.5**. Terminal value at year 5 = 30 × 1.03 ÷ (0.125 − 0.03) = **325.3**; PV = **180.5**. **Total PV of synergies = ₹254.0 crore, or ₹25.4 per Target share.**

Maximum price for Pioneer = standalone ₹80 + synergies ₹25.4 = **₹105.4 per share**. Paying more than that destroys Pioneer's value.

Negotiation. Suppose Pioneer offers **₹95 per share** (₹950 crore): the premium over standalone value is 18.75%; Target's shareholders gain ₹15 per share (₹150 crore) over standalone value; Pioneer gains 105.4 − 95 = ₹10.4 per share (₹104 crore) — about 41% of the synergy.

25.4 Valuing the target by the book's methods

Method	Use in an acquisition
Net realisable value	Seller's floor; relevant if assets will be sold
Replaceable value	Buyer's ceiling — rebuilding is the alternative
Earnings capitalisation/P/E	Benchmark for acquisition multiple
DCF	Standalone and with synergies
EVA/MVA	Whether the target creates value for its capital
Relative (EV/EBITDA, precedent transactions)	Market check; includes control premia paid in other deals
Chop-shop	If the buyer will separate businesses

25.5 Earnings per share: accretion and dilution

Boards also look at the effect on **EPS**. It is a market-optics test, not a value test: a deal can be EPS-accretive and value-destroying, or EPS-dilutive and value-creating.

25.5.1 Worked example 25.2

Pioneer: net income ₹600 crore; 100 crore shares; EPS ₹6.00; P/E 20× → price ₹120. Target: net income ₹70 crore. Offer: ₹950 crore (₹95 per share; P/E paid = 950 ÷ 70 = 13.6×).

Financing	Combined net income	Shares	EPS	Change
All cash (debt at 9%; after-tax interest 950 × 0.09 × 0.75 = ₹64.1 crore)	600 + 70 − 64.1 = 605.9	100	₹6.06	**+1.0%**
All stock (950 ÷ 120 = 7.92 crore new shares)	670	107.92	₹6.21	**+3.5%**
All cash, with ₹30 crore post-tax synergy	635.9	100	₹6.36	**+6.0%**

Both are accretive because Pioneer pays a lower P/E (13.6×) than its own (20×). The stock deal is more accretive because the "cost" of equity issued at a P/E of 20 (earnings yield 5%) is lower than the after-tax cost of debt (6.75%). Accretion says nothing about whether the price is justified by value (₹95 < ₹105.4: yes in this example).

25.6 The exchange ratio in a share-swap merger

In a merger where the acquirer pays in its own shares, the **exchange ratio** is the number of acquirer shares given for each target share.

Exchange ratio = Value per share of target ÷ Value per share of acquirer

Both values must be measured **on the same basis** (the same methods and weights). The syllabus questions (Chapter 21, Q4 and Q5) show the method: compute each company's value per share by net assets and by earnings capitalisation, weight them (1:3), and divide.

25.6.1 Worked example 25.3

Using Chapter 22's problem P8: A: NAV ₹120, earnings ₹200; B: NAV ₹45, earnings ₹70; weights 1 : 2 → A = ₹173.33; B = ₹61.67; **ratio 0.3558**. A holder of 1,000 B shares receives 355.8 A shares.

25.6.2 Fairness of the ratio

The ratio is fair if each group's shareholders receive value equal to what they give up, measured by consistent valuation. Tests: (i) compare the ratio with the market-price ratio; (ii) compare relative contributions to combined earnings,

assets and cash flow; (iii) check the pro-forma ownership of the merged entity against each side's contribution.

Pro-forma ownership. If A has 100 crore shares and B 10 crore, at 0.3558 B's holders receive 3.558 crore A shares: **3.4%** of the merged company (3.558 ÷ 103.558). If B contributes 5% of combined earnings, the ratio looks low for B's holders — a prompt to re-examine the weights.

25.7 Cash, stock or mixed consideration

Consideration	Seller's risk	Buyer's position
Cash	None after closing	Keeps all synergy upside; needs financing; leverage rises
Stock	Shares post-merger risk and synergy	Conserves cash; shares gains and risks; dilution
Mixed / earn-out	Partial	Contingent payments align incentives; need valuation of contingent claim

25.8 Other deal valuation issues

1. **Control premium:** the extra paid over the unaffected price for control; typically comes from synergies and better management; observed premiums range widely.
2. **Contingent liabilities** (as in H Ltd. — Q4): deducted in net assets; assume full materialisation unless stated.
3. **Partly paid shares** (as in XY Ltd. — Q5): allocate consideration on fully-paid equivalents.
4. **Valuation date and subsequent events.**
5. **Tax and legal structure:** share vs asset purchase; merger relief; stamp duty.
6. **Purchase price allocation** after the deal (Ind AS 103): price allocated to identifiable assets and goodwill (Chapter 27).

25.9 Checklist for the buyer

1. Standalone value by two or three methods.
2. Synergies itemised, owned, costed, risk-weighted.
3. Walk-away price = standalone + a share of risk-weighted synergies.
4. Sensitivity of value to synergy delivery and WACC.
5. Funding, leverage, covenants.
6. EPS effect as a secondary test.
7. Integration plan with accountability.

Chapter 25 summary

- Deal analysis separates standalone value, synergies and price.
- Synergies are valued as a project; example: ₹254 crore (₹25.4 per share); maximum price ₹105.4.
- EPS accretion is an optical test; accretive deals can destroy value.
- Exchange ratio = target value per share ÷ acquirer value per share on a consistent basis.

Check your understanding

1. Define standalone value and value to the acquirer.
2. Why are revenue synergies riskier than cost synergies?
3. Compute the exchange ratio: target ₹90, acquirer ₹360.
4. If a buyer pays the full synergy value to the seller, what return does it earn?
5. Why can an EPS-accretive deal still destroy value?

Answers. (1) What the target is worth under present ownership; standalone plus PV of synergies less costs. (2) They depend on customers and competitors; cost savings are within management's control. (3) 90 ÷ 360 = **0.25**. (4) Exactly its cost of capital (zero NPV). (5) Accretion can result from cheap financing or a high acquirer P/E regardless of whether the price exceeds value.

> **Exam focus.** Questions on mergers combine NAV and earnings values with weights to get an exchange ratio; practice Q4 and Q5.

Chapter 26 — Building Free Cash Flow from the Accounts

Every cash-flow valuation starts with the same task: turn a set of financial statements into **free cash flow**. The syllabus illustration in Chapter 5 starts from a one-line free cash flow; real questions give an income statement, a balance sheet and a set of ratios. This chapter shows how to do the conversion carefully, how to cross-check it, and how to forecast the ingredients.

26.1 The ingredients of free cash flow

FCFF = EBIT(1 − t) + Depreciation − Capital expenditure − Increase in net working capital

Ingredient	Where to find it	Common trap
EBIT	Income statement: revenue − operating costs − depreciation	Including non-operating income or one-offs
Tax on EBIT	EBIT × tax rate	Using actual tax (which is reduced by interest)
Depreciation	Income statement/notes	Adding back more than the non-cash charge
Capital expenditure	Change in net fixed assets + depreciation	Forgetting that net fixed assets are *after* depreciation
Net working capital	Balance sheet: inventory + receivables + other operating current assets − payables − other operating current liabilities	Including cash and short-term loans

26.2 Worked example: Veda Pharma Ltd.

Veda Pharma Ltd. (fictional; ₹ crore). Income statement:

	Year 0	Year 1
Revenue	1,500	1,740
Cost of goods sold	(900)	(1,032)
Selling, general and administrative	(200)	(230)
Depreciation	(60)	(70)
EBIT	**340**	**408**
Interest	(40)	(44)
PBT	300	364
Tax @ 25%	(75)	(91)
Net income	**225**	**273**

Working-capital policy: receivable days 60 (on revenue); inventory days 75 and payable days 45 (both on COGS). **Fixed assets:** net fixed assets at the beginning of year 1: ₹520 crore; capital expenditure in year 1: ₹130 crore. **Debt:** net borrowing in year 1: ₹25 crore.

Step 1 — Net working capital

	Year 0	Year 1
Receivables (revenue × 60/365)	246.6	286.0
Inventory (COGS × 75/365)	184.9	212.1
Payables (COGS × 45/365)	(111.0)	(127.2)
Net working capital	**320.5**	**370.9**

Increase in NWC (year 1) = 370.85 − 320.55 = ₹50.3 crore (a cash outflow).

Step 2 — Net fixed assets

Closing net fixed assets = 520 + 130 − 70 = **₹580 crore**. (Capex 130 less depreciation 70 = net investment of 60.)

Step 3 — FCFF by three routes

Route	Calculation	Year-1 FCFF
From EBIT	408 × 0.75 = 306.0; + 70 − 130 − 50.3	**195.7**
From EBITDA	(408 + 70) × 0.75 + 0.25 × 70 − 130 − 50.3 = 358.5 + 17.5 − 130 − 50.3	**195.7**
From net income	273 + 44 × 0.75 + 70 − 130 − 50.3 = 273 + 33 + 70 − 130 − 50.3	**195.7**

All three agree — a strong check on the arithmetic.

Step 4 — FCFE

FCFE = Net income + Depreciation − Capex − ΔNWC + Net borrowing = 273 + 70 − 130 − 50.3 + 25 = **₹187.7 crore**.

Check: FCFF − after-tax interest + net borrowing = 195.7 − 33 + 25 = 187.7 ✔.

Step 5 — Reconcile to a cash-flow statement

₹ crore	
Net income	273.0
Add: depreciation	70.0
Less: increase in NWC	(50.3)
Cash from operations	**292.7**
Less: capital expenditure	(130.0)
Free cash flow after capex (before debt flows)	**162.7**
Add: net borrowing	25.0
FCFE	**187.7**
To go to FCFF: add after-tax interest 33.0, subtract net borrowing 25.0	**195.7**

26.3 Forecasting the ingredients

A valuer rarely forecasts FCFF directly; he forecasts the drivers:

Driver	Typical method
Revenue	Growth rates by segment; volume × price; market share
Operating margin	Cost ratios to revenue; operating leverage; mix
Depreciation	% of revenue or of opening net fixed assets (e.g., 10% as in Question 3)
Capex	% of revenue; capacity plans; maintenance + growth
Working capital	Days (receivable, inventory, payable) or % of revenue
Tax	Effective rate; marginal rate in terminal year

26.3.1 Forecast for Veda (illustrative)

Assume revenue growth 14%, 12%, 10%; cost ratios (COGS 59.3%, SG&A 13.2%, depreciation 4.0% of revenue) unchanged, so EBIT margin 23.4%; capex 7.5% of revenue; working-capital days unchanged; tax 25%; WACC 11%.

₹ crore	Year 1	Year 2	Year 3
Revenue	1,983.6	2,221.6	2,443.8
EBIT	465.1	520.9	573.0
NOPAT	348.8	390.7	429.8
+ Depreciation	79.8	89.4	98.3
− Capex	(148.8)	(166.6)	(183.3)
− Increase in NWC	(51.9)	(50.7)	(47.4)
FCFF	**228.0**	**262.7**	**297.4**

PV at 11% = **636.1**. *Terminal value:* growth 5% and ROIC 16% ⇒ reinvestment rate 31.25%; NOPAT in year 4 = 451.3; FCFF = 310.2; TV = 310.2 ÷ 0.06 = 5,170.7; PV = 3,780.8. **Enterprise value = ₹4,416.8 crore**; with net debt of ₹480 crore, equity value ≈ ₹3,937 crore, or ₹196.8 per share on 20 crore shares.

Observation. Free cash flow grows faster than revenue in the early years because the working-capital and capex needs scale with *growth*, not with revenue: as growth slows from 16% to 10%, reinvestment falls, lifting FCFF.

26.4 Terminal-year consistency

In the terminal year a growing company still needs reinvestment. The textbook questions (Q7, Q8 in Chapter 21) capture this by two assumptions: **capex is offset by depreciation** (no *net* capex) and **working capital grows with revenue** (ΔNWC = 25% × Δrevenue). A fuller approach ties reinvestment to the growth rate and return on capital: **Reinvestment rate = g ÷ ROIC**. Failing to reinvest while assuming growth overstates value.

26.5 Common errors

1. **Using actual taxes rather than tax on EBIT** in FCFF.
2. **Adding back all non-cash items**, including provisions that represent real costs.
3. **Taking net fixed asset change as capex** (forgetting depreciation).
4. **Including cash and borrowings in working capital.**
5. **Using the wrong sign for ΔNWC** (an increase is an outflow).
6. **Forgetting to separate non-operating income** (it is valued in the bridge).
7. **Mixing FCFF with cost of equity or FCFE with WACC.**

26.6 Worked example: from a balance sheet change

Closing and opening balance sheet (₹ lakh). Net fixed assets: opening 800; closing 900; depreciation for the year 100. Inventories: 300 → 330; receivables: 250 → 290; payables: 180 → 200; cash 50 → 70 (ignore). EBIT 300; tax 30%.

- Capex = 900 − 800 + 100 = **200**.
- ΔNWC = (330 + 290 − 200) − (300 + 250 − 180) = 420 − 370 = **50**.
- NOPAT = 300 × 0.7 = **210**.
- **FCFF = 210 + 100 − 200 − 50 = ₹60 lakh.**

Chapter 26 summary

- FCFF = EBIT(1 − t) + depreciation − capex − ΔNWC; compute by two or three routes to check.
- Capex = Δ net fixed assets + depreciation; ΔNWC excludes cash and borrowings.
- FCFE adds net borrowing and removes after-tax interest.
- Forecast drivers (revenue, margin, days, capex ratio); reinvest in the terminal year.
- Veda Pharma: FCFF 195.7, FCFE 187.7; forecast EV ₹4,417 crore.

Check your understanding

1. Compute capex: opening NFA 500, closing 560, depreciation 80.
2. NWC rose from 200 to 230; NOPAT 150; depreciation 40; capex 90. Compute FCFF.
3. Why use tax on EBIT in FCFF?
4. Convert FCFF 120 to FCFE: after-tax interest 30, net borrowing 20.
5. Why is NWC defined excluding cash?

Answers. (1) 560 − 500 + 80 = **140**. (2) 150 + 40 − 90 − 30 = **70**. (3) The interest shield is captured in WACC; using actual tax would double-count it. (4) 120 − 30 + 20 = **110**. (5) Cash and borrowings are financing items; cash is a non-operating asset valued separately.

> **Exam focus.** Show each ingredient separately; examiners award marks for each of NOPAT, depreciation add-back, capex and ΔNWC.

Chapter 27 — Valuing Intangibles, Brands and Goodwill

Two themes recur in the source chapter: **goodwill** (in the contemporary approaches) and **unrecorded intangible assets** (the patents in the EVA questions Q15 and Q16, and the licences in the NRV example). This chapter shows how intangibles are valued, how goodwill arises, and why the balance sheet understates the real capital employed by knowledge-based firms.

27.1 Why intangibles matter

In a modern enterprise much of the value lies in assets that do not appear on the balance sheet: **brands, patents, licences, customer relationships, technology, human capital, data**. Accounting rules generally require internally generated intangibles to be expensed, so they are absent from the books; acquired intangibles appear at cost.

Consequences for valuation:

Approach	Effect of unrecorded intangibles
Asset-based	Book NAV understates value (licences in the NRV example are worth three times their book value)
Income-based	Earnings capture the benefit of intangibles; value is in the earnings
EVA	Invested capital is understated unless intangibles are added (Q15, Q16) — EVA is overstated
Relative	P/B ratios are very high for intangible-heavy firms

27.2 The three families of intangible valuation

Approach	Idea	Typical techniques
Income approach	Value = PV of the economic benefits attributable to the asset	Relief-from-royalty; multi-period excess earnings; with-and-without
Market approach	Prices of comparable transactions	Royalty rates; transaction multiples
Cost approach	Cost to recreate or replace	Cost-to-duplicate, reproduction/replacement cost less obsolescence

27.3 Relief-from-royalty (brands, trademarks, technology)

The asset is valued as the **royalty the owner is relieved from paying** because it owns the asset rather than licensing it.

1. Forecast the **revenue** attributable to the asset.
2. Select an **arm's-length royalty rate** (from comparable licences).
3. Compute **after-tax royalty savings**.

4. Discount at a rate reflecting the asset's risk (typically WACC plus a premium).
5. Add the **tax amortisation benefit** (TAB) if the asset's cost is tax-deductible over time.

27.3.1 Worked example 27.1

A brand supports revenue of ₹800 crore now, growing 10%, 8%, 6%, 5%, 4%, 4%, 3%, 3% over the next eight years, after which the brand is assumed to lose relevance (finite life). Royalty rate 4%; tax 25%; discount rate 14%.

Year	Revenue	Royalty (4%)	After tax (×0.75)	PV @ 14%
1	880.0	35.2	26.4	23.2
2	950.4	38.0	28.5	21.9
3	1,007.4	40.3	30.2	20.4
4	1,057.8	42.3	31.7	18.8
5	1,100.1	44.0	33.0	17.1
6	1,144.1	45.8	34.3	15.6
7	1,178.4	47.1	35.4	14.2
8	1,213.8	48.6	36.4	12.8
				144.0

PV of after-tax royalties = **₹144.0 crore**. If the brand is amortised over 15 years for tax, the TAB factor at 14% is about 1.114, giving a fair value of **₹160.4 crore**.

Sensitivity: a 3% royalty rate gives about ₹108 crore; 5% gives about ₹180 crore — value is proportional to the royalty rate, which is therefore the key judgement and must be supported by comparable licensing data.

27.4 Multi-period excess earnings (customer relationships)

For an asset that is the *primary* driver of earnings (e.g., customer relationships), the value is the PV of the **earnings attributable to it after deducting a return on all other contributory assets** (working capital, fixed assets, workforce, brand).

Worked example 27.2. Existing customers generate revenue of ₹500 crore now, declining 15% a year through churn. EBIT margin on this revenue 18%; contributory-asset charges 6% of revenue; tax 25%; discount rate 15%.

After-tax excess earnings = revenue × (18% − 6%) × (1 − 0.25) = revenue × 9%.

Year	Revenue	Excess earnings	PV @ 15%
1	425.0	38.3	33.3
2	361.3	32.5	24.6

Year	Revenue	Excess earnings	PV @ 15%
3	307.1	27.6	18.2
4	261.0	23.5	13.4
5	221.9	20.0	9.9
6	188.6	17.0	7.3
7	160.3	14.4	5.4
			112.1

Value of the customer relationships ≈ **₹112 crore** (ignoring years beyond 7).

27.5 With-and-without (non-compete agreements, key contracts)

Value = PV of cash flows **with** the asset in place − PV of cash flows **without** it. For a non-compete agreement: compare the business with a competitor entering (lost sales, lower margins) against the business protected. If the with-scenario value is ₹900 crore and the without-scenario ₹830 crore, and the probability that the person would actually compete is 50%, the non-compete is worth 0.5 × 70 = **₹35 crore**.

27.6 Cost approach

Appropriate for assets with no income stream of their own (assembled workforce, in-house software):
Value = Cost to recreate − Obsolescence.
A software platform that would cost ₹25 crore to rewrite with today's tools, but is a legacy architecture with 40% obsolescence, has a cost-approach value of 25 × 0.60 = **₹15 crore**.

27.7 Goodwill

Goodwill is the excess of the value of a business over the fair value of its identifiable net assets.

27.7.1 Accounting goodwill

On acquisition: Goodwill = Consideration − Fair value of identifiable net assets acquired (Ind AS 103). It is tested for impairment annually (Ind AS 36).

27.7.2 Valuation goodwill: the super-profit method

Goodwill = (Maintainable profit − Normal return on net assets) ÷ Capitalisation rate.

Worked example 27.3. Net assets ₹590 crore; normal return 12% → ₹70.8 crore; maintainable profit ₹81.6 crore; super-profit ₹10.8 crore; capitalisation rate 14% → **goodwill ₹77.1 crore**; total value = **₹667 crore** (as in Chapter 19).

27.7.3 The goodwill-based approach (retail hotspot)

Goodwill valued by a multiple of annual sales or footfall (Chapter 9): suitable where location is the key asset.

27.7.4 Components of goodwill

Going-concern value; workforce; synergies; location; reputation. Under Ind AS 103 identifiable intangibles (customer relationships, brands) are separated from goodwill, so the residual goodwill is smaller.

27.8 Purchase price allocation (PPA): an example

Pioneer acquires *Target* for ₹950 crore. Fair values of identifiable net assets: tangible net assets ₹560 crore; brand ₹160 crore; customer relationships ₹112 crore; technology ₹40 crore; deferred tax liability on the intangibles (25% × 312) ₹78 crore.

	₹ crore
Tangible net assets at fair value	560
Brand	160
Customer relationships	112
Technology	40
Less: deferred tax liability (25% of 312)	(78)
Fair value of identifiable net assets	**794**
Consideration	950
Goodwill	**156**

The PPA shows why intangible valuation matters for reporting: a higher brand value reduces goodwill (not tested for amortisation but for impairment) and increases amortisation charges on identifiable intangibles.

27.9 Including intangibles in EVA: Herbal Gyan revisited

In Question 15 (Chapter 21) the patent of ₹40 lakh was unrecorded; adding it raised invested capital from 100 to 140, turning EVA from ₹−3 lakh [12 − 0.15 × 100 = −3] to **−₹9 lakh**. The general rule: **invested capital for EVA should include the market value (opportunity cost) of unrecorded intangibles**; otherwise returns on capital are overstated.

27.10 Practical guidance

1. **Identify** all material intangible assets (brands, customer lists, technology, licences).
2. **Choose** the technique to match the asset (royalty for brands; excess earnings for customer relationships).
3. **Support** royalty rates and lives with market data.
4. **Avoid double-counting**: the same cash flows cannot be attributed to two intangibles; use contributory-asset charges.
5. **Test reasonableness**: the sum of values of all assets should reconcile to the enterprise value (with goodwill as residual).

Chapter 27 summary

- Intangibles are largely absent from the balance sheet but central to value.
- Techniques: relief-from-royalty, multi-period excess earnings, with-and-without, cost, market.
- Goodwill = excess of value (or price) over fair value of identifiable net assets; super-profit method: (profit − normal return) ÷ rate.
- EVA invested capital must include unrecorded intangibles.

Check your understanding

1. Value a trademark: after-tax royalty ₹10 crore a year for 5 years, discount rate 12%.
2. Compute goodwill: net assets 400; normal return 10%; maintainable profit 62; capitalisation rate 12%.
3. Why does the balance sheet understate the capital employed of a patent-driven firm?
4. What is a contributory-asset charge?
5. Distinguish accounting goodwill from valuation goodwill.

Answers. (1) 10 × 3.6048 = **₹36.05 crore**. (2) (62 − 40) ÷ 0.12 = **₹183.3**. (3) Internally generated intangibles are expensed. (4) A charge for the return required by other assets that contribute to earnings, deducted so the intangible gets only its own share. (5) Accounting goodwill is a purchase-price residual; valuation goodwill is the capitalised super-profit.

> **Exam focus.** Remember the two EVA questions with unrecorded patents and the super-profit formula for goodwill.

Chapter 28 — The Indian Context: Standards, Regulation and Practice

The syllabus chapter is written for Indian professionals; its illustrations refer to ₹, to SEBI's ESG proposal and to the work of chartered accountants, merchant bankers and registered valuers. This chapter gives an orientation to the Indian framework in which valuations are commissioned and reviewed, and to the practical issues that arise in India — in estimating inputs, in dealing with promoter-controlled companies, and in documenting the valuation.

> **Important.** The laws, rules and standards mentioned here are amended from time to time. This chapter explains the *structure* and the *principles*; before relying on any specific provision for an examination or an engagement, consult the current text and the latest amendments to the syllabus.

28.1 Who values, and why

Context	Typical valuer	Purpose
Mergers, demergers, schemes of arrangement	Registered valuer; merchant banker (fairness opinion)	Share-exchange ratio; reporting to the tribunal and shareholders
Preferential allotments, issues for consideration other than cash (unlisted companies)	Registered valuer	Pricing compliance
Foreign investment and transfers involving non-residents	Chartered accountant, merchant banker or cost accountant	Pricing at or above/below fair value
Listed-company transactions (preferential issue, takeover, delisting, buy-back)	Independent valuer/merchant banker per SEBI rules	Minimum price; fairness
Financial reporting	Valuation specialists	Fair value (Ind AS 113), impairment (Ind AS 36), purchase price allocation (Ind AS 103), share-based payment (Ind AS 102)
Taxation	Chartered accountant/merchant banker	Fair market value of unquoted shares under the Income-tax Rules
Insolvency	Registered valuers	Fair value and liquidation value for the committee of creditors
Disputes	Independent expert	Fair value in oppression/mismanagement or appraisal proceedings

28.2 The professional framework

28.2.1 Registered valuers

Under the Companies Act, 2013 and the **Companies (Registered Valuers and Valuation) Rules, 2017**, valuations required under the Act must be done by a **registered valuer** registered for the relevant **asset class**: *Securities or Financial Assets*, *Land and Building*, or *Plant and Machinery*. A business valuation (shares or financial assets) falls in the first class. Registered valuers must meet qualification and experience requirements, belong to a registered valuers' organisation, observe a code of conduct and declare independence.

28.2.2 Valuation standards

The **ICAI Valuation Standards** (issued by the Institute of Chartered Accountants of India) and the **International Valuation Standards (IVS)** set out the vocabulary, the bases of value, the approaches and methods (asset, income and market), the scope of work, documentation and reporting. The standards echo the structure of this book: the three approaches; the need to state the *basis* and *premise* of value; the requirement to consider and reconcile more than one approach; and the content of the report.

28.2.3 Independence and documentation

A valuation report should state the purpose, the valuation date, the basis and premise of value, the information relied upon, the methods used (and any considered but rejected, with reasons), the assumptions, the limitations, and a declaration of independence. The report should allow a reader to *follow and replicate* the numbers.

28.3 Fair value under Ind AS 113

Ind AS 113 defines **fair value** as the price that would be received to sell an asset or paid to transfer a liability in an **orderly transaction between market participants at the measurement date** — an *exit price* from the viewpoint of market participants.

28.3.1 The fair-value hierarchy

Level	Inputs	Example
Level 1	Quoted prices in active markets for identical assets	Listed shares
Level 2	Observable inputs other than Level 1	Prices of similar assets; interest rates and yield curves
Level 3	Unobservable inputs	DCF of an unlisted company using management forecasts and a judgemental discount rate

Valuations of unlisted businesses are typically **Level 3**; disclosure includes the valuation technique and the key unobservable inputs and a sensitivity.

28.3.2 Market-participant assumptions

Fair value uses the assumptions that **market participants** would use — not the entity's own synergies, unless market participants would also benefit from them. This is the *fair value* standard, and it differs from *investment value* (value to a specific buyer).

28.4 Impairment: value in use (Ind AS 36)

An asset or cash-generating unit (CGU) is impaired if its carrying amount exceeds its **recoverable amount** — the higher of **fair value less costs of disposal** and **value in use**. Value in use is a DCF with prescribed features:

- cash flows from reasonable and supportable assumptions, generally based on the **most recent budgets** (normally up to five years), extrapolated with a steady or declining growth rate;
- exclude **future restructurings** not yet committed and **enhancements** of the asset's performance;
- exclude **financing** cash flows and **income-tax** flows (in principle, a **pre-tax** discount rate is used);
- the discount rate reflects the market's current assessment of the time value of money and the risks specific to the asset.

Illustration. A CGU has a carrying amount of ₹800 crore. Value in use is ₹760 crore; fair value less costs of disposal ₹740 crore. Recoverable amount = ₹760 crore (the higher); impairment loss = 800 − 760 = **₹40 crore**.

28.5 Purchase price allocation (Ind AS 103) and share-based payment (Ind AS 102)

- **Ind AS 103:** identifiable assets acquired and liabilities assumed are measured at fair value; the residual is goodwill (Chapter 27).
- **Ind AS 102:** options granted to employees are measured at **grant-date fair value** (option-pricing model — Black–Scholes or binomial) and expensed over the vesting period; for an unlisted company the underlying share value comes from a valuation.

28.6 Valuation in specific Indian transactions (overview)

28.6.1 Schemes of arrangement and share-exchange ratios

A scheme under the Companies Act normally requires a **valuation report** from a registered valuer and a **fairness opinion** from a merchant banker on the share-exchange ratio. The report weighs asset, income and market approaches (with weights explained) — as in Q4 (H Ltd./B Ltd.) and Q5 (AB Ltd./XY Ltd.) in Chapter 21. For listed companies, SEBI requirements on pricing and on the role of the audit committee and independent directors also apply.

28.6.2 Foreign investment and FEMA pricing

Where an Indian company issues shares to, or transfers shares from, non-residents, the price must satisfy FEMA pricing guidelines — broadly, the price should be determined using an **internationally accepted pricing methodology on an arm's length basis**, certified by a chartered accountant, merchant banker or cost accountant. DCF is widely accepted as such a methodology.

28.6.3 Taxation of unquoted shares

The Income-tax Rules prescribe methods for determining the **fair market value of unquoted equity shares** for specified purposes; for certain transactions, a **DCF-based valuation** by a merchant banker or an accountant is permitted alongside a net-asset-value formula. Check which method applies to the section concerned.

28.6.4 Insolvency

Under the Insolvency and Bankruptcy Code, 2016, **registered valuers** determine the **fair value** and **liquidation value** of the corporate debtor's assets; these are used by the committee of creditors when evaluating resolution plans (Chapter 14).

28.7 Estimating inputs in an Indian context

Input	Practical points
Risk-free rate	Yield on government securities of comparable (long) maturity on the valuation date; rupee flows → rupee yield
Equity risk premium	Estimated from long-run Indian returns, implied premiums or surveys; include country risk if starting from a mature-market premium
Beta	Regression against NSE/BSE indices; Blume adjustment; bottom-up from peers for unlisted firms (Chapter 7)
Cost of debt	Borrowing rate of comparable credit; $K_d(1 - t)$ with the marginal tax rate
Tax rate	Statutory regime elected (old regime with deductions vs concessional regime), incentives and their expiry; terminal at marginal

Input	Practical points
Growth	Inflation plus real growth; terminal ≤ long-run nominal growth of the economy and ≤ the risk-free rate
Currency and inflation	Match currency of flows and discount rate; avoid discounting rupee flows at a dollar rate

28.8 Promoter-controlled and family-run companies

Many Indian companies are controlled by promoters or families. Valuation implications:

1. **Related-party transactions** may not be at arm's length; normalise (Chapter 19) and consider *leakage*.
2. **Governance risk:** reflect specific risks in cash-flow scenarios rather than as a vague discount.
3. **Minority discounts** are context-dependent: in oppression or appraisal cases a minority holder is often entitled to a pro-rata share of the company's value (no minority discount), whereas in a negotiated sale of a small unlisted stake discounts for lack of control or marketability may be argued.
4. **Pledged shares, cross-holdings and group guarantees** affect risk and should be disclosed.
5. **Key-person risk** (Chapter 17).

28.9 Marketability and control in practice

Adjustment	Typical contexts	Caution
Control premium	Acquisition of control from a minority base value	Premiums are deal-specific; include synergies
Discount for lack of control (DLOC)	Valuing a minority stake from a control value	DLOC = 1 − 1/(1 + control premium)
Discount for lack of marketability (DLOM)	Unlisted minority stakes	Not for valuing the whole company for a strategic buyer; support with evidence

Illustration. DCF equity value (control basis) ₹1,000 crore; a 10% unlisted stake. With a 20% DLOC and 15% DLOM: 1,000 × 10% × 0.80 × 0.85 = **₹68 crore** (compared with ₹100 crore pro-rata). Applying both discounts multiplicatively (not additively) is the usual convention.

28.10 Documentation checklist for an Indian valuation report

1. Engagement terms and purpose; valuation date; premise and standard of value.
2. Sources of information and management representations.
3. Business and industry overview.

4. Historical financial analysis and normalisation.
5. Methodology: approaches considered, adopted and rejected, with reasons.
6. Assumptions: forecast, WACC build-up (Rf, ERP, beta, Kd, weights), terminal value.
7. Calculations and results; sensitivity; cross-checks.
8. Reconciliation and conclusion; range and point estimate.
9. Limitations; independence declaration; registration details.
10. Subsequent events considered.

Chapter 28 summary

- Indian valuations are shaped by the Companies Act (registered valuers), SEBI rules, FEMA pricing, Income-tax Rules, Ind AS 113/36/103/102 and the IBC.
- Ind AS 113 fair value: market-participant exit price; Level 1–3 hierarchy; most business valuations are Level 3.
- Ind AS 36 value in use: DCF with restrictions (pre-tax rate; no future restructuring or enhancement).
- Promoter-controlled firms need normalisation of related-party flows and scenario treatment of governance risk.
- Discounts and premiums depend on the standard of value; apply with support.

Check your understanding

1. Name the three asset classes of registered valuers.
2. Define Level 3 inputs with an example.
3. A CGU: carrying amount 500; value in use 470; FVLCD 480. Compute the impairment.
4. Compute: DCF equity value 600; 20% stake; DLOC 15%; DLOM 20%.
5. Why must market-participant assumptions be used in fair-value measurement?

Answers. (1) Securities or financial assets; land and building; plant and machinery. (2) Unobservable inputs, e.g., management forecasts in an unlisted DCF. (3) Recoverable = 480; impairment = **20**. (4) 600 × 0.20 × 0.85 × 0.80 = **₹81.6**. (5) Fair value is an exit price from the market's viewpoint, not the entity's.

> **Exam focus.** Know the three-level hierarchy, the value-in-use features, and the registered-valuer asset classes.

Chapter 29 — Scenarios, Sensitivity and the Valuation Report

A valuation is only useful if its reader understands how much confidence to place in it. This final chapter covers the tools for expressing uncertainty — sensitivity, scenarios, reverse valuation — and the structure of a good valuation report. It returns to *Sunrise Textiles Ltd* (Chapter 19) as the worked example.

29.1 Why express uncertainty?

Every input of a valuation is an estimate: growth, margin, reinvestment, WACC, terminal growth. In Chapter 5 the same cash flows gave values from about ₹8,755 to ₹19,916 thousand across plausible WACC and growth pairs — a spread of more than 100%. A single number hides this. The source chapter's closing advice — a **range of values** from the seller's minimum to the buyer's maximum — is the practical expression of uncertainty.

29.2 Sensitivity analysis

One-way sensitivity changes one input at a time and records the change in value. The inputs with the largest effect deserve the most research.

29.2.1 Sunrise Textiles: one-way sensitivities (equity value per share, base ₹76.1)

Input changed	Low case	Value	High case	Value
WACC	13%	62.8	11%	93.8
Terminal ROIC	12% (= WACC)	72.1	15%	82.5
NOPAT growth, years 1–5 (all years)	−2 points	72.9	+2 points	79.4
Incremental ROIC, years 1–5	12%	72.5	16%	78.8
Terminal growth	4%	74.2	6%	78.4

Reading. WACC dominates: a one-point change moves value by roughly −17% / +23%. Terminal ROIC comes next (−5% / +8%). Terminal growth matters little *because* terminal ROIC (13%) is close to WACC (12%): growth at returns near the cost of capital adds little value (an insight that follows from the value-driver logic of Chapter 5).

Two-way tables vary two inputs together (WACC × growth, as in Chapters 5 and 19).

29.3 Scenario analysis

Scenarios change several inputs together to represent coherent states of the world.

29.3.1 Sunrise Textiles scenarios

Scenario	NOPAT growth (Y1–Y5)	Terminal g / ROIC	Narrative
Bear	3% a year	4% / 12%	Weak demand and margin pressure; growth barely beats inflation; no excess returns in the long run
Base	8, 8, 7, 6, 6%	5% / 13%	Plan: modest expansion, ROIC slightly above WACC
Bull	11, 10, 9, 8, 7%	5% / 15%	Export upturn and mix improvement; stronger returns on new capital

	Bear	Base	Bull
Enterprise value (₹ crore)	917	1,021	1,132
Equity value (EV − 300 + 40)	657	761	872
Per share (₹)	**65.7**	**76.1**	**87.2**
Probability	25%	50%	25%

Probability-weighted value = 0.25 × 65.7 + 0.50 × 76.1 + 0.25 × 87.2 = **₹76.3 per share**.

Observation. The scenarios are fairly symmetric around the base because the business is a mature manufacturer with modest growth. For a growth business the range would be wider and asymmetric. The probabilities are judgements: say how they were chosen.

29.3.2 Why scenarios beat a higher discount rate for risk

Adding a "safety margin" to WACC to reflect specific risks is crude: it penalises distant cash flows heavily, cannot distinguish systematic from specific risk, and gives no explanation. Scenarios express *what* might happen, *when*, and *how large* — the same logic as the ESG approach in Chapter 18.

29.3.3 Decision trees

For discrete events (regulatory approval, loss of a customer, success of a launch), use a tree with probabilities and outcome values. The distressed-valuation formulas (Chapter 14) are decision trees with two branches.

29.4 Reverse valuation (what the price implies)

A **reverse DCF** takes a price as given and solves for the assumption that reconciles it.

Illustration. Suppose an acquirer asks ₹90 per share for 100% of Sunrise (equity ₹900 crore; EV ₹1,160 crore). Holding other inputs at base, the price implies **any one** of:

Implied variable	Base	Implied by ₹90
WACC	12.0%	**11.2%**
Terminal ROIC	13%	**18.3%**
NOPAT growth in each of years 1–5	8, 8, 7, 6, 6%	**+8 percentage points a year**

Reading: to justify ₹90 the buyer must believe in a materially lower required return, a much stronger competitive position after year 5 (ROIC 18%), or growth of 14–16% a year — each demanding. The reverse DCF turns a vague "seems expensive" into a testable statement, and it is a useful negotiation tool for the buyer: it shows what the seller's price asks one to believe.

29.5 The football field

A **football-field chart** presents the ranges from all methods on one axis, with the price (or offer) marked.

Method	Low (₹/share)	High (₹/share)
Net assets: book to replacement	59	83
Earnings capitalisation (13–15%)	54	63
P/E (peer 12×, 15–25% non-listing discount)	73	83
EV/EBITDA 5–6×	69	88
DCF (WACC 11–13%, g 4–6%)	62	99
DCF scenarios	66	87
EVA-based	78	78

The **overlap zone** of the market-, DCF- and EVA-based methods (≈ ₹73–₹83) is where most methods agree; the conclusion in Chapter 19 (₹70–₹80, point ₹75) sits at its lower half, reflecting the asset-based floor and the cautious capitalisation value. The picture also shows outliers (earnings capitalisation is low because it ignores growth) and prompts the reader to ask why.

29.6 The valuation report

A professional report is judged on whether a competent third party could follow and challenge it.

Section	Content
1. Executive summary	Conclusion of value, standard and premise, date, purpose, range and key drivers
2. Scope and engagement	Subject, interest valued, sources, limitations, independence
3. Business and industry	Description, drivers, risks
4. Financial analysis	History, ratios, normalisation adjustments

Section	Content
5. Methodology	Methods adopted, rejected, and why
6. Assumptions	Forecast, WACC build-up, terminal value — each with source
7. Calculations	Exhibits: asset values, capitalisation, DCF, EVA, multiples
8. Sensitivity and scenarios	Tables, tornado, scenario weights
9. Reconciliation	Football field; comments on each method; conclusion
10. Appendices	Data, workings, glossary, declarations

29.6.1 A sample executive summary (Sunrise Textiles)

> *We have estimated the fair value of 100% of the equity of Sunrise Textiles Ltd as at [date], on a going-concern basis for the purpose of a potential acquisition, at* ***₹700–₹800 crore (₹70–₹80 per share)****, with a point estimate of ₹750 crore. The range reflects asset-based values (₹590–₹830 crore), income-based values (₹544–₹783 crore), a discounted-cash-flow value of ₹761 crore (sensitivity ₹622–₹994 crore) and market multiples (₹690–₹880 crore). The terminal value forms 75% of the DCF enterprise value, and the principal sensitivity is the cost of capital: each percentage point changes equity value by roughly 17–23%. The business currently earns approximately its cost of capital (economic value added of ₹3 crore); value above invested capital depends on achieving the forecast improvement. No synergies are included. This report should be read with the assumptions and limitations set out in Sections 2 and 6.*

29.7 Common weaknesses in valuation reports

1. Single-point conclusion without a range.
2. Assumptions without sources.
3. Methods averaged mechanically without comment.
4. No sensitivity of terminal value.
5. No discussion of why methods disagree.
6. Inconsistent standard of value (e.g., control value used for a minority stake).
7. Failure to state the valuation date and subsequent events.
8. Hiding judgement in a single "adjustment" line.

29.8 A final word

Valuation is the disciplined estimate of what a business is worth, stated with the humility its uncertainty deserves. The syllabus teaches methods — asset, income, cash flow, relative, EVA, SVA — and special contexts — distress, start-ups, platforms, professional firms, ESG. The skill is in choosing among them, applying them consistently, testing the answers against each other and the market, and explaining the result in a range that a decision-maker can use. As

the source chapter concludes about each special situation: *the fundamentals of valuation remain the same* — understanding the business, the revenue model, the quality of management and the risk–reward parameters.

Chapter 29 summary

- Sensitivity ranks drivers; scenarios give coherent states; reverse valuation reveals implied beliefs.
- Sunrise: bear ₹65.7, base ₹76.1, bull ₹87.2, weighted ₹76.3; ₹90 implies WACC 11.2% or terminal ROIC 18.3%.
- The football field and a transparent report communicate the range and its reasons.

Check your understanding

1. Why are scenarios preferable to adding a premium to WACC?
2. Compute the probability-weighted value: ₹60 (30%), ₹80 (50%), ₹110 (20%).
3. What does a reverse DCF tell a buyer?
4. List five sections of a valuation report.
5. Why does terminal growth matter little when terminal ROIC ≈ WACC?

Answers. (1) They show the nature, timing and size of risks and separate systematic from specific risk. (2) 18 + 40 + 22 = **₹80**. (3) What assumptions must be true to justify a given price. (4) Any five of the ten in §29.6. (5) Growth at a return equal to the cost of capital adds no value.

> **Exam focus.** In evaluation questions, finish with a range and a statement of the key sensitivities; this is what the examiner means by "supplement your conclusion with additional information".

Chapter 30 — A Valuer's Questions and Answers

This chapter collects forty questions that students and practitioners ask again and again, with concise answers that point back to the chapter where the topic is developed. It can serve as a quick-reference review before an examination or an engagement.

A. Concepts and approach

1. Why do different valuers reach different values for the same company?
Because valuation involves judgement about the future (growth, margins, risk) and about methods (which approach, which comparables). A good valuation makes the judgements explicit and shows a range (Chapters 1, 13, 29).

2. Is "value" the same as the price in the market?
No. Price is what the asset trades for; value is a reasoned estimate of worth. The gap between them is what investors try to exploit (Chapter 1).

3. Which approach is best — asset, income or cash flow?
It depends on the business and purpose. Asset-based suits holding companies, break-up and liquidation; income-based suits stable, profitable firms; DCF suits firms with forecastable cash flows. Use several and reconcile (Chapters 3–5, 13).

4. Why not just use the P/E ratio for everything?
P/E is relative, distorted by leverage and one-offs, and unusable when earnings are negative. It embeds the market's assumptions about growth and risk (Chapters 4, 8).

5. When is the asset approach inappropriate?
For knowledge-based firms (IT, consulting) and for any firm whose value lies in earning power rather than in assets (Chapter 3).

6. What is the difference between fair value and fair market value?
Fair value (reporting) is an exit price between market participants; fair market value is the price between a willing buyer and seller, neither compelled and both informed. The definitions overlap but are used in different contexts (Chapters 1, 13, 28).

7. What is the minimum I should accept as a seller, and the maximum I should pay as a buyer?
Seller's minimum = the higher of liquidation value and standalone going-

concern value (or the next-best alternative). Buyer's maximum = the lowest of replacement cost, standalone value plus retainable synergies, and financial capacity (Chapter 13).

B. Cash flows and DCF

8. How many years should the explicit forecast cover?
Long enough for growth, margins and returns to approach steady state — often 5 years for stable businesses, 7–10 for growth businesses. A short horizon puts more weight on the terminal value (Chapter 5).

9. What growth rate should I use in the terminal value?
No more than the long-run growth of the economy in the currency of the cash flows, and no more than the risk-free rate; for mature firms often inflation plus modest real growth. Tie it to reinvestment (Chapters 2, 5, 26).

10. Why is the terminal value 70–90% of my DCF? Is that wrong?
It is typical, because a going concern's cash flows beyond the forecast are valued in the terminal value. It signals that the terminal assumptions deserve the most scrutiny and sensitivity analysis (Chapter 5).

11. Do I use FCFF or FCFE?
FCFF discounted at WACC gives enterprise value; FCFE discounted at the cost of equity gives equity value directly. Never mix them (Chapter 5, 26).

12. Should I use actual tax or tax on EBIT in FCFF?
Tax on EBIT; the interest tax shield is in the after-tax cost of debt in WACC (Chapter 26).

13. How should I treat one-time items?
Remove one-time incomes and add back one-time expenses (after tax) to reach maintainable cash flows, as the syllabus illustration does (Chapters 5, 19).

14. How do I deal with working capital?
Forecast net operating working capital (excluding cash and borrowings) from days or a percentage of revenue and deduct its *increase* (Chapter 26).

15. What if cash flows are negative in the early years?
That is common for start-ups and platforms. Include them; the PV of early losses reduces value; ensure the terminal value reflects a plausible mature state and consider survival probability (Chapters 15, 16).

16. Do I discount the terminal value?
Yes — from the horizon back to today (Chapter 2, 5). In the syllabus illustration the figure labelled TV is already a present value (Chapter 5).

C. Discount rates

17. How do I choose the risk-free rate?
A long-dated government bond yield in the same currency as the cash flows on the valuation date (Chapter 6).

18. What is the market risk premium and where does it come from?
The extra return of the market over the risk-free rate, estimated from long-run history, implied premiums or surveys; expect a range (Chapter 6).

19. How do I get a beta for a private company?
Take pure-play peers' equity betas, un-lever to asset betas, average, and re-lever to the target's capital structure (Chapter 7).

20. What is the difference between asset beta and equity beta?
Asset beta reflects only business risk; equity beta adds financial risk from debt; equity beta ≥ asset beta (Chapter 7).

21. Do I use book or market values in WACC weights?
Market values (Hansel Ltd., Chapter 21, Q6). If market values are unknown, use target weights.

22. Should I add a "safety margin" to the discount rate?
Not if the cash flows already reflect risk; use scenarios to express risk explicitly (Chapters 6, 29).

23. Why does WACC change between high-growth and stable phases?
Beta, capital structure and cost of debt change as the business matures (Chapter 21, Q7).

24. When should I use a company-specific risk premium?
For firms with risks not captured by beta (nascent platforms, key-person dependence) when scenario analysis is not feasible; disclose and justify (Chapter 16).

D. Special situations

25. Can I use DCF for a loss-making start-up?
Yes, with adaptation: forecast the path to a mature state, discount at a suitable rate and weight by survival; supplement with VC, scorecard, comparable-transaction and First Chicago methods (Chapter 15).

26. How do I value a distressed company?
Use modified DCF, DCF plus distress value, APV or adjusted relative multiples; liquidation value is a floor; equity behaves like an option (Chapter 14).

27. What is the best way to value a conglomerate?
Sum-of-the-parts/chop-shop: value each business by its own peers and add; deduct corporate costs and net debt; compare with market value to see the conglomerate discount (Chapter 10).

28. How do I value a professional firm?
Normalise partners' pay to market rates and one-offs; capitalise or discount normalised earnings; cross-check with fee multiples; adjust for key-person and concentration risk (Chapter 17).

29. How do I include ESG?
Preferably in the cash flows: carbon cost and scenario analysis for E, welfare and CSR costs for S, penalties for G; avoid double-counting in the discount rate (Chapter 18).

30. How do I value brands and patents?
Relief-from-royalty for brands and technology; multi-period excess earnings for customer relationships; add unrecorded intangibles to invested capital in EVA (Chapter 27).

31. What is the role of the going-concern assumption?
Most valuations assume it; if the business has no value as a going concern, use liquidation value (Chapter 13).

E. Deals and reporting

32. How are synergies treated in price?
Value them separately, risk-adjust, and pay the seller only a share; maximum price = standalone + synergies (Chapter 25).

33. How do I compute an exchange ratio?
Compute value per share of both companies on the same basis (e.g., weights on NAV and earnings) and divide target by acquirer (Chapter 21, Q4–Q5; Chapter 25).

34. What are control and marketability discounts?
Control premium is the extra paid for control; DLOC = 1 − 1/(1 + premium); DLOM reflects illiquidity of unlisted shares. Apply only where the standard of value requires and with evidence (Chapter 28).

35. What does Ind AS 113 require?
Fair value as an exit price using market-participant assumptions, with a three-level input hierarchy; most unlisted DCFs are Level 3 (Chapter 28).

36. What goes into a valuation report?
Purpose, standard and premise of value, valuation date, sources, methods, assumptions, calculations, sensitivity, reconciliation, conclusion, limitations

(Chapter 29).

F. Examination technique

37. How should I present a computational answer?
Formula, labelled steps, figures with units, one-line interpretation (Chapter 23).

38. What if the question does not specify a method?
Value by as many methods as the data allow, comment on each and conclude with a range (Chapter 13, 23).

39. Which topics are most commonly examined?
EVA/MVA, DCF (two-stage), exchange ratio in mergers, WACC with beta re-levering, and short notes on distressed firms, start-ups, platforms, ESG, going vs non-going concern (Chapters 11, 21, 23).

40. How should I revise in the final week?
Use the plan in Chapter 23: formulas, then worked questions (Chapters 21–22, Appendix D), then theory notes and the self-test (Appendix E).

Appendices

Appendix A — Formula Sheet

A.1 Basics

Item	Formula
Present value	$PV = FV \div (1 + r)^n$
NPV	$\Sigma\ CF_t \div (1 + r)^t - \text{Investment}$
IRR	Rate at which NPV = 0
ROI	Return ÷ Investment (use average investment if capital changes)
Perpetuity	$C \div r$
Gordon model	$P_0 = D_1 \div (Ke - g)$
Terminal value	$TV_n = CF_n \times (1 + g) \div (r - g)$

A.2 Asset-based

Item	Formula
Net asset value (book)	Fixed assets + Net current assets − Long-term debt
Net realisable value	Realisable value of assets − Liquidation expenses − Liabilities
Replaceable value	Cost of acquiring equivalent assets − Liabilities
Goodwill (super profit)	(Maintainable profit − Normal return on net assets) ÷ Capitalisation rate

A.3 Income-based

Item	Formula
Price per share	EPS × P/E
Unlisted: adjusted P/E	Peer P/E × (1 − discount for non-listing)
Capitalised value	Maintainable profit ÷ Capitalisation rate
Capitalisation rate	EPS ÷ Price = 1 ÷ P/E (or CAPM)
Weighted average profit	Σ (profit × weight) ÷ Σ weights
Gross-up	PBT = PAT ÷ (1 − t)

A.4 Cash flow and cost of capital

Item	Formula
FCFF	EBIT(1 − t) + Depreciation − Capex − ΔNWC
FCFE	Net income + Depreciation − Capex − ΔNWC + Net borrowing
DCF enterprise value	$\Sigma\ FCFF_t \div (1 + WACC)^t + TV_n \div (1 + WACC)^n$
Equity value	EV − Debt − Preference + Cash/non-operating assets
CAPM	$R = Rf + \beta(Rm - Rf)$
APT	$R = Rf + \beta_1RP_1 + \beta_2RP_2 + \ldots + \beta_nRP_n$
Asset beta	$\beta_a = \beta_e \times E/[E + D(1 - t)] + \beta_d \times D(1 - t)/[E + D(1 - t)]$
Asset beta (β_d = 0)	$\beta_a = \beta_e \times E/[E + D(1 - t)]$
Re-gearing	$\beta_e = \beta_a \times [E + D(1 - t)]/E = \beta_a \times [1 + (1 - t)D/E]$

Item	Formula
WACC	Ke × E/(D + E) + Kd(1 − t) × D/(D + E)

A.5 Relative and other approaches

Item	Formula
Enterprise value	Market capitalisation + Debt − Cash (+ preference + minorities)
PEG	P/E ÷ Growth rate
Chop-shop value	Average of segment values from capitalisation/sales, /assets, /operating income
NOPAT	EBIT − Tax = EBIT(1 − t)
Invested capital	Total assets − Non-interest-bearing current liabilities (≈ Equity + Long-term debt)
EVA	NOPAT − (Invested capital × WACC)
EVA dividend per share	EVA ÷ Number of shares
MVA	Market value of firm − Invested capital
Financial leverage	PBIT ÷ PBT
SVA equity value	PV of FCFs + PV of TV + Non-core assets − Debt

A.6 Distressed, start-ups, ESG

Item	Formula
Expected cash flow	CF × (1 − Probability of distress)
Equity (DCF + distress)	DCF equity × (1 − P) + Distress equity × P
APV	Unlevered value + Tax benefits of debt − Expected bankruptcy cost
Expected bankruptcy cost	(Unlevered value − Distress sale value) × P
VC post-money	Exit value ÷ Required multiple (or ÷ $(1 + \text{IRR})^n$)
VC pre-money	Post-money − Investment
Scorecard value	Average pre-money valuation × Σ (weight × comparison %)
ESG-adjusted cash flow	FCFF − after-tax ESG cost (carbon, welfare, penalties)

A.7 Exchange ratio

Exchange ratio = Fair value per share of target ÷ Fair value per share of acquirer, where fair value per share = weighted average of values on different bases (e.g., weights 1 for NAV and 3 for earnings capitalisation).

Appendix B — Glossary

Adjusted present value (APV). Value of the unlevered firm plus tax benefits of debt minus expected bankruptcy costs.

Arm's length. A transaction between independent parties acting in their own interest.

Arbitrage Pricing Theory (APT). A multi-factor model of expected return: Rf plus the sum of factor betas times factor premia.

Asset beta (ungeared beta). Beta of a business's assets, reflecting business risk only.

Beta. Measure of a security's sensitivity to market movements (systematic risk).

Berkus approach. A start-up valuation method assigning value to five success factors, capped at $2 million pre-revenue and $2.5 million post-revenue.

Capitalisation rate. The rate at which maintainable earnings are converted into value; approximately EPS/price or 1/PE.

CAPM. Capital Asset Pricing Model: $R = Rf + \beta(Rm - Rf)$.

Chop-shop method. Valuation of a multi-industry company by valuing its segments separately using pure-play capitalisation ratios and averaging.

Company-specific risk premium (CSRP). An add-on to the cost of equity for risks not captured by beta (alpha).

Cost-to-duplicate approach. Start-up value as the cost of replicating the business, including development and assets.

De-gearing / re-gearing. Removing / re-applying the effect of financial leverage on beta.

Digital platform. Software-based online infrastructure that facilitates many-to-many interactions.

Distress sale value. Amount realised if a distressed firm's assets are sold in stress.

DCF. Discounted cash flow valuation.

EBITDA. Earnings before interest, tax, depreciation and amortisation.

Enterprise value (EV). Market capitalisation plus debt minus cash.

ESG. Environmental, Social and Governance factors.

EVA. Economic Value Added: NOPAT minus capital charge.

Fair value. Price in an orderly transaction between market participants; for an acquirer, the price "no less, no more".

First Chicago method. A start-up valuation combining DCF and market approaches across worst, normal and best scenarios with probabilities.

Free cash flow. Cash available to providers of capital after operating costs, taxes and reinvestment.

Going concern. Assumption that an enterprise will continue operating for the foreseeable future.

Gordon model. Valuation assuming dividends grow at a constant rate for ever: $P_0 = D_1/(Ke - g)$.

Goodwill. Value of a business above the value of its identifiable net assets.

Horizon value. Another name for terminal value.

Invested capital. Total assets less non-interest-bearing current liabilities (equity plus long-term debt).

IRR. Internal rate of return: the discount rate at which NPV is zero.

Leveraged buy-out (LBO). Acquisition financed mainly with debt (typically ~70:30 debt to equity), repaid from the target's cash flows.

Liquidation value. Net value realised after disposing of all assets and discharging liabilities.

Modified DCF. DCF using probability-weighted cash flows that reflect the likelihood of distress.

MVA. Market Value Added: market value of the firm minus invested capital.

Net asset value (NAV). Assets minus liabilities at book value.

Net realisable value (NRV). Realisable value of assets less liquidation expenses and liabilities.

NOPAT. Net operating profit after tax: $EBIT(1 - t)$.

Non-going-concern valuation. Liquidation valuation.

Normalisation. Adjusting reported figures to maintainable levels.

P/E ratio. Price per share divided by earnings per share.

PEG ratio. P/E divided by the expected growth rate.

Present value. Value today of a future cash flow, discounted at a rate reflecting time and risk.

Pure-play firm. A company engaged entirely in one business, used as a proxy for beta or multiples.

Relative valuation. Valuation by comparison with peers using multiples.

Replaceable value. Cost of acquiring equivalent assets in the open market.

ROI. Return on investment.

Scorecard method. Pre-revenue start-up valuation adjusting the average valuation of comparables by weighted factors.

SVA. Shareholder Value Analysis: DCF organised around value drivers.

Terminal value. Value of the business at the end of the explicit forecast period.

Value drivers. Factors that determine value (sales growth, margin, investment, cost of capital, forecast period).

Venture capital method. Start-up valuation by discounting expected exit value at the investor's required multiple or IRR.

WACC. Weighted average cost of capital.

Appendix C — Discount Factor Tables

Table C1 gives the present-value factor $1/(1+r)^n$; Table C2 the annuity factor $[1 - (1+r)^{-n}]/r$.

Table C1: Present value of ₹1

Year	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	18%	20%
1	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.862	0.847	0.833
2	0.890	0.873	0.857	0.842	0.826	0.812	0.797	0.783	0.769	0.756	0.743	0.718	0.694
3	0.840	0.816	0.794	0.772	0.751	0.731	0.712	0.693	0.675	0.658	0.641	0.609	0.579
4	0.792	0.763	0.735	0.708	0.683	0.659	0.636	0.613	0.592	0.572	0.552	0.516	0.482
5	0.747	0.713	0.681	0.650	0.621	0.593	0.567	0.543	0.519	0.497	0.476	0.437	0.402
6	0.705	0.666	0.630	0.596	0.564	0.535	0.507	0.480	0.456	0.432	0.410	0.370	0.335
7	0.665	0.623	0.583	0.547	0.513	0.482	0.452	0.425	0.400	0.376	0.354	0.314	0.279
8	0.627	0.582	0.540	0.502	0.467	0.434	0.404	0.376	0.351	0.327	0.305	0.266	0.233
9	0.592	0.544	0.500	0.460	0.424	0.391	0.361	0.333	0.308	0.284	0.263	0.225	0.194
10	0.558	0.508	0.463	0.422	0.386	0.352	0.322	0.295	0.270	0.247	0.227	0.191	0.162
11	0.527	0.475	0.429	0.388	0.350	0.317	0.287	0.261	0.237	0.215	0.195	0.162	0.135
12	0.497	0.444	0.397	0.356	0.319	0.286	0.257	0.231	0.208	0.187	0.168	0.137	0.112
13	0.469	0.415	0.368	0.326	0.290	0.258	0.229	0.204	0.182	0.163	0.145	0.116	0.093
14	0.442	0.388	0.340	0.299	0.263	0.232	0.205	0.181	0.160	0.141	0.125	0.099	0.078
15	0.417	0.362	0.315	0.275	0.239	0.209	0.183	0.160	0.140	0.123	0.108	0.084	0.065

Table C2: Present value of an annuity of ₹1 per year

Year	6%	7%	8%	9%	10%	11%	12%	13%	14%	15%	16%	18%	20%
1	0.943	0.935	0.926	0.917	0.909	0.901	0.893	0.885	0.877	0.870	0.862	0.847	0.833
2	1.833	1.808	1.783	1.759	1.736	1.713	1.690	1.668	1.647	1.626	1.605	1.566	1.528
3	2.673	2.624	2.577	2.531	2.487	2.444	2.402	2.361	2.322	2.283	2.246	2.174	2.106
4	3.465	3.387	3.312	3.240	3.170	3.102	3.037	2.974	2.914	2.855	2.798	2.690	2.589
5	4.212	4.100	3.993	3.890	3.791	3.696	3.605	3.517	3.433	3.352	3.274	3.127	2.991
6	4.917	4.767	4.623	4.486	4.355	4.231	4.111	3.998	3.889	3.784	3.685	3.498	3.326
7	5.582	5.389	5.206	5.033	4.868	4.712	4.564	4.423	4.288	4.160	4.039	3.812	3.605
8	6.210	5.971	5.747	5.535	5.335	5.146	4.968	4.799	4.639	4.487	4.344	4.078	3.837
9	6.802	6.515	6.247	5.995	5.759	5.537	5.328	5.132	4.946	4.772	4.607	4.303	4.031
10	7.360	7.024	6.710	6.418	6.145	5.889	5.650	5.426	5.216	5.019	4.833	4.494	4.192
11	7.887	7.499	7.139	6.805	6.495	6.207	5.938	5.687	5.453	5.234	5.029	4.656	4.327
12	8.384	7.943	7.536	7.161	6.814	6.492	6.194	5.918	5.660	5.421	5.197	4.793	4.439
13	8.853	8.358	7.904	7.487	7.103	6.750	6.424	6.122	5.842	5.583	5.342	4.910	4.533
14	9.295	8.745	8.244	7.786	7.367	6.982	6.628	6.302	6.002	5.724	5.468	5.008	4.611
15	9.712	9.108	8.559	8.061	7.606	7.191	6.811	6.462	6.142	5.847	5.575	5.092	4.675

Appendix D — A Second Problem Set (Exam-Style) with Solutions

Fifteen further problems, slightly longer than those of Chapter 22, in the manner of the syllabus questions. Amounts in ₹ crore unless stated. Attempt each before reading the solution.

Problems

S1 — Asset values. Fixed assets ₹1,200; net current assets ₹500; long-term debt ₹700; 20 crore shares. Realisable values: fixed assets ₹1,100; net current assets ₹430; liquidation expenses ₹40. Replacement cost of fixed assets ₹1,400; net current assets at book. Compute value per share on book, net realisable and replaceable bases.

S2 — Earnings approach with adjustments. Reported profit after tax ₹90 crore (tax 30%) includes extraordinary income of ₹14 crore and an extraordinary loss of ₹6 crore (both pre-tax). A new product will add ₹25 crore of profit before tax. Preference dividend ₹12 crore; 10 crore equity shares; capitalisation rate 15%; P/E for equity 9. Compute (a) the value of the business; (b) the market price per equity share.

S3 — Two-stage FCFF. Base year: revenue ₹1,000; EBIT ₹160; capex ₹120; depreciation ₹80. For 3 years revenue, EBIT, capex and depreciation grow 15%; thereafter 6% with capex equal to depreciation. Working capital is 20% of revenue; tax 30%. WACC 12% in the high-growth years and 10% in the stable phase. Compute the value of the firm.

S4 — Beta, WACC and value. Asset beta 0.8; target D:E = 30:70; tax 25%; risk-free 6%; market return 12.5%; pre-tax cost of debt 11%. EBITDA ₹120 includes an extraordinary gain of ₹8 and a pending write-off of ₹12; EV multiple 6. Future free cash flows: ₹50, ₹60, ₹75. Compute β_e, Ke, WACC, adjusted EBITDA, EV by multiple and the PV of the three flows.

S5 — EVA. Revenue ₹2,000; direct costs ₹900; SG&A ₹400 (including a bad-debt provision of ₹30); interest ₹40; tax 30%. Total assets ₹3,000; non-interest-bearing current liabilities ₹700; debt ₹600; equity ₹1,700 (the same provision reduces receivables). Ke 14%; pre-tax Kd 10%. Compute EVA.

S6 — Exchange ratio with three methods. Company A: NAV per share ₹150; earnings value ₹300; market price ₹360. Company B: ₹60; ₹120; ₹150. Weights 1 : 2 : 3. Compute the exchange ratio.

S7 — Chop-shop by EV/EBITDA. Segment EBITDAs: A ₹200 (peer multiple 7×); B ₹150 (9×); C ₹90 (11×). Unallocated corporate costs ₹30 capitalised at 8×. Net debt ₹600; 25 crore shares. Compute value per share.

S8 — Distressed company (modified DCF). Cash flows if surviving: ₹80, ₹100, ₹130 plus a terminal value of ₹900 at year 3. Cumulative probabilities of distress by year 1, 2, 3: 10%, 25%, 40%. Discount rate 13%. Compute the expected value and compare with the value if survival were certain.

S9 — Scorecard. Average pre-money valuation of comparable start-ups ₹20 crore. Comparison percentages: team 120% (weight 30%), opportunity 100% (25%), product 110% (15%), competition 90% (10%), marketing 80% (10%), funding need 100% (5%), others 100% (5%). Compute the pre-money value.

S10 — LBO. Buy a business with EBITDA ₹100 at 6×; 70% debt. EBITDA grows 10% a year for 4 years; debt repaid ₹200 over the period; exit at 6×. Compute sponsor equity, exit equity, money multiple and IRR.

S11 — Digital platform. A comparable platform was acquired at ₹350 per active user. The target has 15 lakh active users and deserves a 10% premium for faster growth. Value it.

S12 — ESG. FCFF next year ₹150; WACC 11%; growth 4%. (a) A social-compliance cost of ₹10 pre-tax a year (tax 30%, growing 4%) will arise. (b) A governance penalty of ₹6 pre-tax is expected in each of the next two years. Compute the effect on EV of each.

S13 — Enterprise value. 5 crore shares at ₹90; debt ₹150; cash ₹60; EBITDA ₹70. Compute market capitalisation, EV and EV/EBITDA.

S14 — Going versus liquidation. DCF value ₹700. Assets could be sold for ₹920, selling costs ₹40, tax on gains ₹30; liabilities settled separately (ignore). What is the better course and what is the going-concern premium or discount?

S15 — SVA. FCFs: ₹40, ₹45, ₹50, then ₹50 a year for ever from year 4; WACC 12%; non-core investments ₹30; debt ₹60. Compute the equity value (exact discounting of the perpetuity).

Solutions

S1. *Book:* 1,200 + 500 − 700 = **₹1,000** → **₹50** per share. *NRV:* 1,100 + 430 − 40 − 700 = **₹790** → **₹39.5**. *Replaceable:* 1,400 + 500 − 700 = **₹1,200** → **₹60**. Here NRV is *below* book because debtors/stock are written down and costs deducted; replacement is the highest.

S2. (a) PBT = 90 ÷ 0.7 = 128.57; remove the extraordinary income (−14) and the extraordinary loss (+6): 120.57; add new product 25: **145.57**; tax 30%: PAT = **101.9**; value = 101.9 ÷ 0.15 = **₹679.3 crore**. (b) Earnings for equity = 101.9

− 12 = **89.9**; EPS = 8.99; price = 8.99 × 9 = **₹80.9**.

S3. *High growth:* FCFF = EBIT × 0.7 − (capex − depreciation) − 20% × Δrevenue.
Year 1: 184 × 0.7 = 128.8 − (138 − 92 = 46) − 0.2 × 150 = 30 → **52.8**. Year 2: 211.6 × 0.7 = 148.1 − 52.9 − 34.5 = **60.72**. Year 3: 243.3 × 0.7 = 170.3 − 60.8 − 39.7 = **69.83**.
PV at 12% = 47.14 + 48.40 + 49.70 = **145.25**.
Terminal year (growth 6%): $EBIT_4$ = 243.34 × 1.06 = 257.94; EAT = 180.56; net capex 0; $revenue_3$ = 1,520.9 and $revenue_4$ = 1,520.9 × 1.06 = 1,612.2, so Δrevenue = 91.3 and ΔNWC = 18.26; $FCFF_4$ = 180.56 − 18.26 = **162.31**. TV = 162.31 ÷ (0.10 − 0.06) = **4,057.7**; PV (÷ 1.12^3) = **2,888.2**.
Value of the firm = 145.25 + 2,888.19 = ₹3,033.4 crore (terminal value 95%).

S4. β_e = 0.8 × [1 + 0.75 × 30/70] = 0.8 × 1.3214 = **1.057**; Ke = 6% + 1.057 × 6.5% = **12.87%**; WACC = 12.87% × 0.70 + 11% × 0.75 × 0.30 = 9.01% + 2.475% = **11.485%**. Adjusted EBITDA = 120 − 8 − 12 = **100**; EV = 6 × 100 = **600**. Discount factors 0.897, 0.805, 0.722 → PVs 44.85 + 48.27 + 54.13 = **147.25** (three-year flows only; no terminal value).

S5. NOPAT = (2,000 − 900 − 400) × 0.7 + 30 = 490 + 30 = **520**. Invested capital = 3,000 − 700 + 30 = **2,330**. WACC = (1,700 × 14% + 600 × 10% × 0.7) ÷ 2,300 = (238 + 42) ÷ 2,300 = **12.17%**. Capital charge = 283.65. **EVA = 520 − 283.65 = ₹236.35 crore.**

S6. A = (150×1 + 300×2 + 360×3) ÷ 6 = (150 + 600 + 1,080) ÷ 6 = **₹305**. B = (60 + 240 + 450) ÷ 6 = **₹125**. Ratio = 125 ÷ 305 = **0.4098** A shares per B share.

S7. A: 200 × 7 = 1,400; B: 150 × 9 = 1,350; C: 90 × 11 = 990; total 3,740; less capitalised corporate costs 30 × 8 = 240 → **EV 3,500**. Equity = 3,500 − 600 = **2,900**; per share = **₹116**.

S8. Expected cash flows: Year 1: 80 × 0.90 = 72; Year 2: 100 × 0.75 = 75; Year 3: (130 + 900) × 0.60 = 618. PV at 13%: 63.7 + 58.7 + 428.3 = **₹550.8**. Survival certain: 70.8 + 78.3 + 714.0 = **₹863.0**. Distress probabilities remove 36% of value.

S9. Factors: 0.36 + 0.25 + 0.165 + 0.09 + 0.08 + 0.05 + 0.05 = **1.045**; value = 1.045 × 20 = **₹20.9 crore**.

S10. Entry EV = 600; debt = 420; equity = **180**. EBITDA in year 4 = 100 × 1.1^4 = 146.4; exit EV = 878.5; debt at exit = 420 − 200 = 220; **exit equity = 658.5**. Multiple = 3.66×; IRR = $3.66^{1/4}$ − 1 = **38.3%**. Source of return: EBITDA growth (46.4 × 6 = 278), debt paydown (200) and no multiple expansion; equity gain = 478.5 = 278.5 + 200 ✔.

S11. 15 lakh × ₹350 = ₹52.5 crore; with a 10% premium: **₹57.75 crore**.

S12. Baseline EV = 150 ÷ 0.07 = **₹2,142.9**. (a) After-tax social cost = 10 × 0.7 = 7; EV = (150 − 7) ÷ 0.07 = **₹2,042.9** → −₹100 (**−4.7%**). (b) Penalty after tax = 4.2 in each of two years: PV = 4.2 ÷ 1.11 + 4.2 ÷ 1.2321 = 3.78 + 3.41 = **₹7.2 crore** (−0.3%) — material to the board but small in value.

S13. Market cap = 5 crore × ₹90 = **₹450 crore**; EV = 450 + 150 − 60 = **₹540 crore**; EV/EBITDA = 540 ÷ 70 = **7.7×**.

S14. Net liquidation proceeds = 920 − 40 − 30 = **₹850** > going-concern value 700. The business is worth **₹150 crore more liquidated** (going-concern *discount* of 150, or 17.6% below liquidation value). Rational course: sell/liquidate (or restructure to raise going-concern value above 850). Valuation on a going-concern basis is justified only if value exceeds liquidation value.

S15. PV of years 1–3: 40 ÷ 1.12 + 45 ÷ 1.2544 + 50 ÷ 1.4049 = 35.71 + 35.87 + 35.59 = **107.18**. Perpetuity of 50 from year 4: value at the end of year 3 = 50 ÷ 0.12 = 416.67; PV = 416.67 ÷ 1.4049 = **296.58**. Total PV = 403.75; **equity = 403.75 + 30 − 60 = ₹373.75 crore.**

Appendix E — Self-Test Question Bank

Fifty short questions covering the whole book. Cover the answers and test yourself; each question refers to the chapter where the topic is taught.

Part 1 — True or false? (Answers follow)

1. The market price of a listed company is always equal to its intrinsic value. *(Ch. 1)*
2. Net asset value on book basis provides a lower limit to the value of the shares. *(Ch. 3)*
3. Replaceable value is generally slightly higher than net realisable value. *(Ch. 3)*
4. For an unlisted company valued by the P/E method, the peer's P/E should be adjusted upward. *(Ch. 4)*
5. The capitalisation rate can be approximated by the reciprocal of the P/E ratio. *(Ch. 4)*
6. In DCF, the terminal value must be discounted back to the valuation date. *(Ch. 2, 5)*
7. CAPM divides risk into a risk-free part and a part relative to the market. *(Ch. 6)*
8. APT specifies the macro-economic factors the analyst must use. *(Ch. 6)*
9. Equity beta is always less than asset beta for a levered company. *(Ch. 7)*
10. Debt beta is generally assumed to be zero. *(Ch. 7)*
11. EV/Invested capital is most appropriate for asset-light firms. *(Ch. 8)*
12. PEG ratio is suited to high-growth or sunrise industries. *(Ch. 8)*
13. In an LBO, the typical debt-to-equity mix is about 70:30. *(Ch. 9)*
14. In the chop-shop method one uses the capitalisation ratios of the target company itself. *(Ch. 10)*
15. EVA is computed after deducting a charge for both debt and equity capital. *(Ch. 11)*
16. MVA equals the market value of the firm plus invested capital. *(Ch. 11)*
17. SVA ignores the cost of incremental capital. *(Ch. 12)*
18. Going-concern value is generally higher than liquidation value. *(Ch. 13)*
19. For a distressed firm, P/E is the best multiple. *(Ch. 14)*
20. In APV, expected bankruptcy cost = (unlevered value − distress sale value) × probability of distress. *(Ch. 14)*
21. The Berkus approach caps pre-revenue valuations at $2 million. *(Ch. 15)*

22. The scorecard method gives the highest weight to the strength of the team. *(Ch. 15)*
23. For digital platforms, the market approach emphasises revenue drivers. *(Ch. 16)*
24. In the top-down approach, SOM is the total addressable market. *(Ch. 16)*
25. For professional firms, normalisation of net income and cash flows allows comparison on equal footing. *(Ch. 17)*
26. Incorporating ESG through the discount rate is more explicit than through cash flows. *(Ch. 18)*
27. In EVA, unrecorded patents should be included in invested capital. *(Ch. 11, 21)*
28. The exchange ratio is the acquirer's value per share divided by the target's. *(Ch. 25)*
29. Capex equals the change in net fixed assets plus depreciation. *(Ch. 26)*
30. Under Ind AS 113, most DCF valuations of unlisted companies use Level 3 inputs. *(Ch. 28)*

Part 1 answers

1. **False** — price may differ from value. 2. **True.** 3. **True.** 4. **False** — adjust *downward* for non-listing. 5. **True.** 6. **True.** 7. **True.** 8. **False** — the analyst must choose the factors. 9. **False** — equity beta ≥ asset beta. 10. **True** (usual assumption). 11. **False** — capital-intensive firms. 12. **True.** 13. **True.** 14. **False** — pure-play peers' ratios. 15. **True.** 16. **False** — minus invested capital. 17. **False** — it explicitly includes it. 18. **True.** 19. **False** — revenue/EBITDA multiples are preferred. 20. **True.** 21. **True.** 22. **True** (0–30%). 23. **True.** 24. **False** — SOM is the serviceable *obtainable* market. 25. **True.** 26. **False** — cash flows are more explicit. 27. **True.** 28. **False** — target ÷ acquirer. 29. **True.** 30. **True.**

Part 2 — Short numerical questions

31. PV of ₹2,000 receivable in 2 years at 10%.
32. Value: profit ₹60 crore; capitalisation rate 12%.
33. Price: EPS ₹7; P/E 11 adjusted 20% for non-listing.
34. Unlevered beta 0.9; D/E 40%; tax 25%; levered beta?
35. WACC: Ke 15%, Kd 8% pre-tax, tax 25%, D/V 25%.
36. NOPAT 200; invested capital 1,000; WACC 12%; EVA?
37. Market value 1,800; invested capital 1,400; MVA?
38. Equity: DCF 400; distress 50; P = 20%.

39. VC: exit 120 in 4 years; investor wants 6×; investment 10. Pre-money?
40. EV: market cap 700; debt 200; cash 90.

Part 2 answers

31. 2,000 ÷ 1.21 = **₹1,652.9**. 32. 60 ÷ 0.12 = **₹500 crore**. 33. 7 × 11 × 0.8 = **₹61.6**. 34. 0.9 × (1 + 0.75 × 0.4) = **1.17**. 35. 0.75 × 15 + 0.25 × 8 × 0.75 = 11.25 + 1.5 = **12.75%**. 36. 200 − 120 = **₹80**. 37. **₹400**. 38. 320 + 10 = **₹330**. 39. Post = 120 ÷ 6 = 20; pre = **₹10**. 40. 700 + 200 − 90 = **₹810**.

Part 3 — Short answers

41. Name the three core approaches to valuation.
42. State two limitations of the asset-based approach.
43. What are the steps for valuing an unlisted company by P/E?
44. Why is the terminal value so important in DCF?
45. Differentiate asset beta and equity beta.
46. Write one reason why the chop-shop method may undervalue or overvalue a firm.
47. What does a negative EVA signify?
48. List the four methods for valuing distressed companies.
49. Name three value drivers for start-ups.
50. How is ESG incorporated into cash flows for E, S and G?

Part 3 answers

41. Asset-based, earnings-based (income) and cash-flow-based (DCF). 42. Ignores earning power and future cash flows; uses historical cost; omits intangibles. 43. Choose the equivalent quoted company's P/E; adjust downward for non-listing; determine maintainable EPS; multiply. 44. It represents the value of all cash flows beyond the explicit period and is often 70–90% of the value; sensitive to growth and discount rate. 45. Asset beta measures only business risk (no leverage); equity beta adds financial risk and is higher for levered firms. 46. Segment peers may not be comparable, allocation of costs is arbitrary, and separation costs/synergy losses are ignored. 47. The firm earns less than its cost of capital; it destroys value. 48. Modified DCF; DCF plus distress value; adjusted present value; relative valuation. 49. Any three of product, management, traction, revenue, industry attractiveness, demand–supply, competitiveness. 50. E: 2-degree scenario analysis and carbon points; S: cost of social measures on revenue (labour conditions, CSR); G: penalties, fines and taxes from poor governance.

How to score

Score (of 50)	Interpretation
45–50	Excellent; proceed to timed practice
38–44	Good; revisit the chapters for the misses
30–37	Fair; re-read Parts II–IV and redo worked examples
< 30	Re-read from Chapter 1 and work the "Check your understanding" boxes

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